Uncertainty and the Shadow Banking Crisis: Estimates from a Dynamic Model *

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November 2020

Abstract

Shadow banks play an important role in the modern financial system and are arguably the source of key vulnerabilities that led to the 2007-2009 financial crisis. I develop a quantitative framework with uncertainty fluctuations and endogenous bank default to study the dynamics of shadow banking. I argue that the increase in asset return uncertainty during the crisis results in a spread spike, making it more costly for shadow banks to roll over their debt in the short-term debt market. As a result, these financial institutions are forced to deleverage, leading to a decrease in credit intermediation. The model is estimated using a bank-level dataset of shadow banks in the United States. The parameter estimates imply that uncertainty shocks can explain 72% of asset contraction and 70% of deleveraging in the shadow banking sector. Maturity mismatch and asset fire-sales amplify the impact of the uncertainty shocks. First-moment shocks to bank asset return, financial shocks, or fire-sale cost shocks alone can not reproduce the large interbank spread spike, dramatic deleveraging or contraction in the U.S. shadow banking sector during the crisis. The model also allows for policy experiments. I analyze how unconventional monetary policies can help to counter the rise in the interbank spread, thus stabilizing the credit supply. Taking bank moral hazard into consideration, I find that government bailout might be counterproductive as it might result in more aggressive risk-taking among shadow banks, especially when bailout decisions are based on bank characteristics.

Keywords: shadow banking, uncertainty, maturity mismatch, fire-sale, unconventional monetary policy.
JEL Classification: D81, E32, E44, E50, G18, G20.

*Contact Information: Xu Tian, 150 St. George Street, University of Toronto, Toronto, Ontario, M5S 3G7. Email: xu.tian@utoronto.ca. I would like to thank the editor Gustavo Manso, an anonymous associate editor, and two anonymous referees for their great comments that helped me improve the paper. I am grateful to my advisors Yan Bai and Toni Whited for their continuous guidance and support. I would also like to thank Victor Aguirregabiria, George Alessandria, Mark Bils, Nick Bloom, Brent Bundick, Charles Calomiris, Yongsung Chang, Dong Beom Choi, Stephan Danninger, Steven Davis, Tuo Deng, Olga Itenberg, Dan Lu, Tommaso Mancini-Griffoli, Ryan Michaels, Gabriel Mihalache, Kris James Mitchener, Yena Park, Jianping Qi, Vincenzo Quadrini, Robert Ready, Ivan Werning, Stephen Terry, and Xiaodong Zhu for their great comments for this paper. I would also like to thank discussants and other participants in the Midwest Macro Meeting 2015, Georgetown GCER 2015, Money, Macro and Finance Conference 2015, FMA 2015 Annual Meeting, FIRS 2016, RES PhD meeting 2016, Stanford Institute for Theoretical Economics (SITE) Workshop 2017, and seminar participants at the University of Rochester, Bank of Canada, University of Toronto, Georgetown University, Temple University, Peking University, Baylor University, CKGSB, and the International Monetary Fund for their helpful comments and discussions. Special thanks to WestGrid Computing and Center for Integrated Research Computing(CIRC) at the University of Rochester for computational support. All remaining errors are my own.

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1 Introduction

The 2007-2009 financial crisis is distinguished from previous crises by the role the shadow banking sector played. Shadow banks are financial intermediaries that conduct maturity, credit, and liquidity transformation without explicit access to central bank liquidity or public sector credit guarantees. They play a major role in global finance and are arguably the source of key vulnerabilities that led to the recent crisis. Different from traditional banks whose funding mainly depends on customer deposits, shadow banks finance their asset purchases primarily by means of collateralized debt with very short maturity, such as repurchase agreements (repo) or asset-backed commercial papers (ABCP). The markets for these financial instruments are typically highly liquid. However, during the recent financial crisis, the initial loss of some of the assets (mainly subprime mortgage-backed securities) that serve as collateral in the repo or ABCP transactions[1], together with the uncertainty surrounding individual exposures to such assets, led financial institutions to largely stop exchanging liquidities in these markets. The interbank spread spiked and market liquidity shrank dramatically, forcing shadow banks to deleverage and ultimately resulting in a contraction in credit intermediation. Despite the substantial and growing literature on uncertainty shocks and shadow banks, little work has been done to quantify the extent to which uncertainty shocks could explain the large deleveraging and asset contraction in the shadow banking industry and study the impact of alternative shocks in a comprehensive manner. This paper aims to fill the gap by providing a tractable quantitative framework to dissect the shadow banking dynamics.

To what extent do uncertainty shocks matter for the dramatic asset contraction and deleveraging in the shadow banking industry? Do certain features in the shadow banking industry, such as high maturity mismatch and fire-sale cost, play any role in amplifying the impacts of uncertainty shocks? If yes, how much? Could first-moment shocks to asset returns, financial shocks, or asset liquidation shocks explain the observed shadow banking dynamics during the 2007-2009 financial crisis? Answering these questions in a purely empirical setting is challenging. First, multiple shocks hit the economy in the crisis, and these shocks are not directly observable. To identify the key underlying shocks caused the asset contraction and deleveraging, we need to observe what would have happened in parallel, counterfactual worlds in which some shocks are absent. Second, to assess to what extent

[1] Covitz, Liang, and Suarez (2014) argue that contraction in the ABCP market played a central role in transforming concerns about the credit quality of mortgage-related assets into a global financial crisis.

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maturity mismatch and fire-sale cost amplify the adverse shocks, we need to observe what would have happened counterfactually if there were no maturity mismatch or fire-sale costs. Evaluating these counterfactuals is difficult, because it is hard to find exogenous variations that eliminate other shocks or remove maturity mismatch and fire-sale costs. Even if there were such a shock, it is likely to be limited in scope, raising concerns about external validity. Overall, it is unclear how to answer these quantitative questions without a model.

In this paper, I overcome these challenges by developing and structurally estimating a quantitative model with uncertainty fluctuations and endogenous bank default to explain the increase in the interbank spread, the deleveraging, and the contraction of shadow banking credit intermediation during the recent financial crisis. The increase in bank asset return uncertainty raises the bank default probability, leading to an increase in the interbank market spread.[2] Facing a higher cost of financing, banks are forced to deleverage, thus decreasing credit intermediation. Using this framework, I also analyze how unconventional monetary policy (debt guarantees and the Troubled Asset Relief Program (TARP)) can help to reduce the interbank spread and stimulate the supply of credit. I also show that government bailout might be counterproductive as it might result in more aggressive risk-taking among shadow banks, especially when bailout decisions are based on bank characteristics. The anticipation of such bailouts might distort ex ante borrowing incentives: an individual bank has an incentive to borrow more and increase its size more aggressively to take advantage of the government bailout.

Using a comprehensive bank-level data set, I first present some motivating empirical evidence regarding the U.S. shadow banking industry. I find that during the 2007-2009 financial crisis, the asset return uncertainty of shadow banks increased by 147%; shadow banks deleveraged much more than traditional banks and shrank their asset size dramatically; they depend heavily on short-term funding rather than customer deposits and tend to choose riskier asset portfolios than traditional banks; the asset contractions and deleveraging are stronger in shadow banks with higher fire-sale costs and higher maturity mismatch, implying that fire-sale cost and maturity mismatch are potentially important attributes which might amplify the impact of adverse shocks.

This paper's contribution is mainly theoretical and quantitative. To shed light on the key mechanisms that underlie the observed empirical facts, I build a quantitative model with uncertainty

[2]Since the major participants in the short-term collateralized debt market are financial institutions, in this paper, I use the term "interbank market" and "short-term collateralized debt market" interchangeably. See Pozsar, Adrian, Ashcraft, and Boesky (2013) for an in-depth analysis about shadow banking and short-term collateralized debt markets.

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fluctuations and endogenous bank default. In the model, shadow banks borrow from the interbank market using short-term risky debt and invest in a long-term risky asset. In each period, every bank receives an idiosyncratic shock to its asset return. Depending on the realization of asset return and debt due, each bank decides whether to default, how much new debt, equity, and assets to issue, and how much long-term asset to fire-sale (if necessary) to pay its debts due. Following the literature, assuming that the idiosyncratic asset return shock is drawn from a log-normal distribution, when the standard deviation of the distribution increases, the bank's asset return becomes riskier and its expected default probability would increase; thus it would be charged a higher interest rate in the interbank market. Facing a higher cost of financing and a higher probability of costly asset fire-sales, shadow banks reduce their leverage and assets.

Using simulated method of moments (SMM), the model parameters are estimated to best fit a wide-ranging set of facts about the U.S. shadow banking industry, such as moments from the distribution of leverage, new asset issuance, default probability, and dividend payouts. Using the estimated parameters, I document the model's implications for the shadow banking dynamics by conducting an event study on the 2007-2009 financial crisis and several counterfactual experiments in the spirit of recent microeconometric analyses. These out-of-sample tests reveal that uncertainty shocks can generate statistics closely match the data and explain 72% of asset contraction and 70% of deleveraging in the U.S shadow banking industry. In particular, when running the same regressions using the model-generated data, the estimated coefficient for the impact of the asset return uncertainty on shadow bank leverage is quantitatively similar to its empirical counterpart even though it is not targeted in the estimation. I also find that high maturity mismatch and fire-sale costs amplify the impact of uncertainty shocks. When a bank is hit by uncertainty shocks, the higher is the degree of maturity mismatch, the more likely the bank will be forced to fire-sale its long-term assets. The higher is the fire-sale cost, the more likely the bank will fail to roll over its debt and go bankrupt. Alternative models with only first-moment shocks (shocks to the level of asset return), models without asset fire-sale costs or maturity mismatch, cannot reproduce the large spike in the interbank spread, dramatic deleveraging, or contraction in the U.S. shadow banking industry in the recent financial crisis.

I further extend the baseline model to study comprehensively the impact of alternative shocks that could have explained the contraction of credit in the shadow banking sector. Specifically, I study the joint contributions of uncertainty shocks, first-moment shocks to asset return, shocks to the

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liquidation/resale value of assets, and financial shocks. I find that financial shocks and uncertainty shocks together can explain around 76% of asset contraction and 77% of delveraging. By comparison, financial shocks and first-moment shocks together can only explain 25% of asset contraction and 26% of deleveraging. Without uncertainty shocks and first-moment shocks, financial shocks alone can only explain 11% of asset contraction and 12% deleveraging. Fire-sale cost shocks and uncertainty shocks together can explain around 78% of asset contraction and 77% of deleveraging. Fire-sale cost shocks and first-moment shocks together can only explain 27% of asset contraction and deleveraging. If we shut down uncertainty shock and first-moment shocks, fire-sale cost shocks alone in the extended model can only explain 14% of asset contraction and 13% of deleveraging.

This paper contributes to the body of literature on the shadow banking crisis. Following Gorton and Metrick (2012), who argue that the panic of 2007-2008 was a run on the short-term collateralized debt market, various studies have emphasized the role of the shadow banking sector in the origin and propagation of the financial turmoil. Pozsar, Adrian, Ashcraft, and Boesky (2013) maintain that the shadow banking sector played a major role in the 2007-2009 financial crisis. Krishnamurthy, Nagel, and Orlov (2014) document how the collapse of the ABCP market led to the implosion of the intermediation chain within the shadow banking sector. Adrian and Shin (2010) argue that due to the limited liability of the short debt contracts of shadow banks, during the crisis, creditors in the interbank market were less willing to roll over their lending, resulting in the procyclicality of financial intermediaries' leverage and credit supply. Chodorow-Reich (2014) argues that the contraction of the credit supply matters more for the decline in output than the decrease in credit demand, and interbank market disruption played a crucial role in the decrease in credit intermediation. Schroth, Suarez, and Taylor (2013) use the 2007 ABCP crisis as a laboratory to study the determinants of debt runs. Corbae and D'Erasmo (2018) develop a model of banking industry dynamics to study the relation between commercial bank market structure, entry and exit along the business cycle. The shocks in these papers are all first-moment shocks (TFP shocks or investment quality shocks). Different from these papers, I focus on the impact of uncertainty shocks (also called second-moment shocks) on the financing and investment decisions of shadow banks. Moreira and Savov (2016) build a macro-finance model to study shadow banking dynamics. Buchak, Matvos, Piskorski, and Seru (2018) use a model to decompose the relative contribution of technology and regulation to the rise of shadow banks. Nuno and Thomas (2017) also build a model to analyze the impact of uncertainty shocks on banks' leverage

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choices. They use log-linear approximations to study how financial frictions amplify shocks near the steady state of the economic system. The quantification based on the perturbation method is more precise if the economy is not knocked too far away from the steady state. By comparison, my model characterizes financial intermediary dynamics in a fully nonlinear manner. Such nonlinearities can generate rich and interesting dynamics. Also, in their model, banks do not make endogenous strategic default decisions, whereas my model allows for such a possibility. Although the literature on shadow banking is growing, existing literature only studies shadow banking activity empirically or analyzes commercial banks' behaviors using calibrated models.[3] To the best of my knowledge, no previous studies have used comprehensive micro-level shadow banking data to structurally estimate the impacts of time-varying uncertainty and financial frictions in this sector. This paper fills the gap.

This paper is complementary to the growing body of literature studying the impact of uncertainty on the real economy, such as Gilchrist, Sim, and Zakrajsek (2014), Christiano, Motto, and Rostagno (2014), Bloom, Floetotto, Jaimovich, Saporta-Eksten, and Terry (2019), Arellano, Bai, and Kehoe (2018), and Alfaro, Bloom, and Lin (2019). All these papers focus on the impact of uncertainty on non-financial firms' choice about investment or employment, whereas I focus on the decisions of financial intermediaries. In the earlier studies, the marginal rate of substitution of financial intermediaries is often implicitly assumed to be equal to that of households or firms. Thus, financial intermediaries always act on behalf of households or firms and their decisions are often not modeled. However, one key feature of the recent crisis is the disruption of financial intermediation. Without explicitly modeling the behaviors of financial institutions, we might not be able to fully understand the cause of this crisis.

Finally, this paper also adds to the literature concerning interbank markets and unconventional monetary policy, such as Gertler and Kiyotaki (2010). Different from Gertler and Kiyotaki (2010), in my model, a bank's debt is risky and the bank might choose to default in equilibrium, whereas in Gertler and Kiyotaki (2010), interbank loans are fully collateralized and there is no default risk in

[3]Calibration often only tries to match a few stylized facts with many model parameters, and whenever there are too many degrees of freedom, inference is impossible. In contrast, structural estimation matches at least as many stylized facts as model parameters, and therefore, standard errors can be calculated in the same way as GMM standard errors. It tries to find the unique set of model parameters that reconcile the model with the data. It is thus an explicitly empirical exercise that attempts to "stress test" a single model to compare the performance of competing models. See Strebulaev and Whited (2012) for a detailed comparison of calibration and structural estimation. My study adds to the emerging literature that employs structural model estimation to answer standard corporate finance questions in investment, leverage choice, dividend payout, liquidity management, CEO compensation and turnover (See Hennessy and Whited (2005), Hennessy and Whited (2007), Rampini and Viswanathan (2010), Korteweg (2010), Taylor (2010), Bolton, Chen, and Wang (2011), Morellec, Nikolov, and Schurhoff (2012), Glover and Levine (2017),Terry (2017), Wu (2018), and Nikolov, Schmid, and Steri (2018) among others).

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the equilibrium. Moreover, in this strand of literature, crises are often driven by first-moment (levels) shocks such as investment quality shock, while I argue that even if the expected level of asset return shock remains unchanged, an increase in the uncertainty of asset return can result in a freeze in the interbank market and the reduction in the supply of credit. Furthermore, I take into account the moral hazard behavior of shadow banks by endogenizing the bailout probability as a function of the bank size. Researchers have typically analyzed how credit policy can mitigate a credit crunch ex-post or theoretically investigated the moral hazard of government bailout. This paper contributes to fill this gap by developing a quantitative model to assess the interaction between ex-post interventions and the build-up of risk ex-ante in a tractable unified framework.

The paper is organized as follows. In Section 2, I document several stylized facts of the U.S. shadow banking industry. In Sections 3 and 4, I construct and estimate a quantitative model with uncertainty fluctuations and endogenous bank default to study the shadow banking dynamics. I check the cross-sectional and time-series fitness of the model in Section 5 by estimating the model based on different subsamples. In Section 6, I further extend the baseline model to study in a comprehensive manner the impact of alternative shocks that could have explained the asset contraction and deleveraging in the shadow banking sector. I also consider alternative extensions of the model with asset portfolio choice, systemic risk, time-varying aggregate productivity volatility, and a traditional banking sector. Concluding remarks are provided in Section 7.

2 Motivating Evidence

2.1 Data Source

In this section, I document several stylized facts about the U.S. banking industry which motivates my model. Since Gorton and Metrick (2012), there has been an outpouring of theoretical work on the shadow banking sector but the empirical work using micro-level data of non-bank financial institutions is only slowly catching up. I obtain yearly bank-level panel data from the Bankscope and Worldscope databases. Bankscope and Worldscope contain comprehensive information on financial institutions across the globe. I choose real estate and mortgage banks, investment banks, securities firms, private banking and asset management companies, investment and trust corporations, finance companies and

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other nonbank financial institutions in the U.S. as representatives of shadow banks.[4] I also choose commercial banks, savings banks, cooperative banks, and bank holding companies as representatives of traditional banks. The final panel data sample contains 281 shadow banks and 9554 traditional banks from 1998 to 2013, covering 49 states in the U.S.

2.2 Stylized Facts

Stylized fact 1: The asset return uncertainty in the shadow banking sector increased in the 2007-2009 financial crisis.

The left panel of Figure 1 illustrates the kernel density of return on assets (ROA) [5] for shadow banks during non-crisis periods and crisis periods.[6] The standard deviation of shadow banks' ROA increases by 48% during the crisis. To obtain a clearer idea about the magnitude of the change, I also show the change in traditional banks' ROA standard deviation for reference. By comparison, the ROA standard deviation of traditional banks increases by only 22%, less than one-half that of shadow banks. Explaining the differences between shadow banks and commercial banks is not the focus of this paper. All the empirical analyses for traditional banks are mainly shown for comparison and to provide a better understanding of shadow banks.[7]

To measure the uncertainty of banks' ROA, I follow a standard approach from Foster, Haltiwanger, and Krizan (2001). I assume the bank gross return on assets follows an AR(1) process. To ease notation, I define the bank gross asset return rate as $groa_{it} = 1 + ROA_{it}$. Then I fit the gross asset return rate into an AR(1) process, controlling for time and bank-level fixed effects, as follows:

$$log\left(groa_{it}\right) = \rho log\left(groa_{it-1}\right) + \psi_t + \mu_i + X_{it} + \epsilon_{it}, \tag{1}$$

where ψ_t is the year fixed effect (controlling for cyclical shocks) and μ_i is bank-level fixed effect

[4] My classification of shadow banks is similar to the Financial Stability Board's measure of shadow banking activity based on nonbank financial intermediaries (NBFIs) engaged in credit intermediation activities. Only less than 3% of shadow banks changed their specializations in our sample. Including or excluding these institutions would not significantly change the estimation results.

[5] ROA is bank's return on assets, which is one of the standard measures in the banking literature. ROA is define as (net income +interest payment)/total assets. Following the literature, I add back the interest payment so that the return rate is the rate before interest payment and the return's definition is consistent with the model in Section 3.

[6] Following Bloom, Floetotto, Jaimovich, Saporta-Eksten, and Terry (2019), non-crisis periods are defined as from the 2005 to 2006, whereas crisis periods are defined as from 2008 to 2009. According to the NBER, the recession began in 12/2007, so 2007 is not a clean "before" or "during" recession year.

[7] I focus on exploring what shocks and features of shadow banks led to the observed stylized facts in shadow banking sector in the main text of this paper and relegate the comparison between these two sectors to Section 6.

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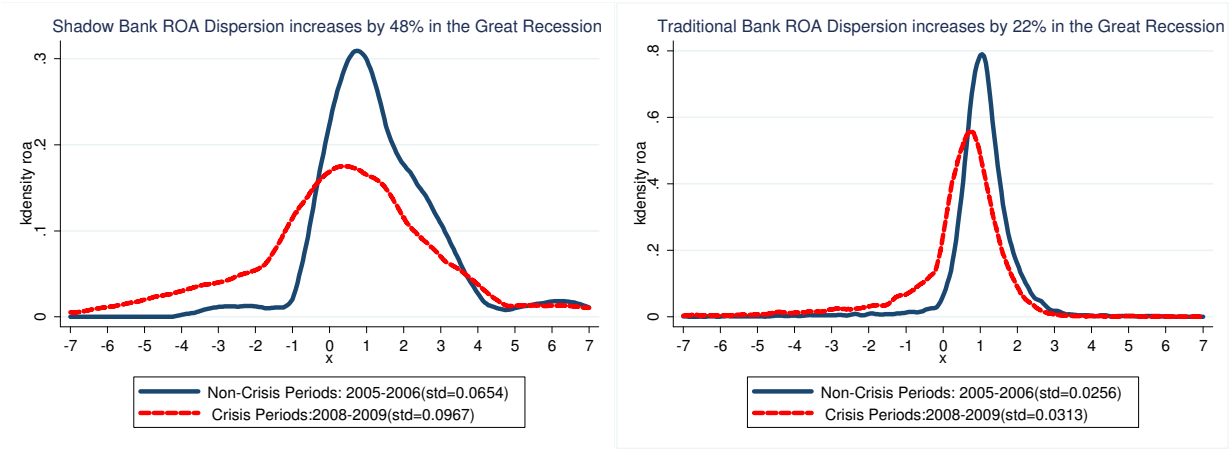


Figure 1: **ROA Dispersion of Banks**. This figure depicts the kernel density of ROA for shadow banks and traditional banks in crisis periods and non-crisis periods. Non-crisis periods are defined as from 2005 to 2006, whereas crisis periods are defined as from 2008 to 2009. According to the NBER, the recession began in December 2007, so 2007 is not a clean "before" or "during" recession year.

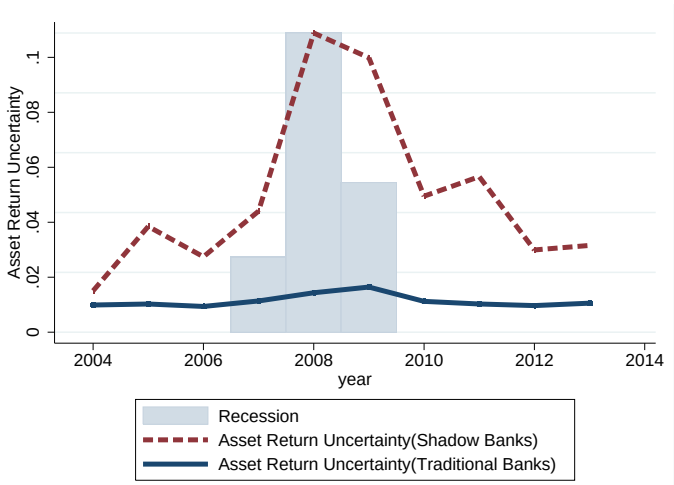


Figure 2: **Bank Asset Return Uncertainty**. This figure depicts the asset return uncertainty of shadow banks and traditional banks from 2004 to 2013. Asset return uncertainty is defined as the standard deviation of the residual term after fitting the panel data of individual bank gross ROA into an AR(1) process, controlling for time and bank level fixed effects and other bank-level heterogeneities. The classification of shadow banks and traditional banks is shown in the Appendix. The length of grey bar represent how much fraction of any particular year is defined as recession period according to NBER. One quarter of 2007 is in recession, while all of 2008 and two quarters of 2009 are in recession.

(controlling for bank-level differences). X_{it} are other control variables including bank size and asset riskiness.

The uncertainty on bank asset return σ_t is then defined as the cross-sectional dispersion of the residual ϵ_{it}. The evolution of σ_t is then shown in the Figure 2. The figure shows that the uncertainty in shadow bank asset return increases from 0.044 to 0.109 during the 2007-2009 recessions. The percentage change is 147%, much higher than that of the traditional bank asset return uncertainty,

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which is only 56%.[8]

I also construct two alternative measures for the asset return uncertainty.[9] The first approach follows the two-step procedure proposed by Gilchrist, Sim, and Zakrajsek (2014) using stock price information. The pattern of this alternative measure is very similar to the baseline measure. The correlation between these two measures is 0.9846. The details for this alternative measure are shown in the Appendix B.3. Note that I couldn't observe the daily stock price information for all banks in my sample. Due to the data limitation, I choose to follow the standard approaches as in Bloom (2009), Bloom, Floetotto, Jaimovich, Saporta-Eksten, and Terry (2019) and Arellano, Bai, and Kehoe (2018) to construct the uncertainty shock measures in this paper. The other advantage of the standard approach is that it provides a very tight link between the model and the data. The gross ROA is assumed to follow an AR(1) process, both in the model and in the data. And asset return uncertainty are measured as the standard deviations of the residual terms, both in the model and in the data. This tight mapping provides much cleaner identification in the parameter estimation.

I also use option prices information of non-bank financial institutions in the Option Metric database and back out the implied uncertainty using Black-Scholes formula. Uncertainty is measured as the annual average of daily (365-day) implied volatility of at-the-money-forward call options following Alfaro, Bloom, and Lin (2019). The patterns of estimated changes in "perceived" uncertainty are also very similar to the measured "realized" uncertainty. The correlation between these two measures is 0.9668. This is unsurprising. As shown in the literature (see Baker, Bloom, and Davis (2016) among others), many measures of uncertainty are highly correlated.

Stylized fact 2: Interbank market spreads spiked up during the 2007-2009 financial crisis.

I use the TED spread as the indicator of the interbank market spread. The TED spread is the difference between the annualized interest rates on three-month interbank loans and on three-month U.S. T-bills. It is a commonly used indicator of perceived credit risk in the economy. An increase

[8] One explanation for this is illustrated in Stylized Fact 5: traditional banks and shadow banks has very different asset composition and funding structure. The exposure to risky asset is much higher for shadow banks. And therefore their asset return uncertainty is much more volatile compared with traditional banks. The measures of asset return uncertainty are robust after controlling for more bank level or industry level heterogeneities, and across different subsamples divided by bank sizes or specializations. The larger change of asset return uncertainty is also not driven by the differences in sample sizes. We could get similar results for traditional banks if we draw random samples with similar sizes from the current sample of traditional banks.

[9] Detailed results and discussions regarding the construction of alternative measures for asset return uncertainty are in Online Appendix B.3.

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in the TED spread is a sign that lenders believe the risk of default on interbank loans (also known as counterparty risk) is increasing and thus they charge a higher interest rate when lending in the interbank market. I obtain the daily TED spread data from the FRED database and plot it in Figure 3. The average TED spread is approximately 0.4% from the fourth quarter of 2001 until August of 2007 when BNP Paribas suspended the valuation of three of its hedge funds related to the U.S. asset-backed securities. The market uncertainty increased dramatically and the average interbank credit spread increased sharply to 2.5%. The TED spread continued to increase following Lehman Brothers' bankruptcy on September 15, 2008 and reached its peak of 4.58% on October 10, 2008, 25 days after Lehman's bankruptcy. The increase in TED spread indicates an increase in bank funding cost.[10]

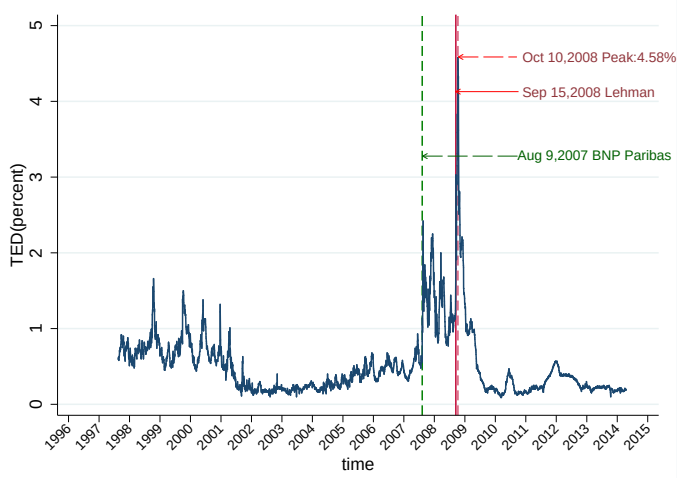


Figure 3: **TED Spread**. This figure depicts the change in the interbank market spread based on daily TED spread data obtained from the FRED Database. The TED spread is the difference between the interest rates on interbank loans and short-term U.S. government debt (T-bills). It is a commonly used indicator of perceived credit risk in the economy.

Stylized fact 3: Asset size in the shadow banking sector shrank dramatically during the 2007-2009 financial crisis.

I obtain the bank-level data from the Bankscope database and Worldscope database and calculate the total assets for the U.S. shadow banking sector. The dashed line in Figure 4 depicts the total assets of shadow banks. [11] Figure 4 shows that the shadow banking sector has a positive trend in assets.

[10] To check the robustness of this result, I also construct an alternative measure of bank funding cost in Online Appendix B.7.

[11] Traditional banks also engage in shadow banking activities. They set up conduits and the activity of these conduits are not recorded on their balance sheet. In the empirical analysis of this paper, I only focus on the shadow banking activities observable in the data. My estimation of shadow bank size is close to FSB's estimation using aggregate data.

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During the 2007-2009 recession, the asset size of the shadow banking sector shrank dramatically.[12] By comparison, traditional banking sector asset size increased during the 2007-2009 crisis, although with a lower growth rate.

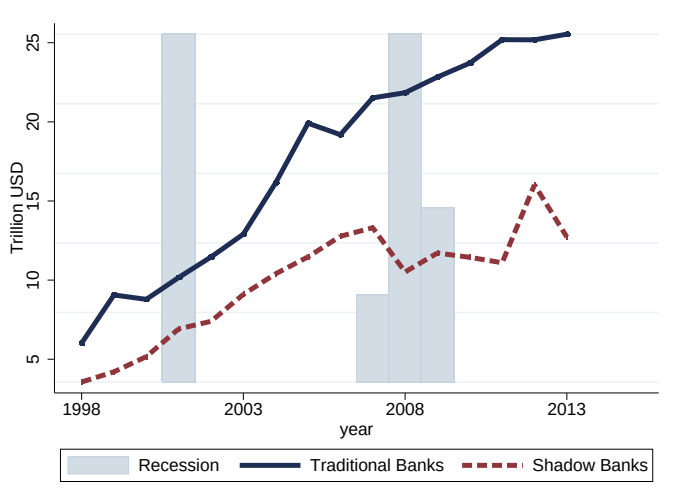


Figure 4: **Total Assets of Banks**. This figure depicts the change in total asset size of the U.S. traditional and shadow banking sector respectively using bank-level data set from the Bankscope and Worldscope databases. The classification of shadow banks and traditional banks is shown in the appendix. The units of the Y-axis are trillions of 2005 US dollars. The grey bar is the NBER recession. Since we only have yearly data, the recession periods are plotted such that the length of grey bar represents the fraction of any particular year that is defined as a recession period according to NBER.

Stylized fact 4: The leverage of shadow banks decreased during the 2007-2009 financial crisis. Bigger shadow banks tended to have higher leverage. The policy function margin contributed more than the distribution margin to the change in aggregate leverage of shadow banks.

I calculate the aggregate leverage for shadow banks and traditional banks in non-crisis periods and crisis periods. The leverage is defined as total liabilities divided by total assets. The results are shown in Table 1. The leverage of shadow banks decreases by 0.074. Traditional bank leverage decreases by merely 0.005.

To obtain a clearer idea factors contributing to the aggregate leverage change, I decompose the aggregate leverage change into the two margins: policy function margin and distribution margin. Specifically, I divide the bank asset size into 10 asset quantiles. At each asset quantile i, banks have a mean asset size k_i. Denote b_{it} as the mean debt held by banks at asset quantile i at period t, and ω_{it} as the measure of banks within asset quantile i at period t. B_t and K_t are aggregate liabilities and assets,

[12] A relatively smaller fraction of asset contraction results from the banks that exit the market. Only 14% of the decline in asset size comes from shadow banks' defaults and the remaining 86% comes from the shrinkage of assets in existing banks.

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Table 1: Leverage Change Decomposition

Shadow Banks	non-crisis periods	crisis periods	leverage change
Aggregate leverage	0.969	0.895	-0.074
Distribution margin		61%	
Policy function margin		39%	
Traditional Banks	**non-crisis periods**	**crisis periods**	**leverage change**
Aggregate leverage	0.896	0.891	-0.005
Distribution margin		7%	
Policy function margin		93%	

Notes: The distribution margin is defined as the change in aggregate leverage resulting from a change in the bank's asset distribution given a fixed leverage policy function. The policy function margin is defined as the change in aggregate leverage resulting from banks across all asset quantiles choosing a different leverage ratio, given a fixed asset distribution. Non-crisis periods are defined as from 2005 to 2006, whereas crisis periods are defined as from 2008 to 2009.

respectively. Then the change in aggregate leverage

$$\triangle L_t \equiv \triangle \frac{B_t}{K_t} = \frac{B_t}{K_t} - \frac{B_{t-1}}{K_{t-1}} = \frac{\sum \omega_{it} b_{it}}{\sum \omega_{it} k_i} - \frac{\sum \omega_{it-1} b_{it-1}}{\sum \omega_{it-1} k_i} \tag{2}$$

Denote the total assets of banks within asset quantile i as a fraction of aggregate assets as $\alpha_{it} \equiv \frac{\omega_{it} k_i}{\sum \omega_{it} k_i} = \frac{\omega_{it} k_i}{K_{it}}$, and the leverage of banks at asset quantile i at time t as $l_{it} \equiv \frac{b_{it}}{k_i}$. Then the change in aggregate leverage can be decomposed to the following two components:

$$\triangle L_t = \sum \alpha_{it} l_{it} - \sum \alpha_{it-1} l_{it-1} = \underbrace{\sum (l_{it} - l_{it-1}) \alpha_{it-1}}_{\text{policy function margin}} + \underbrace{\sum (\alpha_{it} - \alpha_{it-1}) l_{it}}_{\text{distribution margin}} \tag{3}$$

The first term of the equation captures the policy function margin: the change in aggregate leverage resulting from banks of all asset quantiles taking a lower leverage, given a fixed asset distribution. The second term captures the distribution margin: the change of aggregate leverage resulting from the change in the bank's asset distribution given a fixed leverage policy. The contributions of each margin are reported in Table 1. The distribution and policy function margins contribute to 61% and 39%, respectively, of the aggregate leverage change in shadow banks. By comparison, the change in aggregate leverage of traditional banks almost exclusively comes from the policy function margin.

Figure B7 in Appendix B.8 illustrates the leverage for different asset quantiles of banks using the bank-level data. Shadow banks of different asset sizes on average choose a lower leverage during the crisis. This captures the policy function margin. Figure B7 also shows that bigger banks tend to have

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higher leverage; this positive correlation is robust for both types of banks. This result is consistent with Adrian and Shin (2010). As Adrian and Shin (2010) argued, if banks target a leverage ratio, the optimal leverage will not increase with asset values. However, if banks target a level of risk exposure, leverage will be positively correlated with assets values. My finding supports the conjecture that banks target a certain level of risk exposure. Figure B8 in Appendix B.8 depicts the change in bank asset distribution. Shadow banks choose a smaller asset size and the whole asset distribution becomes more skewed toward the lower asset level region. This captures the distribution margin.

Stylized fact 5: Shadow banks are significantly different from traditional banks in funding structure and exposure to risky assets. Deleveraging and asset contraction are larger in bank subsamples with higher fire-sale costs and higher maturity mismatch.

Shadow banks depend heavily on short-term funding, while traditional banks mainly depend on customer deposits. Figure 5 illustrates the funding structures of shadow banks and traditional banks. Traditional banks highly depend on customer deposits which account for more than 60% of traditional banks liability for almost all years in the sample. By comparison, shadow banks highly depend on funds provided by the short-term collateralized debt market. Before 2008, short-term funding accounts for more than 70% of the funding for shadow banks. Since 2007, the funding provided by the short-term collateralized market has declined, and shadow banks gradually switched to rely more on long-term funding after 2011. Customer deposits have always been an almost negligible funding source for shadow banks. Its proportion in shadow banks' total funding is less than 10% for every year of the sample. Given this feature, when modeling the liability side of shadow banks, I abstract from customer deposits.

Shadow banks are also significantly different from traditional banks in their exposure to risky assets. Figure 6 illustrates that shadow banks invest more than 60% of their assets in risky financial assets (derivative and other securities), while traditional banks invest less than 30% of their assets in these risky products. Traditional banks and shadow banks have very different asset composition and funding structure. The exposure to risky asset is much higher for shadow banks. This partially explains why shadow banks' asset return uncertainty is much more volatile compared with traditional banks. [13]

[13]Note that shadow banks did not return to their pre-crisis levels of short-term funding and exposure to risky assets. This is likely due to the fact that Congress passed some regulatory reform of the financial sector in the Dodd-Frank Wall Street Reform and Consumer Protection Act, which gradually took place and might have affected the operating cost and

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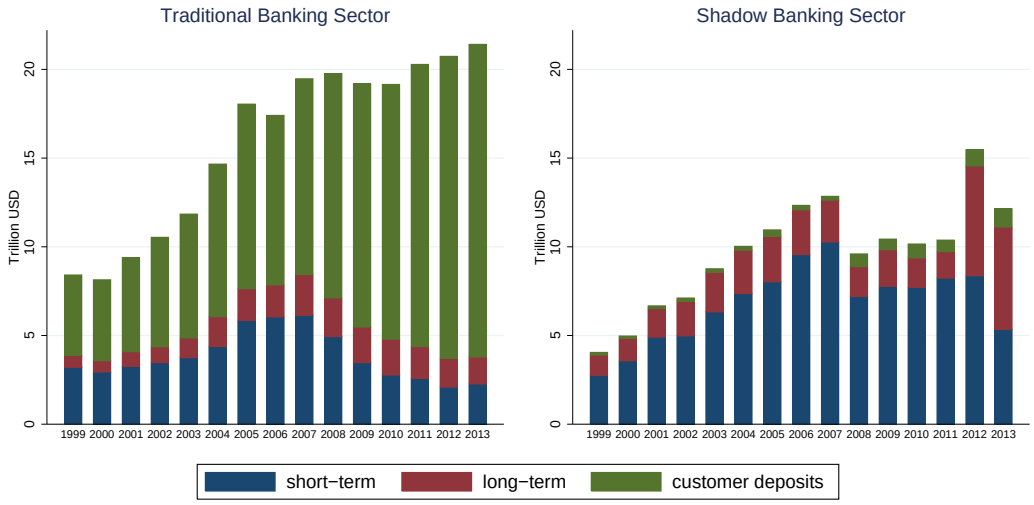


Figure 5: **Funding Structure**. This figure depicts the change in funding structure of shadow banks and traditional banks. The unit on the y-axis is 2005 trillion USD. Short-term and long-term refers to the short-term debt and long-term debt the banks hold, respectively. The classification of shadow banks and traditional banks is shown in the appendix.

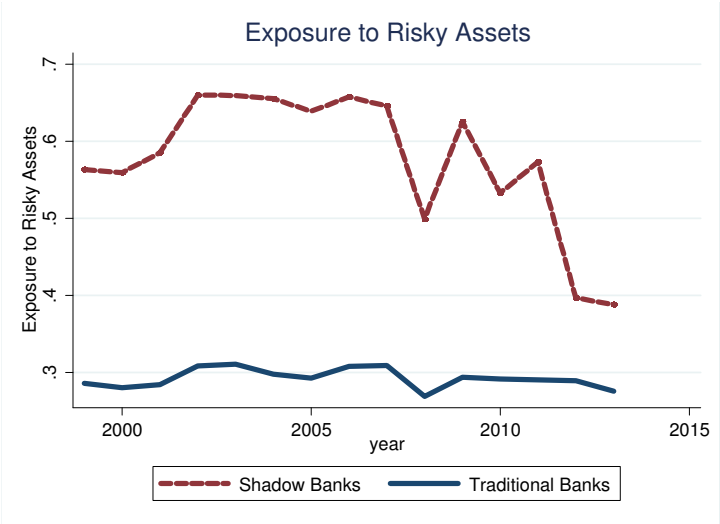


Figure 6: **Risk Exposure**. This figure depicts the change in the asset risk exposure of shadow banks and traditional banks. Risk exposure is defined as the fraction of total assets invested in derivative and other securities. The classification of shadow banks and traditional banks is shown in the appendix.

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To check the cross-sectional variations of asset contraction and deleveraging in the shadow banking sector, I sort the shadow banks based on the measures of fire-sale costs or maturity mismatch and split the full sample into two subsamples.[14] Table 2 shows that deleveraging and asset contraction are much stronger in bank subsamples with higher fire-sale costs and higher maturity mismatch. These results are consistent with a rich body of literature (see Brunnermeier, Gorton, and Krishnamurthy (2014), Farhi and Tirole (2012), Krishnamurthy (2010) among others) which argue that maturity mismatch and associated fire-sale cost play an important role in amplifying the contraction of the economy.

Table 2: Cross-sectional Variation in Asset Contraction and Deleveraging

	deleveraging	asset contraction
A: Split by asset fire-sale cost		
high fire-sale cost	8.63%	25.17%
low fire-sale cost	5.87%	18.06%
B: Split by maturity mismatch		
high maturity mismatch	9.72%	29.08%
low maturity mismatch	4.49%	12.01%

Notes: This table reports the cross-sectional variations of asset contraction and deleveraging in the shadow banking sector. I sort the shadow banks based on the measures of fire-sale costs or maturity mismatch and split the full sample into two subsamples.

In sum, based on the first four stylized facts, we can see that the asset return uncertainty of shadow bank increases dramatically during the 2007-2009 crisis; interbank spread also increases, forcing shadow banks to deleverage and reduce their asset sizes. Since shadow banks contributed more to the total banking credit contraction in the recent crisis compared with traditional banks, in this paper, I focus on the shadow banks.[15] Motivated by these stylized facts, I build and estimate a quantitative model with heterogeneous banks, uncertainty fluctuations, and endogenous bank default to explain

risk-taking motives of shadow banks after 2010. In this paper, I focus on exploring what are the most important underlying shocks behind the contraction and deleveraging of the shadow banking sector during the 2007-2009 financial crisis. The results of the model are robust when estimating the model based on the sample excluding the years after 2010. Despite the complex setting and rich dynamics, the model remains tractable and could be further extended to study what caused the lack of recovery in risk-taking post crisis. For instance, mapping to the model, the changes in financial regulations could be thought as the change in the operating cost of shadow banking, or imposing a risk-weighted capital constraint restrictions in the extended model with asset portfolio choices (see Appendix C.6). There could be multiple theoretical mechanisms contributing to these changes. And identification of different mechanisms behind these changes requires more detailed micro data on shadow bank operation and financial regulation. This highlights a direction for future empirical and structural research that can uncover what affect shadow banks risk-taking incentives after the 2007-2009 financial crisis.

[14] See Appendix B.2 for the details about measurement construction. See Appendix B.9 for a more granular split.

[15] Given the limited role traditional banks played in the recent financial crisis (mainly due to their lower leverage, different funding structure, and smaller changes in asset return volatility), to keep the main model as concise and tractable as possible, I focus on exploring what shocks and features of shadow banks led to the observed stylized facts in shadow banking sector in the main text of this paper and relegate the comparison between these two sectors to Appendix C.9.

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the increase in the interbank spread, the deleveraging, and the contraction of shadow banking credit intermediation during the recent financial crisis. The model characterizes key features of shadow banking dynamics in a full nonlinear manner, and it provides a micro-foundation for the impact of uncertainty on the financing and investment decisions of shadow banks. It embeds realistic features of the shadow banking industry that are distinct from traditional banks. Since shadow banks have very different asset and liability composition than traditional banks, as illustrated in Stylized Fact 5, different from conventional modeling about traditional banks, I abstract from customer deposits in the model. Also, the model features maturity mismatch and fire-sale cost. As shown in the stylized fact 5, the asset contractions and deleveraging are stronger in subsamples with higher fire-sale costs and higher maturity mismatch, implying that fire-sale cost and maturity mismatch are potentially important attributes which might amplify the impact of adverse shocks. Despite the complex setting and rich dynamics, the model remains tractable, thus avoiding the problem of a "black box" model with many indistinguishable moving internal parts. This facilitates a better understanding of the key mechanisms at play.

3 The Model

In this section, I develop a dynamic model of shadow banking with uncertainty fluctuations and endogenous bank default to rationalize the empirical findings. The structural model is useful for two reasons. First, it provides a close connection between theory and data and helps us quantify the magnitude of each proposed channel. Second, the structural approach allows counterfactual simulations which are useful for examining policy implications.

3.1 Environment

Time is discrete and denoted by $t \in \{0, 1, 2, ...\}$. There is a continuum of shadow banks borrow from the interbank market using risky debt. A clearing house takes funds from some banks and lends to other banks. In each period, depending on the idiosyncratic shock to its asset return, a bank would choose to borrow from the clearing house or save in the clearing house. For simplicity, assume that when a bank chooses to save ($b < 0$), it only receives risk-free rate r_t as the rate of return. Each bank invests in a risky loan. The risky loan gives the bank the gross return $z_t x_t k_t$. z_t is the idiosyncratic

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gross rate of return on the bank's assets, x_t captures the aggregate state of the economy.

Assume that the idiosyncratic bank asset return has two components: a permanent component a and a time-varying component v.[16] In particular,

$$z_{it} = a_i v_{it}. \tag{4}$$

Denote $H(z'|z)$ as the conditional probability of z_{it}. Following Arellano, Bai, and Zhang (2012) and Midrigan and Xu (2014), assume that the permanent component follows a Pareto distribution with an upper bound A and a shape parameter μ:

$$Pr(a_i \leq \omega) = \frac{1-\omega^{-\mu}}{1-A^{-\mu}}. \tag{5}$$

The time-varying component v_{it} follows an AR(1) process:

$$log\left(v_{it}\right) = \rho_v log\left(v_{it-1}\right) + \psi_t + \sigma_{t-1}\epsilon_{it} \tag{6}$$

where $\epsilon_{it} \sim N(0,1)$. I impose $\psi_t = -\frac{\sigma_{t-1}^2}{2}$ to keep the mean level of v_{it} across banks unchanged as σ_{t-1} varies.

σ_{t-1} is the aggregate shock to the standard deviation of idiosyncratic asset return shocks. The variance of innovations σ_{t-1} moves over time, generating periods of low and high asset return uncertainty. Following Bloom, Floetotto, Jaimovich, Saporta-Eksten, and Terry (2019) and Arellano, Bai, and Kehoe (2018), assume that σ_t takes on two values, a high value, σ_H, and a low value, σ_L, with transition probabilities determined by the probabilities of remaining in the high and low uncertainty states, π_{HH} and π_{LL}:

$$\sigma_t \in \{\sigma_L, \sigma_H\}, \text{ where } Pr(\sigma_{t+1} = \sigma_j | \sigma_t = \sigma_k) = \pi^{\sigma}_{k,j}. \tag{7}$$

The aggregate productivity x_t follows an AR(1) process:

$$log\left(x_t\right) = \rho_x log\left(x_{t-1}\right) + \sigma^x \epsilon^x_t \tag{8}$$

[16] In the baseline model, the bank assets as a whole have varying returns. These returns can be thought as the weighted average returns of assets with fixed rate return and risky assets with floating rate returns. The model's mechanism is robust in the extended model with asset portfolio choice where I allow for the shadow banks to hold assets with fixed return rates. See Appendix C.6 for more details.

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where ϵ_t^x is an i.i.d standard normal variable.

Finally, following Zhang (2005) and Koh (2016), I directly specify the stochastic discount factor without explicitly modeling the consumer's problem. The stochastic discount factor (SDF) is given by:

$$log\left(m_{t+1}\right) = log\beta - \gamma_x\left(\tilde{x}_{t+1} - \tilde{x}_t\right) - \gamma_\sigma\left(\sigma_{t+1} - \sigma_t\right) \tag{9}$$

where $\tilde{x}_t = log(x_t)$, $0 < \beta < 1$, $\gamma_x > 0$, and $\gamma_\sigma < 0$ are constant parameters.

The SDF is a function of two aggregate shocks: aggregate productivity shocks and uncertainty shocks. In recessions, uncertainty tends to increase and aggregate productivity tends to decrease. Setting $\gamma_x > 0$ and $\gamma_\sigma < 0$ results in a countercyclical SDF: it is high in recessions and low in booms, which leads to a risk-free rate that is high in booms and low in recessions like in the data.[17]

3.2 Maturity Mismatch and Asset Fire-sale

Every period, each bank invests in a risky loan. Different from a conventional one-period loan, the risky loan in this model features a long-term risky financial asset. In each period, only a fraction λ of the asset matures.[18] The rest of the asset $(1-\lambda)k_t$ is rolled over to the next period. The parameter $\lambda < 1$ captures the maturity mismatch between the bank asset and liability. The smaller is λ, the longer the average maturity of the asset, the greater is the degree of maturity mismatch.[19]

Table 3: Balance Sheet of Banks

Assets	Liabilities
Long-term asset $(1-\lambda)k_t$	Net worth n_t
Current asset $\pi_t + \lambda k_t$	Risky debt b_t

Notes: This table shows the balance sheet of an incumbent bank when it needs to repay its debt. On the assets side, the total gross assets return consists of two parts: current assets and long-term assets. In each period, only λ fraction of the loan matures. $z_t x_t$ is the gross return on bank assets. Denote $\pi_t \equiv (z_t x_t - 1)k_t$ as the bank's profit. The current asset of banks can be written as $z_t x_t k_t - (1-\lambda)k_t = \pi_t + \lambda k_t$, where λk_t is the principal payment.

[17] This is similar to the specifications in Koh (2016) and Begenau and Salomao (2019).

[18] Assets and liabilities exhibit a maturity mismatch, which is a characteristic of the shadow banks. These institutions invest in long-term risky financial assets largely funded by short-term debt that must be rolled over frequently. The model is one of shadow banking instead of commercial banking because there is no government insurance present on any portion of the bank's assets, nor is there any funding via deposits.

[19] Maturity mismatch in this model is defined as the ratio between average asset maturity and liability maturity. The (weighted) average maturity of existing loans (assets) at date t , assuming the bank neither defaults nor makes any adjustments on the current investment in loans, is $\sum_{s=0}^{\infty} \frac{(s+1)\lambda k_{t+s}}{k_t} = \sum_{s=0}^{\infty} (s+1)\lambda(1-\lambda)^s = \frac{1}{\lambda}$, as the residual loan outstanding at date $t+s$, $s \geq 0$, is $k_{t+s} = k_t(1-\lambda)^s$. Liability maturity is normalized to 1. Thus, the degree of maturity mismatch equals $\frac{1}{\lambda}$.

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The balance sheet of an incumbent bank when it needs to repay its debt is shown in Table 3. This table provides a snapshot of the balance sheet of the bank before the tax payment, fixed operating cost, equity issuance/dividend payout and asset fire-sale. On the assets side, the total gross assets return consists of two parts: current assets and long-term assets. In each period, only λ fraction of the loan matures. $z_t x_t$ is the gross return rate on bank assets. Denote $\pi_t \equiv (z_t x_t - 1)k_t$ as the bank's profit. The current asset of banks can be written as $z_t x_t k_t - (1-\lambda)k_t = \pi_t + \lambda k_t$, where λk_t is the principal payment.[20] I allow the bank to issue new debt to pay old debt. Hence, in each period, the net debt the bank needs to repay can be written as $b_t - q_t b_{t+1}$. Denote the cash balance as c, which is the current asset minus the amount of debt due plus the new debt issued in each period:

$$c \equiv (1-\tau_c)\pi + \lambda k - b + qb' + \frac{\tau_c(b' - qb')}{1+r} - \xi \tag{10}$$

where ξ is the fixed cost of operation and $1/(1+r)$ is the discount factor.[21] $\tau_c(b' - qb')/(1+r)$ is the present value of interest tax deduction.[22]

When the cash balance becomes negative, the bank fire-sales part of its long-term assets. Assume that when the bank fire-sales its long-term assets, it can only get ζ fraction of the total value, where $0 < \zeta < 1$. The lower is the fire-sale parameter ζ, the higher is the fire-sale cost. Note that when $c < 0$, the bank does not have enough cash flow to repay the debt. It only needs to fire-sale part of its long-term assets. The amount of long term assets the bank needs to fire-sale equals $|c|$ divided by the fire-sale parameter ζ.

Both maturity mismatch and fire-sale costs amplify the impact of adverse shocks. When the bank is hit by adverse shocks, the higher the degree of maturity mismatch, the smaller is the fraction of current asset, the more likely the bank will be forced to fire-sale its long-term assets. The higher is the fire-sale cost, the more long-term asset the bank needs to fire-sale to repay the debt, the more likely the bank will not be able to roll over its debt and go bankrupt. This is consistent with a rich body of literature (see Brunnermeier, Gorton, and Krishnamurthy (2014), Farhi and Tirole (2012),

[20] zx is the gross return rate on bank assets. One can denote $zx = 1 + r_k$, where r_k is the asset return rate. Then cash flow would be equal to $(r_k + \lambda)k$, in which $r_k k$ is the interest payment, λk is the fraction of principal paid in period t. $(1-\lambda)$ fraction of principal is left to the next period.

[21] Note $\mathbb{E}m = 1/(1+r)$, where m is the stochastic discount factor and r is the risk-free rate.

[22] Following Strebulaev and Whited (2012) and Gourio (2013), I assume the bank takes the present value of the interest tax deduction in the period in which it issues debt. This feature greatly simplifies the determination of the risky interest rate. Otherwise, the tax deduction would depend on last period's shocks, and current profits would then also depend last period's shocks.

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Krishnamurthy (2010) among others) which find that maturity mismatch and associated fire-sale cost play an important role in amplifying the contraction of the economy.

3.3 The Incumbent Bank's Problem

The incumbent bank's problem is as follows. In each period, the bank would choose to default or not to default:

$$V(z,k,b;\mathbf{s}) = \max_{d\in\{0,1\}} (1-d)V^c(z,k,b;\mathbf{s}) + dV^d(z;\mathbf{s}) \tag{11}$$

where d is the default decision of the bank; when $d=1$, the bank chooses to default.

If the bank chooses to continue the operation, based on the idiosyncratic state variables z,k,b and an aggregate state variable vector $\mathbf{s}=\{\sigma,x\}$, it chooses how much to borrow from the interbank market - b' , and how much risky asset to hold next period - k'. The value of continuation is determined by the following Bellman equation:

$$V^c(z,k,b;\mathbf{s}) = \max_{\{b',k'\}} \left\{(1+\gamma_e 1_{\{e<0\}} - \tau_d 1_{\{e>0\}})e + \mathbb{E}_{z,s} m(\mathbf{s},\mathbf{s}')V(z',k',b';\mathbf{s}')\right\} \tag{12}$$

s.t.

$$e = \begin{cases} c+(1-\lambda)k-k'-\Phi(k',k) & \text{if } c\geq 0 \\ (1-\lambda)k+\frac{c}{\zeta}-k'-\Phi(k',k) & \text{if } c<0 \end{cases} \tag{13}$$

$$c = (1-\tau_c)\pi + \lambda k - b + qb' + \frac{\tau_c(b'-qb')}{(1+r)} - \xi \tag{14}$$

where $m(\mathbf{s},\mathbf{s}')$ is the stochastic discount factor; $\Phi(k',k)$ is the quadratic asset adjustment cost: $\Phi(k',k) = \frac{\phi}{2}\left(\frac{k'-(1-\lambda)k}{k}\right)^2 k$; e represents dividend payout when $e>0$ or equity issuance when $e<0$; τ_d denotes the dividend tax; γ_e denotes the equity issuance cost.

The microfoundation of asset size adjustment cost is that when a bank adjusts its asset size, it needs to pay a cost to acquire relevant information about the newly issued asset, look for appropriate buyers or sellers, and pay for the legal fees and possible personnel adjustment costs. This cost is related to the rate of adjustment, such that the cost of investment is higher for more rapid changes. This feature helps me match the average standard deviation of new asset issuance/asset ratio in the data. I also allow for the presence of non-convex adjustment costs (fire-sale costs) in assets. As stated in the literature, the presence of the non-convex adjustment costs can lead to a real options or wait-and-see effect of uncertainty shocks.

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When $e < 0$, the bank issues equity and has to pay an extra equity issuance cost of γ_e. The equity issuance costs are motivated by underwriting fees and adverse selection costs. To keep the model tractable, I do not model costs of external equity as the outcome of an asymmetric information problem. Like Cooley and Quadrini (2001), I capture adverse selection costs and underwriting fees in a reduced form. I adopt a simple formulation by choosing a linear equity payout costs, γ_e, as in Gomes and Schmid (2016) and Begenau and Salomao (2019).[23]

3.4 Default and Debt Pricing

Banks borrow from a debt schedule that depends on their default decisions. For each debt contract, the creditor transfer qb' to the bank today and receive b if the bank repay and a recovery value $R(z', k', b'; x')$ if the bank default next period. Assume the value of the bank when it chooses to default is zero:

$$V^d(z; \mathbf{s}) = 0. \tag{15}$$

The price schedule of risky debt offered by a clearing house can be written as:

$$q(z, k', b'; \mathbf{s}) \quad = \quad \mathbb{E}_{z,s} \left\{ m(\mathbf{s}, \mathbf{s}') \left[1 - d(z', k', b'; \mathbf{s}') + d(z', k', b'; \mathbf{s}') R(z', k', b'; x') \right] \right\} \tag{16}$$

where $R(z, k, b; x)$ is the recovery of the lender when the bank defaults, as in Hennessy and Whited (2007) and Gomes and Schmid (2010):

$$R(z, k, b; x) = \max \left\{ 0, \frac{(1 - \tau_c)\pi + \lambda k + \zeta(1 - \lambda)k - \Phi(k, 0) - \xi}{b} \right\}. \tag{17}$$

3.5 Bank Entry

Entry is required to replace bankrupt banks and ensure an ergodic distribution of banks in equilibrium. I model entry in line with Clementi and Palazzo (2016), Gomes and Schmid (2016), and Begenau and Salomao (2019). Assume that every period a unit mass of potential new entrants arrives in the economy. Potential entrants behave similarly to incumbents but face different initial conditions. Specifically, potential new entrants have no initial level of debt, and receives a signal ν about their

[23] In an extension of the baseline model, I will allow this parameter to be time-varying.

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productivity. This signal follows a Pareto distribution, $\nu \sim \Upsilon(\nu)$ with parameter ω, which enables heterogeneous entrants. Banks have to pay an entry fee ($f_e > 0$) so not all banks find it optimal to enter. These two parameter helps me to pin down the entry rate and size distribution of the entering banks.

The entrant only starts operating in the period after the entry decision, but must decide today on the optimal k' and b' given start-up asset k_0. The value function of the entrant can be written as:

$$V^e(\nu;\mathbf{s}) = \max_{\{b',k'\}} \left\{(1+\gamma_e 1_{\{e<0\}} - \tau_d 1_{\{e>0\}})e + \mathbb{E}_{\nu,s} m(\mathbf{s},\mathbf{s}')V(z',k',b';\mathbf{s}')\right\} \tag{18}$$

s.t.

$$e \quad = \quad qb' - k' - \Phi(k',k_0). \tag{19}$$

A potential entrant will start a new bank if the value of entry V^e higher than entry cost: $V^e \geq f_e$.

3.6 Recursive Competitive Equilibrium

Given an initial bank distribution and a stochastic discount factor $m(\mathbf{s},\mathbf{s}')$, a recursive competitive equilibrium consist of (i) value functions $V(z,k,b;\mathbf{s})$, $V^c(z,k,b;\mathbf{s})$, $V^d(z;\mathbf{s})$, $V^e(\nu;\mathbf{s})$; (ii) policy functions for incumbents $k'(z,k,b;\mathbf{s})$, $b'(z,k,b;\mathbf{s})$, and $d(z,k,b;\mathbf{s})$; policy functions for entrants $k'(\nu;\mathbf{s})$ and $b'(\nu;\mathbf{s})$; (iii) debt pricing function $q(z,k,b;\mathbf{s})$, and recovery function $R(z,k,b;x)$; (iv) bounded sequence of incumbents measures $\{\Gamma_t\}_{t=1}^{\infty}$ and entrants measures $\{\mathcal{E}_t\}_{t=1}^{\infty}$, such that for all $t \geq 0$:

1. Given the stochastic discount factor $m(\mathbf{s},\mathbf{s}')$ and debt pricing function $q(z,k,b;\mathbf{s})$, policy functions $k'(z,k,b;\mathbf{s})$, $b'(z,k,b;\mathbf{s})$, and $d(z,k,b;\mathbf{s})$ and value function $V(z,k,b;\mathbf{s})$, $V^c(z,k,b;\mathbf{s})$, $V^d(z;\mathbf{s})$ solve the incumbents' problem;

2. Given the stochastic discount factor $m(\mathbf{s},\mathbf{s}')$, recovery function $R(z,k,b;x)$, and debt pricing function $q(z,k,b;\mathbf{s})$, policy functions $k'(\nu;\mathbf{s})$, $b'(\nu;\mathbf{s})$ and value function $V^e(\nu;\mathbf{s})$ solve the entrants' problem;

3. Given the stochastic discount factor $m(\mathbf{s},\mathbf{s}')$, recovery function $R(z,k,b;x)$, and debt pricing function $q(z,k,b;\mathbf{s})$, lenders in the interbank market break even.

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4. For all Borel sets $\mathcal{Z} \times \mathcal{K} \times \mathcal{B} \subset \mathcal{R}^+ \times \mathcal{R}^+ \times \mathcal{R}$ and $\forall\, t \geq 0$,

$$\mathcal{E}_{t+1}(\mathcal{Z} \times \mathcal{K} \times \mathcal{B}) = \int_{\mathcal{Z}} \int_{\mathcal{B}_e(\mathcal{K},\mathcal{B},\mathcal{S})} d\Upsilon(\nu) dH(z'|\nu) \tag{20}$$

where $\mathcal{B}_e(\mathcal{K},\mathcal{B},\mathcal{S}) = \{\nu \;\; s.t. \;\; k'(\nu;\mathbf{s}) \in \mathcal{K}, b'(\nu;\mathbf{s}) \in \mathcal{B}, \;\; \text{and} \;\; V^e(\nu;\mathbf{s}) \geq f_e\}$

5. For all Borel sets $\mathcal{Z} \times \mathcal{K} \times \mathcal{B} \subset \mathcal{R}^+ \times \mathcal{R}^+ \times \mathcal{R}$ and $\forall\, t \geq 0$,

$$\Gamma_{t+1}(\mathcal{Z} \times \mathcal{K} \times \mathcal{B}) = \int_{\mathcal{Z}} \int_{\mathcal{B}(\mathcal{K},\mathcal{B},\mathcal{S})} d\Gamma_t(z,k,b) dH(z'|z) + \mathcal{E}_{t+1}(\mathcal{Z} \times \mathcal{K} \times \mathcal{B}) \tag{21}$$

where $\mathcal{B}(\mathcal{K},\mathcal{B},\mathcal{S}) = \{(z,k,b) \;\; s.t. \;\; k'(z,k,b;\mathbf{s}) \in \mathcal{K}, b'(z,k,b;\mathbf{s}) \in \mathcal{B}, \;\; \text{and} \;\; V(z,k,b;\mathbf{s}) > 0\}$

4 Quantitative Analysis

4.1 Estimation

The model has 22 parameters, as shown in Table 4. 8 parameters are calibrated externally and the rest 14 parameters are jointly estimated following a simulated method of moments approach as in Erickson and Whited (2002).

Following Lin and Zhang (2013), I choose the persistence of aggregate productivity shock $\rho_x = 0.95$, and the average volatility of the aggregate productivity, $\sigma^x = 0.007$.[24] The values of subjective discount factor, β, the price of risk of productivity shocks, γ_x , and the price of risk of uncertainty shocks, γ_σ, are chosen to match the average and standard deviation of risk-free rate, and the average Sharpe ratio.[25] Following Graham (2000), corporate tax rate τ_c and payout tax rate are set to 0.3 and 0.12, respectively. The maturity mismatch parameter λ (the fraction of maturing asset at each period) is set to 0.0956 to match the average maturity mismatch index for shadow banks.[26]

The rest 14 parameters include the following: equity issuance cost parameter γ_e, fire-sale cost parameter ζ , parameters governing the value and transition matrix of aggregate uncertainty shocks

[24]The model is simulated at the quarterly frequency, and then the model moments are constructed on the annual frequency based on quarterly simulated data to facilitate the comparison with the data moments.

[25]In the external estimation, I first simulate the aggregate productivity shocks based on the values of ρ_x and σ^x. Then in the estimation outer-loop, for any values of parameters governing the value and transition matrix of aggregate uncertainty shocks, $\pi^\sigma_{H,H}, \pi^\sigma_{L,L}, \sigma_H, \sigma_L$, I simulate the aggregate uncertainty shocks and externally estimate $\beta, \gamma_x, \gamma_\sigma$ to match the average and volatility of risk-free rate, and the average Sharpe ratio.

[26]Details for the construction and calculation of the measurement of maturity mismatch are included in Appendix B.2.

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Table 4: Model Parameters

Parameter	Value	Description
		External Estimation
ρ_x	0.95	Persistence of aggregate productivity shock
σ^x	0.007	Std. dev of aggregate productivity shocks
β	0.9812	Stochastic discount factor parameter
γ_x	10.1783	Stochastic discount factor parameter
γ_σ	-8.7583	Stochastic discount factor parameter
τ_c	0.3	Corporate tax rate
τ_d	0.12	Payout tax rate
λ	0.0956	Degree of maturity mismatch
		Internal Estimation
γ_e	0.0613 (0.0103)	Equity issuance cost
ζ	0.7036 (0.0486)	Fire-sale cost
ξ	0.0061 (0.0011)	Fixed operating cost
k_0	0.0031 (0.0008)	Entrant start-up asset
ϕ	0.1884 (0.0173)	Asset size adjustment cost
ρ_v	0.9202 (0.0203)	Persistence of asset return
$\pi^\sigma_{H,H}$	0.9448 (0.0124)	Probability of staying in the high uncertainty state
$\pi^\sigma_{L,L}$	0.9778 (0.0132)	Probability of staying in the low uncertainty state
σ_H	0.0674 (0.0107)	Level of uncertainty shock (high uncertainty state)
σ_L	0.0168 (0.0092)	Level of uncertainty shock (low uncertainty state)
μ	3.3007 (0.4712)	Pareto distribution shape
A	17.4936 (1.9726)	Pareto distribution upper bound
f_e	0.0032 (0.0006)	Fixed cost of entry
ω	2.9557 (0.3619)	Pareto distribution parameter of entrant productivity

Notes: This table reports the estimated parameters, with clustered standard deviation in the parentheses. The estimation is done with SMM, which chooses model parameters by matching the moments from a simulated panel of banks to the corresponding moments from the data.

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$\pi^{\sigma}_{H,H}, \pi^{\sigma}_{L,L}, \sigma_H, \sigma_L$, the asset adjustment cost parameter ϕ, the persistence of asset return ρ_v, fixed operating cost ξ, entrant start-up asset k_0, entry cost f_e, Pareto distribution parameter of the entrant productivity ω, and two parameters μ and A that govern the Pareto distribution of the permanent productivity. They are jointly estimated internally to match the following 18 moments: the mean of leverage, the mean of default rate, the mean, standard deviation, and the auto-correlation of dividend payout/asset ratio and new asset issuance/asset ratio, the mean of equity issuance, the mean, standard deviation, auto-correlation, and skewness of asset return uncertainty, the average growth rate and relative size of new entrant banks, mean and standard deviation of interest spread, and the slope of the leverage schedule over assets using simulated methods of moments (SMM). The detailed estimation procedure is illustrated in Appendix A. The target moments are constructed using the data from 1998 to 2013.[27]

4.2 Identification

The model is highly nonlinear, and all parameters affect all the moments. Nevertheless, some parameters are more important for certain statistics. The success of SMM estimation depends on model identification, which requires that we choose moments that are sensitive to variations in the structural parameters. I now describe and rationalize the moments that I choose to match.

The fire-sale cost parameter ζ is most relevant for the mean of leverage. The smaller the fire-sale cost parameter ζ, the larger the loss the bank would suffer during financial distress. Hence, the bank would take lower leverage. The asset adjustment cost parameter ϕ is more relevant for the standard deviation and mean of new asset issuance/asset ratio.[28] The higher the adjustment cost ϕ, the less willing the bank is to change its asset level, hence the mean and standard deviation of new asset issuance would be smaller. This also indirectly increases the autocorrelation of new asset issuance/asset ratio. Without asset adjustment cost, the standard deviation of new asset issuance/asset ratio will tend to be too high, whereas the persistence of new asset issuance/asset ratio will tend to be too low. Fixed entry cost parameter f_e affects the average entry rate. Given f_e, the Pareto exponent ω

[27] The model could also fit the data moments well in early subsample (1998-2005) and late subsample (2006-2013).

[28] In the data, total assets is the book value of assets after writing down the impaired loan, and securities and other credit impairment charges: $\text{asset}_{t+1} = \text{asset}_t - \text{maturing asset}_t + \text{new asset issuance}_t - \text{writedown}_{t+1}$. In the data, I could not directly observe what fraction of asset will mature in each period. I also do not have direct bank-level measure for asset issuance. I back out the measure for new asset issuance using data of assets and asset write-down (impaired loan, and securities and other credit impairment charges), assuming in each year λ_a fraction of assets matures: $\text{maturing asset}_t = \lambda_a \cdot \text{asset}_t$, where the annual asset maturing rate $\lambda_a = 1 - (1-\lambda)^4$. The change in new asset issuance measures the increases or decrease in bank investment.

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is estimated to target the relative size of entrants with respect to incumbents. k_0 mainly affects the mean growth rate of entrants. The equity issuance cost parameter γ_e directly affects the mean of equity issuance. The asset return persistence parameter ρ_v is most closely related to the autocorrelation of the dividend-to-asset ratio and new asset issuance/asset ratio.[29] The fixed operating cost ξ mainly affect the average default rate.[30] The higher is the fixed operating cost, more banks will choose to default after receiving a negative shock. ξ, therefore, indirectly affects the interest spread and the mean of leverage as well. Also, since the fixed cost is size-independent, it affects smaller banks more. This makes smaller banks more cautious and choose lower leverage, which, in turn, increases the leverage-asset slope.[31]

Following Arellano, Bai, and Kehoe (2018) and Bloom, Floetotto, Jaimovich, Saporta-Eksten, and Terry (2019), the four parameters governing the value and transition matrix of aggregate uncertainty shocks $\pi^{\sigma}_{H,H}, \pi^{\sigma}_{L,L}, \sigma_H, \sigma_L$ are estimated to match the mean, standard deviation, autocorrelation, and skewness of asset return uncertainty.[32] The mean of asset return uncertainty is largely driven by the mean of uncertainty shocks σ_t. The standard deviation of asset return uncertainty is determined largely by the distance between σ_H and σ_L, and the autocorrelation of asset return uncertainty is determined by the levels of the transition probabilities $\pi^{\sigma}_{H,H}, \pi^{\sigma}_{L,L}$ of these shocks. The skewness of asset return uncertainty is controlled by the difference in these transition probabilities. In the estimation, $\pi^{\sigma}_{L,L}$ is slightly larger than $\pi^{\sigma}_{H,H}$ so that, on average, high uncertainty shocks are realized relatively infrequently. This leads to skewness because the resulting asset return uncertainty reflects the disproportionate probability that is put on the low volatility shocks. These would also indirectly affect the standard deviation of the dividend-to-asset ratio and new asset issuance/asset ratio by changing the standard deviation of the bank's asset return. Given the value of fixed operating cost ξ, the change in these four parameters will also affect the bank's default probability, and the mean and standard deviation of interest spread.

The shape and upper bound parameters of Pareto distribution, μ and A, determine the bank's

[29] Dividend in the data is calculated by adding up both dividend and share repurchase, as in Hennessy and Whited (2007) and Nikolov and Whited (2014). Same as these papers, my model does not distinguish between the two forms of equity payout, I assume a flat payout tax τ_d, leading to payout costs.

[30] The average entry rate is equal to average default probability in the equilibrium. They may not be equal to each other in each economic state. But their average values are equal to each other in the long-run simulation to ensure a ergodic distribution of banks in the equilibrium.

[31] The leverage-asset slope is constructed by regressing leverage on 10 asset quantiles. The coefficient on asset quantile is the leverage-asset slope.

[32] In both the data and the model, I fit the measured gross asset return rate into an AR(1) process. The asset return uncertainty is measured as the cross-sectional standard deviations of the residual terms.

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asset distribution. The smaller the shape parameter, the fatter the right tail of bank distribution[33] and the higher the concentration of market share. The upper bound of the truncated Pareto distribution determines the degree of permanent technology heterogeneity across banks. The shape and upper bound of the Pareto distribution would thus indirectly affect all the other moments through their impacts on the bank's distribution. They help me match the slope of the leverage schedule. The higher the upper bound of the Pareto distribution and the smaller the shape parameter, the more heterogeneous the bank will be. Large heterogeneity in the permanent productivity component will tend to increase the slope of the leverage-asset slope.

4.3 Estimation Results

I solve the model numerically and estimate it using SMM. I report actual data moments, model moments, and t-statistics that test whether data moments and model-generated moments are statistically significantly different. Table 5 summarizes model-generated moments. Overall, the model is able to match the key features of shadow banks in the data well. No simulated moment is statistically significantly different from its actual data counterpart. Table 4 summarizes the estimated parameter values.

The estimated fire-sale parameter $\zeta = 0.7036$. This implies on average around 30% of value will be lost in the fire-sale. This value lies within the broad range of estimates of capital resale discounts in Ramey and Shapiro (2001) and the estimated haircut rates for nonbank financial institutions by BIS Committee for nonbank financial institutions calculated based on the survey by BIS Committee on the Global Financial System. The estimated recovery value (0.70) is lower than the value used in Hennessy and Whited (2007) (0.85) and Davydenko, Strebulaev, and Zhao (2012) (0.78) and higher than the one used in Gilchrist, Sim, and Zakrajsek (2014) (0.5),Gamba and Triantis (2008) (0.5), Glover (2016) (0.55), Hennessy and Whited (2005) (0.592), and Gomes and Schmid (2016) (0.45).

The estimated equity issuance parameter $\gamma_e = 0.0613$. The typical shadow bank acts as if facing an underwriter charging a proportional fee equal to $\$61,300$ on the first million dollars of gross proceeds, close to the estimates in Altinkilic and Hansen (2000)[34] and estimated equity issuance cost in Nikolov and Whited (2014) (0.043). This value is lower than the estimated value of equity issuance in Begenau

[33] Despite temporal variations, however, the general shape of the bank size distribution can be shown to be well described by a lognormal distribution with a Pareto tail (see Janicki and Prescott (2006) and Benito (2008)).

[34] Altinkilic and Hansen (2000) estimate that the average marginal underwriting fee on equity is $\$51,488$ per million.

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Table 5: Target Moments

Target Moments	Data	Model	t-statistics
mean of leverage	0.932	0.930	0.101
mean of new asset issuance/asset ratio	0.437	0.432	0.178
stdv. of new asset issuance/asset ratio	0.152	0.143	0.531
autocorrelation of new asset issuance/asset ratio	0.876	0.865	0.190
mean default rate (= entry rate)	1.10%	1.09%	0.112
mean of dividend/asset ratio	0.021	0.026	-0.603
stdv. of dividend/asset ratio	0.015	0.017	-0.365
autocorrelation of dividend/asset ratio	0.795	0.725	0.897
mean of asset return uncertainty	0.064	0.063	0.121
stdv. of asset return uncertainty	0.045	0.045	-0.015
autocorrelation of asset return uncertainty	0.732	0.711	0.331
skewness of asset return uncertainty	0.931	0.984	-0.438
mean of equity issuance/asset ratio	0.012	0.012	0.115
leverage-asset slope	0.043	0.047	-0.311
mean entrant growth	0.242	0.216	0.352
entrant relative size	0.201	0.211	-0.103
mean of interest spread	0.55%	0.53%	0.153
stdv. of interest spread	0.44%	0.41%	0.175

Notes: This table reports the simulated and actual moments and the clustered t-statistics for the differences between the corresponding moments. Calculations are based on an annual sample of shadow banks from the Bankscope and Worldscope databases. The estimation is done with SMM, which chooses model parameters by matching the moments from a simulated panel of banks to the corresponding moments from the data.

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and Salomao (2019) (0.2), and higher than the estimated value in Gomes and Schmid (2016) (0.025).

The estimated asset adjustment cost parameter $\phi = 0.1884$. Shadow banks invest in long-term illiquid asset such as mortgage-backed securities. The microfoundation of asset size adjustment cost is that when a bank adjusts its asset size, it needs to pay a cost to acquire relevant information, look for appropriate buyers or sellers, and pay for the legal fees and possible personnel adjustment costs. It can also potentially capture the liquidity of the asset market. It inherently captures any real cost or institutional inefficiency which slows down the speed of asset adjustment. It is related to the rate of adjustment, such that the cost of investment is higher for more rapid changes. This feature helps to smooth the problem and match the average standard deviation of new asset issuance/asset ratio in the data. The estimated adjustment cost parameter is smaller than those estimated in non-financial firms (Eisfeldt and Muir (2016) (0.26), Nikolov and Whited (2014) (1.278)) for physical capital investment, implying that the market of financial assets is more liquid than that of physical capital.

4.4 Model Implications

4.4.1 Policy Functions and Interest Spread Schedule

In the left panel of Figure 7, I plot the optimal policy functions for the asset size. The model generates a positive correlation between the current asset size and the future asset size. Banks that have larger asset sizes in the current period tend to choose larger asset sizes in the next period. Moreover, banks choose smaller asset sizes when the uncertainty of asset return is high.

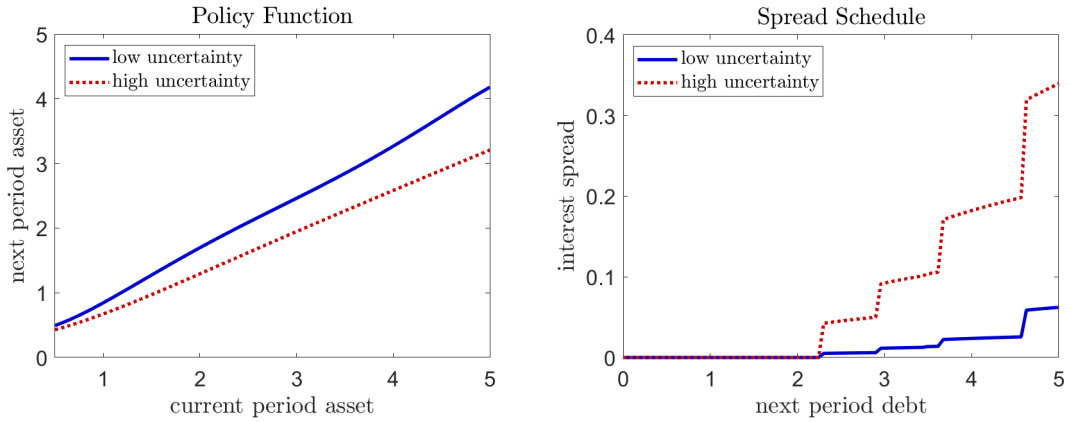


Figure 7: **Policy Function and Interest Spread Schedule**. The left panel of this figure shows the optimal choice of asset size in next period as a function of asset size in the current period for a bank with mean permanent asset return component, stochastic shock, and debt level. The right panel of this figure shows the interbank interest spread as a function of the debt issuance for a bank with mean permanent asset return component, stochastic shock, and asset level.

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The right panel of Figure 7 illustrates the interest spread schedule of risky debt. In the figure, the spread schedule for risky debt under low and high asset return uncertainty for a bank with mean idiosyncratic productivity, mean permanent productivity, and mean asset size is plotted in blue solid and red dashed lines, respectively. When the uncertainty of asset return is high, bank faces higher interbank market spread. This is consistent with what is observed in the data. Large loans are expensive because of higher default risk. When debt level rises, the bank's default probability increases; hence, the spread would also increase. Therefore, the spread schedule is upward sloping when the debt level is high. [35]

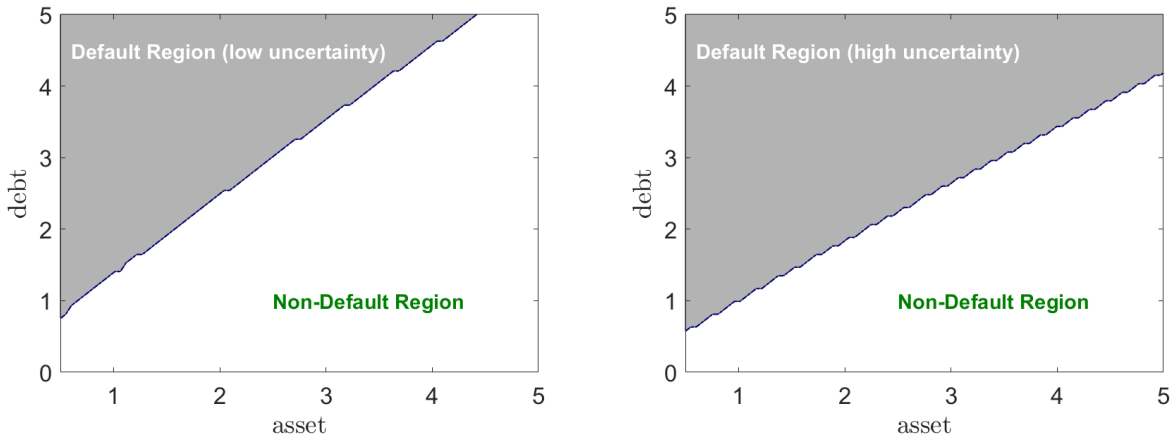


Figure 8: **Default Region**. The figure depicts the default and non-default regions of a bank with median permanent asset return component and stochastic asset return component.

Figure 8 illustrates the default and non-default regions for banks with median permanent asset return component and stochastic asset return component when they choose different asset sizes and debt level. For large banks at this productivity level, even if they choose the highest debt level, they will still generate enough income to repay their debt if the asset return uncertainty is low. Thus they will not choose to default. However, this figure does not imply large banks will never default. If their asset returns drop below a certain threshold, they might still choose to default. On average, banks with higher asset sizes have lower default probabilities. A bank's default probability increases when it issues more debt. Also, bank default probability increases under higher asset uncertainty. As shown in the figure, the default area expands when the uncertainty of asset return increases. This indicates that the banks will be more likely to default in the high uncertainty state.

[35] The characteristics of the spread schedule are similar to those in Arellano, Bai, and Zhang (2012). There is a positive correlation between spread and leverage. There are kinks in the interest rate schedule due to discrete default and fire-sale decisions, and discretization of the processes of bank asset return.

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4.4.2 Regression: Model vs. Data

I simulate the model for 100,000 banks for 2000 periods. The model delivers a simulated bank panel. When constructing the sample, the first 1936 periods are dropped. I only use the last 64 periods (16 years) to draw the sample used for regression.[36] I randomly draw banks sample with the same sample size as in the data from the simulated panel of the last 64 periods. Using the simulated bank panel data, I run the same regression as in the data to check if the model's implications match the data. Both regressions control for bank fixed effects and bank asset return. The results are reported in Table 6. The regression can be viewed as an out-of-sample test of the model given that they not targeted in the estimation. The model generates a similar positive correlation between leverage and asset size, and a negative correlation between asset return uncertainty and leverage. When asset return uncertainty increases by 1%, leverage decreases by 0.011 percentage points in the data, and by 0.013 percentage points in the model. When asset size increases by 1%, leverage increases by 0.017 percentage points in the data and by 0.020 percentage points in the model.

Table 6: Regression: model vs. data

	Model	Data
Variables	leverage$_{it}$	leverage$_{it}$
$log(\text{asset}_{it-1})$	0.020***	0.017***
	(0.006)	(0.004)
$log(\text{asset return uncertainty}_{t-1})$	-0.013***	-0.011***
	(0.004)	(0.003)
Fixed effect	$\surd$	$\surd$
R^2	0.30	0.22

Notes: Standard errors in parentheses. Standard errors of simulated data regression are calculated using the bootstrapping method. $***p < 0.01, **p < 0.05, *p < 0.1$.

4.4.3 Event Study: Modeling the 2007-2009 Financial Crisis

The 2007-2009 financial crisis can be modeled as a decrease in asset returns or an increase in asset return uncertainty. In this paper, I first focus on the impact of an increase in bank asset return uncertainty. In this experiment, following Arellano, Bai, and Kehoe (2018), I choose a sequence of asset return uncertainty shocks so that the measured asset return uncertainty in the model reproduces

[36] The simulated variables are constructed on the annual frequency based on quarterly simulated data to facilitate the comparison with the data moments.

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the corresponding asset return uncertainty path in the data. The responses of the modeled economy are shown in Figure 9. This event study can also be viewed as an out-of-sample test of the model given that the changes in the variable of interests from normal periods to recession periods are not directly targeted in the estimation.

In Figure 9, the data and simulated paths by the baseline model with pure uncertainty shocks are shown in a solid line and a dashed line, respectively. I normalize the total assets by calculating the percentage deviation from its base value at the year 2007 when the crisis started. I first calculate the percentage deviations of assets and leverage in 2008 and 2009 from their base values. Then the mean of these deviations capture the asset contraction and deleveraging in the model and in the data, respectively. Comparing the changes by dividing the mean of deviations in the model by the mean of the deviation in the data, I calculate to what extent the model can explain the data. Uncertainty shocks in the baseline model can explain 72% of asset contraction and 70% of deleveraging in the U.S shadow banking industry. Mean leverage decreases in both the model and the data. The increase in uncertainty raises the external financing cost and the probability of a costly fire-sale; thus, the banks decrease their leverage to lower their expected default probability and financing costs. Figure 9 also shows the simulated path of the interbank spread. The spread increases when uncertainty increases, as observed in the data.[37]

The model could also match some non-targeted moments quite well, as shown in Table 7. This table 7 can also be viewed as out-of-sample tests of the model given that these moments are not directly targeted in the estimation. Panel A of Table 7 shows the correlation between expected default probability[38], leverage, and interbank spread. I find that the interbank market spread is positively correlated with bank default probability and is negatively correlated with bank leverage. The magnitude of correlations generated by the model is close to those observed in the data. Panel B of Table 7 shows the proportion of aggregate leverage change which can be attributed to asset distribution margin (banks are distributed more densely at lower asset levels) and policy function margin (banks on different asset quantiles employ lower leverage). In both the model and the data, a larger portion of aggregate leverage change can be attributed to the distribution margin. The model

[37]The data for interbank spread are converted from daily frequency to annual frequency in this figure. See Appendix B.6 for detailed conversion procedures.

[38]The default probability of banks in the data is estimated by bankruptcy frequency in the data set. The Bankscope and Worldscope databases provide detailed information about when a bank becomes inactive. When a bank files for Chapter 7 or Chapter 11 for bankruptcy, it is labeled as inactive. The bank default probability in a given year is calculated as the number of banks that become inactive divided by the total number of banks in that year.

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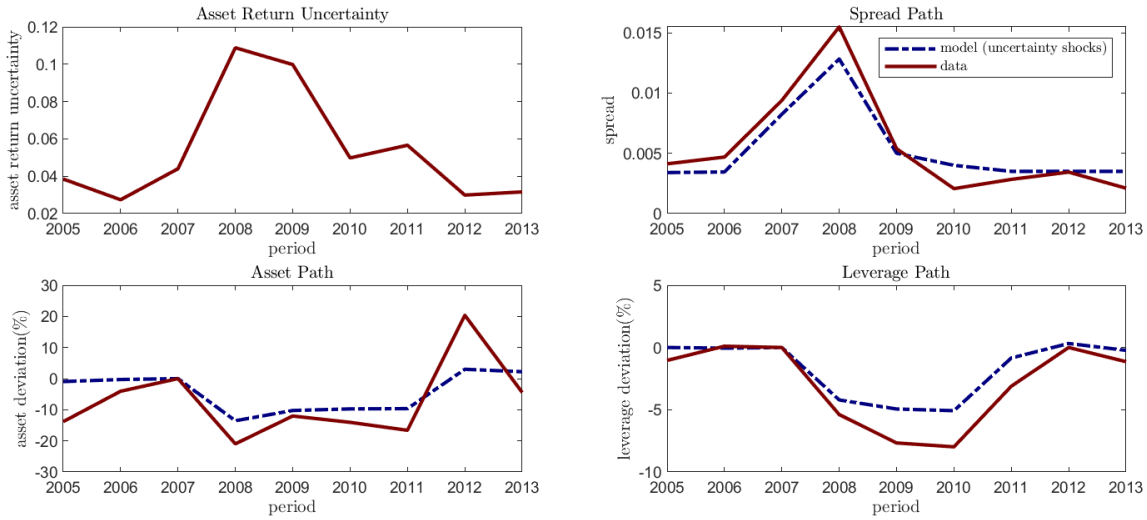


Figure 9: **Event Study: model vs. data**. The figure depicts the simulated evolution paths of total assets, aggregate leverage ratio, and mean interest spread of shadow banks in response to the asset return uncertainty shocks compared with their data counterparts. Leverage is defined as the total-debt-to-total-assets ratio. The data and simulated paths by the benchmark model are shown by the solid line and dashed line respectively.

over-predicts the policy function margin by only 5%. Panel C decomposes the percentage of the decline in asset size comes from shadow banks' defaults and the percentage comes from the shrinkage of assets by existing banks. Overall, the model not only successfully matches the change in average data moments, but also the underlying changes in policy function and bank asset distribution.

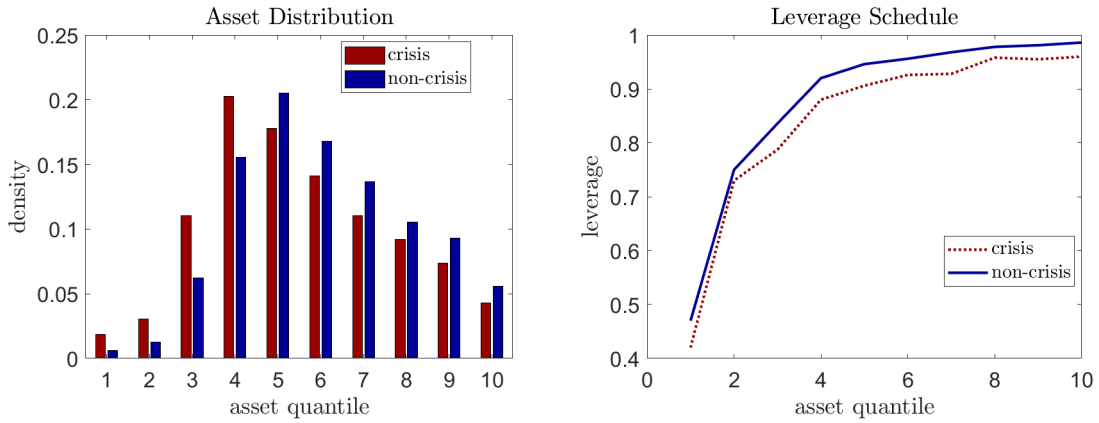


Figure 10: **Asset Distribution and Leverage Over Different Asset Quantiles**. This figure depicts the simulated asset distribution and leverage schedule over different asset quantiles based on the simulated bank panel. Crisis periods are defined as periods with high asset return uncertainty value. The rest of the periods are defined as non-crisis periods.

The model can also generate a similar pattern for the change in the leverage schedule over different asset quantiles and asset distributions, as we observe in the data. The left panel of Figure 10 shows the asset distribution in non-crisis periods and crisis periods. Banks reduce their size such that the whole

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Table 7: Statistics: model vs. data

A. Correlations	Model	Data
Correlation(default probability, interest spread)	0.87	0.91
Correlation(leverage, interest spread)	-0.55	-0.48
B. The Decomposition of Change in Leverage		
Distribution margin	56%	61%
Policy function margin	44%	39%
C. The Decomposition of Change in Assets		
Default banks	14%	11%
Existing banks	86%	89%

Notes: This table shows the simulated statistics of shadow banks compared with their data counterparts. Panel A shows the correlation between expected default probability and interbank spread and the correlation between leverage and spread. Panel B shows the contribution of policy function margin and distribution margin to the change in aggregate leverage respectively. Panel C shows the percentage of the decline in asset size comes from shadow banks' defaults and the percentage comes from the shrinkage of assets in existing banks.

distribution of assets skews more to the left in the crisis, similar to our empirical observation. The right panel of Figure 10 shows the change of leverage over different asset quantiles. Consistent with what is observed in the data, banks with higher asset sizes employ higher leverage. During a crisis, banks of different asset quantiles generally employ lower leverage.

To check whether model without maturity mismatch or fire-sale costs can also generate similar impacts on interbank spreads, bank asset sizes and leverages, I also carry out two counterfactual experiments by re-simulating the baseline model under two alternative specifications: one without maturity mismatch in which I impose the maturity mismatch parameter $\lambda = 1$; one without fire-sale cost in which I impose the fire-sale cost parameter $\zeta = 1$. In both simulation, I fit in the sequence of asset return uncertainty shocks in the baseline model, and the values of other parameters are kept the same as in the baseline model, such that the differences in responses can only result from the differences in maturity mismatch or fire-sale costs. The simulations reveal what would have happened in counterfactual economies if there were no maturity mismatch or fire-sale costs. The simulated pathways of spread, asset, and leverage are shown in Figure C2 in the Appendix. Uncertainty shocks in the model without maturity mismatch or fire-sale costs explain only around 30% of the change in the interbank spread, leverage, and asset of the U.S. shadow banking sector. This implies that maturity mismatch and firesale cost amplify the impact of uncertainty shocks.[39]

[39]In another related quantitative exercise, I shut down maturity mismatch or firesale cost in the model with purely

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The quadratic asset adjustment cost helps smooth the problem and match the mean and standard deviation of new asset issuance/asset ratio in the data. To gauge its effect on model dynamics, I carry out a counterfactual experiment where I shut down the asset adjustment cost by setting the quadratic adjustment cost parameter ϕ to 0 while keeping the values of other parameters the same as the baseline estimates. As expected, without asset adjustment cost, banks are more willing to change their asset sizes in response to uncertainty shocks and the asset contraction is slightly stronger. The absence of costly asset adjustment reduces the bank default probability on average, which leads to smaller changes in interest spread. The changes in leverage are smaller accordingly.

To further test whether first-moment shocks to asset return can also generate similar impacts on interbank spreads, bank asset size, and leverage, following Arellano, Bai, and Kehoe (2018) and Bloom, Floetotto, Jaimovich, Saporta-Eksten, and Terry (2019), I re-simulate the baseline model with purely first-moment shocks, i.e., shocks to the level of asset return. The risky loan gives the bank the gross return $z_t x_t k_t$. z_t is the idiosyncratic gross rate of return on the bank assets. x_t is the aggregate state variable that captures the mean asset return rate of the economy. I choose a sequence of first-moment shocks to mean asset return so that the mean of measured gross return on assets in the model reproduces the path of its data counterpart. The values of other parameters are kept the same as in the baseline model, such that the differences in responses can only result from the differences in the shocks we feed in. The responses of the economy to purely first-moment shocks to asset returns are shown in Figure 11. The model with purely first-moment shocks only explains around 20% of the changes in the interbank spread, leverage, and asset size of the U.S. shadow banking sector.[40] I also re-estimate the model under these alternative specifications and carry out the event studies using these re-estimated parameters. The results of the event studies are quantitatively similar.

To summarize the explanatory power of each model specification, I report in Table 8 to what extent each model specification could explain the asset contraction, deleveraging, and changes in interest spread. Column 1 shows the pure impact of uncertainty shocks in the baseline model. Column 2 shows the impact of uncertainty shocks in a no maturity mismatch model where I set the maturity mismatch

first-moment shocks. Maturity mismatch and firesale cost amplify the impact of first-moment shocks by a similar proportion as they do to the uncertainty shocks.

[40] This is a quantitative result. Both first-moment shocks and second moment shocks could move certain fractions of banks below the default threshold if the banks do not change their investment and leverage choices. The estimation reveals that the uncertainty shocks would move a bigger measure of banks below the default threshold than the first-moment shocks if banks stick to their leverage and investment policies of non-crisis periods in crisis periods. To reduce the probability of default, banks choose to reduce their leverage and investment more dramatically under uncertainty shocks.

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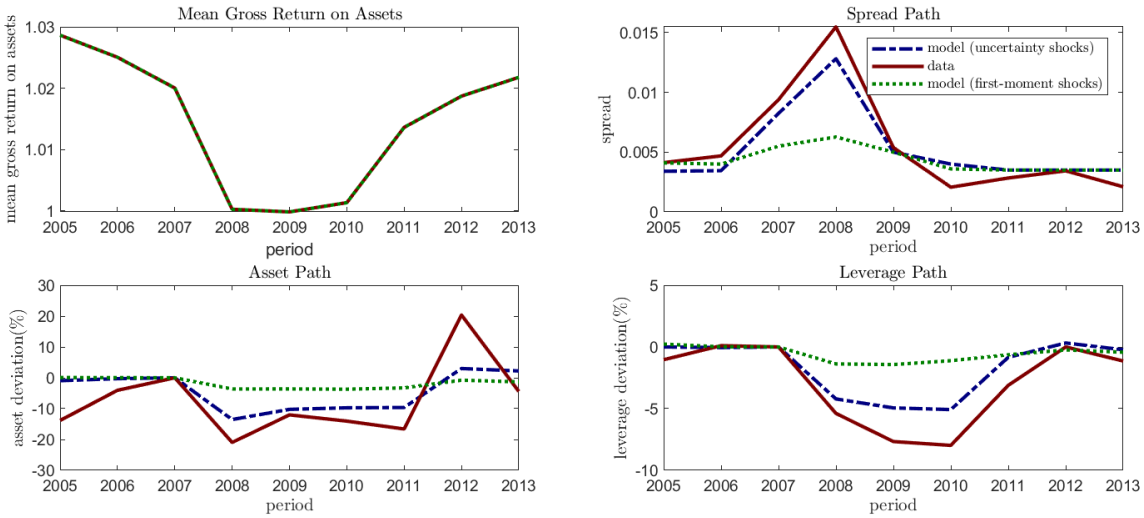


Figure 11: **Event Study: model (pure first-moment shocks) vs. data**. The figure depicts the simulated evolution paths of total assets, aggregate leverage ratio, and mean interest spread of shadow banks in response to the first-moment (level) shocks compared with second-moment (uncertainty) shocks and their data counterparts. Leverage is defined as the total-debt-to-total-assets ratio.

parameter λ to 1 while keeping other parameters the same as the baseline estimates. Column 3 shows the impact of uncertainty shocks in a no-fire-sale cost model where the fire-sale cost parameter ζ is set to 1 while other parameters are kept the same as the baseline estimates. Column 4 shows the impact of uncertainty shocks in a model where the quadratic adjustment cost parameter ϕ is set to 0 while other parameters are kept the same as the baseline estimates. Column 5 shows the pure impact of first-moment shocks to asset return. Column 6 shows the joint impact of first-moment shocks and uncertainty shocks (second-moment shocks) in the baseline model. First-moment shocks and uncertainty shocks together in the baseline model could already explain around 90% deleveraging and asset contraction observed in the data.

These event studies show that maturity mismatch and fire-sale cost significantly amplified the impact of uncertainty shocks. And uncertainty shocks play a more important role than first-moment shocks in the 2007-2009 financial crisis. These results have strong policy implications. First, since maturity mismatch and fire-sale cost amplify the impacts of uncertainty shocks, to counter the rise in interbank spread and stabilize the credit intermediation, certain measures should be taken to restrict the degree of maturity mismatch and reduce the cost of asset fire sales. Also, if the government policies only focus on first-moment shocks, rather than the more severe culprit of the crisis-uncertainty shocks, it can at most reduce the credit contraction by 20%. Uncertainty shocks will still lead to dramatic

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Table 8: Disentangling the Model Dynamics

	Uncertainty Shocks	No Maturity Mismatch	No Fire-sale Cost	No Adjustment Cost	First-Moment Shocks	1st+2nd Moment Shocks
Uncertainty shocks	Yes	Yes	Yes	Yes	No	Yes
First-moment shocks	No	No	No	No	Yes	Yes
Fire-sale cost	Yes	Yes	No	Yes	Yes	Yes
Maturity mismatch	Yes	No	Yes	Yes	Yes	Yes
Adjustment cost	Yes	Yes	Yes	No	Yes	Yes
Asset contraction	72.08%	31.02%	27.90%	81.45%	21.96%	90.44%
Deleveraging	70.12%	29.03%	27.46%	62.14%	21.46%	90.35%
Δ Interest spread	72.66%	31.65%	28.83%	64.03%	22.01%	91.10%

Notes: This table reports to what extent each model specification could explain the asset contraction, deleveraging, and changes in interest spread. Column 1 shows the impact of uncertainty shocks in the baseline model. Column 2 shows the impact of uncertainty in a no maturity mismatch model where I set the maturity mismatch parameter λ to 1 while keeping other parameters the same as the baseline estimates. Column 3 shows the impact of uncertainty in a no-fire-sale cost model where the fire-sale cost parameter ζ is set to 1 while other parameters are kept the same as the baseline estimates. Column 4 shows the impact of uncertainty shocks in a model where the quadratic adjustment cost parameter ϕ is set to 0 while other parameters are kept the same as the baseline estimates. Column 5 shows the impact of first-moment shocks. Column 6 shows the joint impact of first-moment shocks and uncertainty shocks in the baseline model.

credit contractions. To counter credit contraction, only maintaining the average asset return above a certain level is insufficient; certain measures should be taken to contain the asset return uncertainty since it plays a more important role in affecting the size of shadow banking.[41]

I also re-estimate the model under the following four specifications: model with purely first-moment shock,[42] model without fire-sale cost (baseline model with fire-sale parameter ζ being set to 1), model without maturity mismatch (baseline model with maturity mismatch parameter λ being set to 1), and model without asset adjustment cost (baseline model with asset adjustment cost parameter ϕ being set to 0). Table D2 in the Appendix reports the estimation results. Models with the first three alternative specifications tend to underestimate the mean of interest spread and default probability, the standard deviation of dividend/asset ratio and new asset issuance/asset ratio, and overestimate the mean of leverage. The model without asset adjustment cost tends to over-predict the mean and the standard deviation of asset issuance/asset ratio. The over-identification tests fail to reject the baseline model with a p-value of 0.6440. The simulated moments are not statistically significant different from the actual moments, hence the baseline model is valid. The tests reject the model with purely first-moment shock, the model without fire-sale costs, the model without maturity mismatch, and the model without

[41] Potential policy remedies include stress tests on the shadow banking system ex ante to contain the riskiness of the bank asset portfolio and ex post government interventions to restrict asset price movements.

[42] In this model, I impose the uncertainty of the bank return constant by setting $\sigma_t = \overline{\sigma}, \forall t$, where $\overline{\sigma}$ is estimated to match the average asset return uncertainty in the data.

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asset adjustment cost at the significance level of 10%. The estimated parameters for each specification are reported in Table D5 in the Appendix.

4.5 Policy Experiment: Debt Guarantee and TARP

In this section, I extend the baseline model to further study the effects of government intervention. In the 2007-2009 financial crisis, to stimulate the economy, the Federal Reserve took drastic unconventional monetary policies, including direct lending, the troubled Asset Relief Program (TARP), and debt guarantee, among other steps. One strand of literature led by Gertler and Kiyotaki (2010) analyzes the effect of unconventional monetary policies on the real economy ex post. At the same time, a growing body of theoretical literature investigates the moral hazard caused by the government bailout. When considering moral hazard, government intervention can be severely counterproductive. Once forming the expectation of a government bailout, banks might act even more recklessly and this might even lead to a vicious cycle. I focus on the effects of two government policies: the Troubled Asset Relief Program (TARP) and debt guarantee.

First, I consider the impact of Troubled Asset Relief Program (TARP), which the U.S. government announced during the recent financial crisis. Under TARP, the U.S. Treasury provided capital to many financial institutions of all sizes across the country. For shadow banks which received the rescue funds, the total rescue funds amounted to around 5% of their total assets. The Federal Reserve took some nonstandard procedures such as extending the range of collateral a bank can use to borrow from the central bank, and the exchange of illiquid assets for liquid assets. I model TARP in the following way: the government would purchase a certain fraction of a bank's long-term assets at full value, such that the bank can avoid part of the asset fire-sale costs. The balance sheet of the bank under TARP is shown in Table 9.

Table 9: Bank Balance Sheet Under TARP

Assets	Liabilities
Long-term asset $(1-\lambda)k_t - G_t$	Net worth n_t
G_t	
Current asset $\pi_t + \lambda k_t$	Risky debt b_t

Notes: This table shows the bank balance sheet under TARP. G_t is the amount of cash shortage replenished by the government.

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In Table 9, G_t is the amount of cash shortage replenished by the government:

$$G_t = \xi^g \left[b_t - q_t b_{t+1} + \xi - (1-\tau_c)\pi_t - \lambda k_t - \frac{\tau_c(b_{t+1} - q_t b_{t+1})}{1+r_t} \right] \tag{22}$$

where ξ^g is a parameter which governs the fraction of cash shortage replenished by the government, and is calibrated to match the fraction of total government rescue funds in total assets. After government intervention, the bank still needs to fire-sale $\frac{(1-\xi^g)(b_t - q_t b_{t+1} + \xi - (1-\tau_c)\pi_t - \lambda k_t - \tau_c(b_{t+1} - q_t b_{t+1})/(1+r_t))}{\zeta}$ amount of its long-term assets to pay the debt due, the total amount of long-term assets left after the fire-sale is $(1-\lambda)k_t - G_t - \frac{(1-\xi^g)G_t}{\zeta\xi^g}$. The bank constraint becomes:

$$e_t = (1-\lambda)k_t - G_t - \frac{(1-\xi^g)G_t}{\zeta\xi^g} - k_{t+1} - \Phi(k_{t+1}, k_t). \tag{23}$$

Second, I analyze the impact of debt guarantee. During the recent financial crisis, the government also announced a debt guarantee for new short- and medium-term debt issued by eligible institutions in case of bank default. I examine the impact of this program by making the lender in the interbank market take into account the possibility that the government will guarantee the debt of the bank in case of default. This formulation is in the same spirit as the "third-party bailout" in Aguiar and Gopinath (2006).

Then the price schedule for risk debt can be written as follows:

$$\begin{aligned} q(z,k',b';\mathbf{s}) \quad = \quad & \mathbb{E}_{z,s}\left\{ m(\mathbf{s},\mathbf{s}') \left[1 - d(z',k',b';\mathbf{s}') \right] \right\} \\ & + \mathbb{E}_{z,s}\left\{ m(\mathbf{s},\mathbf{s}') \left[(1-p)\, d(z',k',b';\mathbf{s}') R(z',k',b';x') + p\, d(z',k',b';\mathbf{s}') \right] \right\}. \end{aligned} \tag{24}$$

Since the bank can be bailed out by the government after it chooses to default, its default value is no longer zero. After taking into account the government bailout, the default value for the bank becomes:

$$V^d = pV^c(z,k,0;\mathbf{s}). \tag{25}$$

I take into consideration the bank's moral hazard by modeling banks with larger assets having a higher probability of receiving a debt guarantee. I assume the bailout probability p is endogenously determined by the bank's size and asset return. In particular, I assume that the bailout probability is an increasing function of the bank's size and its asset return: $p = \kappa(zx)^\chi k^{1-\chi}$. Then I compare the statistics under endogenous bailout and exogenous bailout. The expected bailout probabilities of

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Table 10: Government Policy Comparison

	No Policy	TARP	Debt Guarantee	
			exogenous bailout	endogenous bailout
mean spread	0.53%	0.48%	0.51%	1.54%
mean leverage	0.930	0.979	0.983	0.991
mean default rate	1.09%	0.80%	1.31%	2.00%
total liquidity	2.916	3.327	3.256	3.114
asset contraction	14.91%	13.52%	17.19%	26.59%

Notes: This table shows the simulated statistics of shadow banks using simulated bank panel under different government policies. TARP is modeled as the government purchases a certain fraction of bank's long-term asset at full value, such that the bank can avoid part of the costs of a fire-sale. Debt guarantee is modeled as the government bails out a bank with a certain probability if a bank defaults. Bailout probability is the same for all banks under the exogenous bailout case and is an increasing function of bank asset size under the endogenous bailout case.

these two types of bailout strategies are the same. The difference is that, in the case of exogenous bailout, the government literally chooses which banks to bailout by lottery, whereas in the case of an endogenous bailout probability, the government bails out the banks with larger assets with higher probability.

I compare the different effects of debt guarantee and TARP on default rate, leverage, interbank spread, market liquidity, and asset contraction.[43] The results are shown in Table 10.[44] In this table, two government interventions are chosen such that the total government resources used in each case are equal. Under a debt guarantee, two bailout probability strategies are chosen such that the expected bailout probabilities are the same in the two cases. The first column shows the statistics under no government policy. The second column corresponds to asset purchases of 5% of total bank assets under TARP. The third and fourth columns correspond to the expected government bailout probability of 21.2% in case the bank defaults. This is obtained when I set $\kappa = 0.121$ and $\chi = 0.358$. [45]

As shown in Table 10, given government intervention strategies, the banks on average would choose higher leverage ex ante. Under TARP, because banks partly avoid a costly fire-sale, the banks' default probability is lower. Thus, they face lower spreads than in the no policy case. However, the

[43]The total liquidity is calculated as the sum of assets of all non-default banks in the market, and the unit is 10^5. It inherently captures the size of credit intermediation by shadow banks. To calculate the asset contraction, I carry out the same event study as in Section 4.4.3.

[44]Before the financial crisis, since shadow banks are not under the regulatory oversight of the central banks and are not protected by the FDIC, the expected probability of government bailout for these financial institutions is on average very low. The results of this quantitative exercise should be interpreted as the counterfactual responses of shadow banks if they had anticipated the government bailout, or the actions of shadow banks in the future given their changed belief resulting from the government intervention in the crisis.

[45]The detailed procedures for the calculation are in Appendix C.4.

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banks would employ a slightly higher leverage than in the no policy case. Under the debt guarantee intervention, when the bailout probability is set as an increasing function of bank size, a bank would employ a higher leverage than in the exogenous bailout case. The reason is that a bank would tend to employ a higher leverage to build a bigger "empire" to take advantage of the government bailout. The implication is that it might be optimal to make uncertainty of government intervention as a commitment, as suggested in Nosal and Ordonez (2016). The uncertainty of government bailout makes the bank more cautious, avoiding being the worst performer and exposing itself to the risk of not getting bailed out. Also, notice that under a debt guarantee, although the default probability of banks is higher, because of government debt guarantee, the spread the bank faces can be lower than in the no policy case. Under a debt guarantee, the total liquidity in the market is higher because banks employ higher leverage. However, since the bank default rate is higher under debt guarantee, the total liquidity of the market is lower than that under TARP. In particular, under the endogenous bailout case, although the total liquidity is higher than the no policy case because banks choose higher leverage, it is lower than the TARP or exogenous bailout case because banks default with higher probability. In terms of asset contraction: under TARP, since banks partly avoid a costly fire-sale, the banks' default probabilities and spreads are lower than in the no policy case, and the asset contraction is dampened; under debt guarantee, banks employ much higher leverage and default more frequently, which leads to a larger asset contraction in response to adverse shocks.[46] Debt guarantees with endogenous government bailout have a much more deleterious effect on bank behaviour because they raise the value of equity in times of default and thus engender risk-shifting activities. This is consistent with Bianchi (2016), who argues that bailouts are idiosyncratic and targeted make the economy significantly more exposed to financial crises. This is because bailouts that depend on idiosyncratic firm decisions lead to significantly more overborrowing.[47]

[46] As a caveat, one should be careful about the interpretation of the asset contraction. On the one hand, to the extent that other unmodelled economic agents (e.g., household) would prefer to smooth their consumption, the large fluctuation in credit could be detrimental to welfare if it leads to large swings in consumption streams (see Krusell, Mukoyama, Sahin, and Smith (2009) and Gelain, Lansing, and Natvik (2017) among others). On the other hand, larger credit contraction could lead to shift to a safer portfolio, which reduces the instability of the financial system. A counterfactually smaller contraction might lead to higher financial risk and lower agents' welfare. An extended general equilibrium model can speak more about the welfare implications of asset contraction. In such a framework, the welfare of the economic agent will be a joint function of risk and the agent's consumption level. Depending on which force dominates, the welfare implications could be different. While the welfare implication of the asset contraction remains outside the scope of this paper, this highlights an interesting direction for future research.

[47] The anticipation of such bailouts distorts ex ante borrowing incentives: an individual bank has an incentive to borrow more because the transfer it will receive in the future is increasing in the banks' individual debt.

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5 Subsample Estimation

Although the model match the full sample very well and could explain a large fraction of deleveraging and asset contraction, there is a wide heterogeneity in terms of maturity mismatch and fire-sale cost. Such heterogeneity provides a natural setting to test my model and examine whether it can generate predictions consistent with the data.

To check the cross-sectional fitness of the model,[48] I sort the shadow banks based on the measures of fire-sale costs or maturity mismatch and split the full sample into two subsamples.[49] I reestimate the model based on these subsamples and report the average deleveraging and asset contraction in each subsample.[50] The results are shown in Table 11. First, the actual data reported in columns 1 and 2 of Table 11 reveals that deleveraging and asset contraction are much stronger in bank subsamples with higher fire-sale cost and higher maturity mismatch. Next, I compare the model-generated moments with those observed in the actual data, and conclude that the model is able to fit the heterogeneous bank characteristics in each subsample. The over-identification tests fail to reject the baseline model in each subsample. The simulated moments are not statistically significantly different from the actual moments. Last, relying on the structural model, I compare the contribution of uncertainty shocks to deleveraging and the asset contraction in each subsample. I find that uncertainty shocks can account for a slightly larger portion of deleveraging and asset contraction in banks with higher fire-sale costs and higher maturity mismatch.

6 Model Extensions and Robustness Checks

I have made several simplifying assumptions to construct my baseline model, and it is possible that my results are sensitive to these assumptions. In addition, it is possible that alternative extensions of the first-moment shock model might be able to fit the data as well as the baseline model. This section addresses these two broad issues, with all of the results I report below documented in detail in the Appendix. I first further extend the model to study comprehensively the impact of alternative shocks that could have explained the contraction of credit in the shadow banking sector. Specifically, I study the joint contributions of uncertainty shocks, first-moment shocks to asset return, shocks to the

[48] I also examine the time-series fitness of the model. For space limitation, the results are relegated to Appendix C.5.
[49] See Appendix B.2 for the details for measurement construction.
[50] The estimation results are reported in Table D6 in the Appendix.

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Table 11: Subsample Results

	deleveraging	asset contraction	$uncertainty_1$%	$uncertainty_2$%	J-stat	p-value
A: Split by asset fire-sale cost						
high fire-sale cost	8.63%	25.17%	75.38%	75.40%	2.11	0.71
low fire-sale cost	5.87%	18.06%	65.44%	65.41%	3.08	0.54
B: Split by maturity mismatch						
high maturity mismatch	9.72%	29.08%	76.18%	75.62%	2.03	0.73
low maturity mismatch	4.49%	12.01%	64.36%	64.89%	3.45	0.48

Notes: This table reports the results for the subsample estimations. I sort the shadow banks based on the measures of fire-sale costs or maturity mismatch and split the full sample into two subsamples. The actual data reported in column one and two of Table 11 reveals that deleveraging and asset contraction are much stronger in bank subsamples with higher fire-sale costs and higher maturity mismatch. Column 3 and 4 compare the model-generated moments with those observed in the actual data. $uncertainty_1$ and $uncertainty_2$% report the contributions of uncertainty shocks to deleveraging and asset contraction in each subsample. The J-statistic and p-value test the overidentification constraint for the moment conditions in the estimation.

liquidation/resale value of assets, and financial shocks. Then I extend the first-moment shock model by incorporating endogenous asset portfolio choices and systemic risk. I also further consider two more extensions of the baseline model by introducing time variations to the volatility of aggregate productivity and incorporating a traditional banking sector to the model.[51]

6.1 Incorporating Financial Shocks

In the first extension, following Eisfeldt and Muir (2016), Nikolov and Whited (2014), and Alfaro, Bloom, and Lin (2019), I incorporate financial shocks to the baseline model by allowing the equity issuance cost parameter γ_e to be time-varying. Specifically, I assume γ_t^e follows an AR(1) process with mean μ_γ, persistence ρ_γ, and standard deviation σ_γ:

$$log\left(\gamma_t^e\right) = (1-\rho_\gamma)\, log\left(\mu_\gamma\right) + \rho_\gamma log\left(\gamma_{t-1}^e\right) + \nu_t,\ \nu_t \sim N(0, \sigma_\gamma^2) \tag{26}$$

Similar to Eisfeldt and Muir (2016), I reestimate the model and choose the values of μ_γ, ρ_γ, and σ_γ such that the mean and standard deviation of equity issuance, and the covariance between equity issuance and new asset issuance in the model match the corresponding data moments. The mean parameter μ_γ mainly affects the average equity issuance. The standard deviation parameter σ_γ mainly affects the standard deviation of equity issuance. The persistence parameter ρ_γ and standard deviation

[51] Due to page limitation, I relegate extra robustness checks regarding endogenous asset portfolio choice, systemic risk, time-varying aggregate productivity volatility, and traditional banking sector to Appendix C.6, C.7, C.8, and C.9.

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parameter σ_γ help match the covariance between new asset issuance and equity issuance. Without the stochastic cost, the correlation between new asset issuance and equity issuance will be quite high since banks tend to raise more equity when there are good productivity shocks and hence good investment opportunities. The stochastic shocks help break this high correlation in the model.

The estimation results for this extended model with financial shocks are reported in Table D7 in the Appendix. When estimating the model, the continuous AR(1) process is discretized into a two-state markov process following the methods of Tauchen and Hussey (1991): $\gamma_t^e \in \{\gamma_h^e, \gamma_l^e\}$. γ_l^e and γ_h^e denote its values in non-crisis and crisis periods, respectively. The estimation reveals that $\gamma_l^e = 0.0461$ and $\gamma_h^e = 0.0807$, which implies the average equity financing cost increased by around 75% for shadow banks in crisis, close to what Eisfeldt and Muir (2016) find for non-financial firms.

Using the estimated extended model, I carry out the same event study as in Section 4.4.3 to gauge the impact of financial shocks and compare its impact with that of the uncertainty shocks. The results are shown in columns 1-4 of Table D3 in the Appendix. Financial shocks and uncertainty shocks together can explain around 76% of asset contraction and 77% of deleveraging. By comparison, financial shocks and first-moment shocks together can only explain 25% of asset contraction and 26% of deleveraging. Without uncertainty shocks and first-moment shocks, financial shocks alone can only explain 11% of asset contraction and 12% deleveraging.[52]

6.2 Incorporating Fire-sale Cost Shocks

Following Gilchrist, Sim, and Zakrajsek (2014), I introduce fire-sale cost (asset liquidation cost) shocks to the baseline model by allowing the fire-sale cost parameter ζ_t to be time-varying. Assume ζ_t follows an AR(1) process:

$$log\left(\zeta_t\right) = \left(1-\rho_\zeta\right) log\left(\mu_\zeta\right) + \rho_\zeta log\left(\zeta_{t-1}\right) + v_t,\ v_t \sim N(0, \sigma_\zeta^2) \tag{27}$$

Due to lack of data source that tracks the resale value of capital, Gilchrist, Sim, and Zakrajsek (2014) estimate an AR(1) specification for capital resale cost using as a proxy the ratio of the price index of used car sales relative to that of new car sales, two components of the monthly CPI published by the BLS. Their parameter estimates are less applicable for my setting since shadow banks hold

[52] I also test the robustness of the results under two alternative specifications in Appendix C.10.

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financial assets rather than physical assets. Due to lack of data that tracks the fire-sale cost in the shadow banking industry over time, I reestimate the model and choose the values of μ_ζ, ρ_ζ, and σ_ζ such that the mean and standard deviation of leverage, and the covariance between leverage and new asset issuance in the model match the corresponding data moments. The mean parameter μ_ζ mainly affects the average leverage. The standard deviation parameter mainly affects the standard deviation of leverage. The persistence parameter ρ_ζ and standard deviation parameter σ_ζ help match the covariance between new asset issuance and leverage. The covariance between new asset issuance and investment is informative about the position of debt in the financing pecking-order, which hinges upon ζ.

The estimation results for this extended model with fire-sale cost shocks are reported in Table D8 in the Appendix. When estimating the model, the continuous AR(1) process is discretized into a two-state Markov process following the methods of Tauchen and Hussey (1991): $\zeta_t \in \{\zeta_h, \zeta_l\}$. ζ_l and ζ_h denote its values in crisis and non-crisis periods, respectively. The estimation reveals $\zeta_l = 0.584$ and $\zeta_h = 0.841$, which implies a haircut rate of 42% in recession and 16% in normal periods. These values lie within the broad range of estimated haircut rates for nonbank financial institutions by BIS Committee on the Global Financial System.[53]

Using the estimated extended model, I carry out the same event study as in Section 4.4.3 to gauge the impact of fire-sale cost shocks and compare its impact with that of the uncertainty shocks. The results are shown in Columns 5-8 of Table D3 in the Appendix. Fire-sale cost shocks and uncertainty shock together can explain around 78% of asset contraction and 77% of deleveraging. Fire-sale cost shocks and first-moment shocks together can only explain 27% of asset contraction and deleveraging. If we shut down uncertainty shock and first-moment shocks, fire-sale cost shocks alone in the extended model can only explain 14% of asset contraction and 13% of deleveraging. These estimates can be viewed as the upper bound for the true effects of fire-sale shocks since relatively small liquidations/resells of assets might not have any fire-sale costs.[54]

I also reestimate a full-blown model where I incorporate both financial shocks and fire-sale cost shocks to the baseline model. The estimation results are reported in Table D9 in the Appendix. Using this full-blown model, I carry out the same event study as in Section 4.4.3. The results are shown in

[53] There are very few empirical surveys of haircuts in the repo and securities lending markets. One of the more comprehensive was conducted by the BIS Committee on the Global Financial System in 2007 and 2009.

[54] I also test the robustness of the results under three alternative specifications in Appendix C.10.

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Columns 9-12 of Table D3 in the Appendix. The full-blown model can explain around 98% of asset contraction and deleveraging, as shown in column 9 of Table D3. Uncertainty shocks still matter more than first-moment shocks in this model. Financial shocks and fire-sale cost shocks together can at most explain 24% of asset contraction and deleveraging we observe in the data.

7 Conclusion

In this paper, I develop and estimate a tractable quantitative framework to study the shadow banking dynamics. I first document several stylized facts of the U.S. shadow banking industry using a detailed micro-level data set. Then I build a quantitative model with heterogeneous banks, uncertainty fluctuations, and endogenous bank default to explain the increase in the interbank spread, the deleveraging, and the contraction of shadow banking credit intermediation during the recent financial crisis. When the uncertainty of bank asset return increases, interbank spread spikes. Shadow banks would employ a lower leverage to reduce their default probabilities and financing costs. This leads to a contraction in the credit intermediation. The model characterizes key features of shadow banking dynamics in a full nonlinear manner.

I estimate the model using a comprehensive micro-level data set to quantify the impacts of time-varying uncertainty in the shadow banking sector. The estimations reveals that uncertainty shocks can generate statistics closely match the data and explain 72% of asset contraction and 70% of deleveraging in the U.S. shadow banking industry. High maturity mismatch and fire-sale costs amplify the impact of uncertainty shocks. Alternative model settings with only pure first-moment shocks, without asset fire-sale costs or without maturity mismatch, can explain around at most one-third of the changes in the interbank spread, leverage, and assets of the U.S. shadow banking sector. Using the quantitative model, I also analyze the impact of unconventional monetary policies. The finding suggests that unconventional monetary policies can help dampen the credit contraction. However, after taking into account bank moral hazard, I find that a government bailout might be counterproductive, especially when bailout decisions are based on bank characteristics. I also further extend the baseline model to study in a comprehensive manner the impact of alternative shocks that could have explained the contraction of credit in the shadow banking sector. The event studies reveal that the baseline model results remain robust in these extended models: uncertainty shocks still play a more important role

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than first-moment shocks; maturity mismatch and fire-sale cost significantly amplified the impact of uncertainty shocks; financial shocks or fire-sale shocks alone can not explain the observed shadow banking dynamics. Despite the complex setting and rich dynamics, the model is highly tractable and can be easily extended to study banks' asset portfolio choices, systemic risk, and the difference between shadow banks and traditional banks.

To keep the analysis tractable, I have focused on a stylized model that abstracts from some realistic features in the actual economy. I intentionally keep the baseline model as parsimonious as possible for tractability and clarity of the theoretical results. However, the key mechanism of the model is likely to carry over to an alternate model with additional features. An important direction for future research is to study optimal fiscal and monetary policy interventions in the presence of bank moral hazard. It would be interesting to explore what policies governments should take ex-ante to avoid excessive risk-taking. A mix of ex-ante macro-prudential policy and ex-post interventions may serve the optimal purpose of banking regulation. The model could also be further extended to study what affect shadow banks' risk-taking incentives after the 2007-2009 financial crisis. Future research along these lines should be both promising and fruitful.

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Online Appendices for "Uncertainty and the Shadow Banking Crisis: Estimates from a Dynamic Model"

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A SMM estimation

The model has 22 parameters, as shown in Table 4. 8 parameters are calibrated externally and the rest 14 parameters are jointly estimated following a simulated method of moments approach as in Erickson and Whited (2002). The 14 parameters include the following: equity issuance cost γ_e, fire-sale cost ζ, parameters governing the value and transition matrix of aggregate uncertainty shocks $\pi^{\sigma}_{H,H}, \pi^{\sigma}_{L,L}, \sigma_H, \sigma_L$, the asset adjustment cost ϕ, persistence of asset return ρ_v, fixed operating cost ξ, entrant start-up asset k_0, entry cost f_e, Pareto distribution parameter of the entrant productivity ω, and two parameters μ and A that govern the Pareto distribution of the permanent productivity. They are jointly estimated to match the following 18 moments: the mean of leverage, the mean of default rate, the mean, standard deviation, and the auto-correlation of dividend payout/asset ratio and new asset issuance/asset ratio, the mean of equity issuance, the mean, standard deviation, auto-correlation, and skewness of asset return uncertainty, the average growth rate and relative size of new entrant banks, mean and standard deviation of interest spread, and the slope of the leverage schedule using simulated methods of moments (SMM), which minimizes a distance criterion between key moments from actual data and simulated data. SMM proceeds in the following way: For an arbitrary value of parameter vector $\theta = \{\gamma_e, \zeta, \pi^{\sigma}_{H,H}, \pi^{\sigma}_{L,L}, \sigma_H, \sigma_L, \phi, \rho_v, \xi, k_0, f_e, \omega, \mu, A\}$, the dynamic problem is solved and the policy functions are generated. Then I use the policy functions to simulate a data panel of $(KN, T+1000)$, where K is a strictly positive integer denoting the number of simulated panel data sets, N is the length of the simulated sample, and T is the time dimension of the simulated data. The first 1000 periods are discarded so as to start from the ergodic distribution.

Let x_{it} be the actual data vector, $i \in \{1, ..., N\}$, $t \in \{1, ..., T\}$, and let $y_{itk}(\theta)$ be the simulated vector from simulation k, $i \in \{1, ..., N\}$, $t \in \{1, ..., T\}$, and $k \in \{1, ..., K\}$. The simulated data vector, $y_{itk}(\theta)$, depends on a vector of structural parameters, θ. Define the moment conditions as:

$$\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left[h(x_{it}) - \frac{1}{K}\sum_{k=1}^{K}h(y_{itk}(\theta))\right] \equiv \Psi^A - \Psi^S(\theta) \tag{28}$$

where $h(y_{itk}(\theta))$ is a vector of simulated moments and $h(x_{it})$ is the actual data moments. $\Psi^A = \frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}h(x_{it})$, $\Psi^S(\theta) = \frac{1}{NTK}\sum_{i=1}^{N}\sum_{t=1}^{T}\sum_{k=1}^{K}h(y_{itk}(\theta))$

The simulated moments estimator is defined as the solution to the minimization of:

$$\hat{\theta} = \underset{\theta}{argmin}\left[\Psi^A - \Psi^S(\theta)\right]' \hat{W}\left[\Psi^A - \Psi^S(\theta)\right] \tag{29}$$

in which $\hat{W}$ is a positive definite matrix that converges in probability to a deterministic positive definite matrix W. It is constructed by calculating the inverse of the variance-covariance matrix of the data moments. Define Ω as the variance covariance matrix of the data moments Ψ^A. Lee and Ingram (2010) show that under the estimating null, the variance covariance of the simulated moments $\Psi^S(\theta)$ is equal to $\frac{1}{K}\Omega$. Since $\Psi^A$and $\Psi^S(\theta)$ are independent by construction, $\hat{W} = \left[(1+\frac{1}{K})\Omega\right]^{-1}$. Ω is calculated using influence function method following Erickson and Whited (2002).

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I use a simulated annealing algorithm to minimize the objective function. This starts with a predefined first and second guess. For the third guess onward, it takes the best prior guess and randomizes from this to generate a new set of parameter guesses. That is, it takes the best-fit parameters and randomly "jumps off" from this point for its next guess. Over time the algorithm "cools", so that the variance of the parameter jumps falls, allowing the estimator to fine-tune its parameter estimates around the global best fit. I restart the program with different initial conditions to ensure the estimator converges to the global minimum. The simulated annealing algorithm is extremely slow, which restricts the size of the parameter space that can be estimated. Nevertheless, I use this because it is robust to the presence of local minima and discontinuities in the objective function across the parameter space.

The simulated moments are asymptotically normal for fixed K. Denote $g(\theta) \equiv \Psi^A - \Psi^S(\theta)$. The asymptotic distribution of θ is given by:

$$\sqrt{n}(\theta - \hat{\theta}) \xrightarrow{d} N(0, avar(\hat{\theta})) \tag{30}$$

in which

$$avar(\hat{\theta}) = (1 + \frac{1}{K}) \left[\frac{\partial g}{\partial \theta} W \frac{\partial g}{\partial \theta'} \right]^{-1} \left[\frac{\partial g}{\partial \theta} W \Omega W \frac{\partial g}{\partial \theta'} \right] \left[\frac{\partial g}{\partial \theta} W \frac{\partial g}{\partial \theta'} \right]^{-1} \tag{31}$$

in which Ω is the probability limit of a consistent estimator of the covariance matrix. I calculate the estimate of this covariance matrix using influence function of the moment vector clustered at bank level following Erickson and Whited (2002).

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B Empirical Appendix

B.1 Data

I now describe in detail how I obtain the data for shadow banks. In the literature, shadow banks are often defined as financial institutions that do not have access to central bank liquidity or public sector credit guarantees. One of the difficulties in analyzing shadow banking is the lack of publicly available micro-level data set. In this paper, I obtain the yearly bank-level panel data from the Bankscope database, complemented with bank-level data from Worldscope database. Bankscope and Worldscope database contain comprehensive information on financial institutions across the globe. I choose real estate and mortgage banks, investment banks, securities firms, private banking and asset management companies, investment and trust corporations, finance companies and other nonbank financial institutions in the U.S. as representatives of shadow banks. My classification of shadow banks is similar to the Financial Stability Board's measure of shadow banking activity based on nonbank financial intermediaries (NBFIs) engaged in credit intermediation activities. I also choose commercial banks, savings banks, cooperative banks, and bank holding companies as representatives of traditional banks. I drop all banks which have negative equity/asset/capital or deposits, and all banks with faulty records such as inconsistent information on any generic variables: date of establishment/type of company, etc. The final panel data sample contains 281 shadow banks and 9554 traditional banks from 1998 to 2013, covering 49 states of the US.

B.2 The Measurements for Maturity Mismatch and Fire-sale Cost

There is no bank-level information available on maturity and fire-sale cost for shadow banks in my dataset. Due to the data limitation, I resort to aggregate data to obtain the estimates for maturity mismatch and fire-sale cost. Following the approach in Hanson, Shleifer, Stein, and Vishny (2015), I assemble data on the assets and liability of various types of financial intermediaries using the Federal Reserve's Financial Accounts of the United States. I use data on intermediary balance sheet. In an effort to avoid subjective judgment, following Hanson, Shleifer, Stein, and Vishny (2015), wherever possible I assign numerical values for liquidity and maturity based on the bank liquidity requirements put forth under Basel III Accord. For each instrument type, I attempt to choose values of these parameters based on the proposed calibration of Basel III's Net Stable Funding Requirement (NSFR) in Basel Committee on Banking Supervision (2010) and the final calibration of the Liquidity Coverage Ratio (LCR) in Basel Committee on Banking Supervision (2013).

Let A_{ji} and L_{ji} denote intermediary j's assets and liabilities of instrument type i. $A_j = \sum_i A_{ji} = \sum_i L_{ji}$ denotes the total assets of intermediary type j. For intermediary type j, its maturity mismatch index is the weighted average contractual maturity of its liabilities versus its assets: $MMI_j = \frac{\sum_i L_{ij} \times Maturity_i}{\sum_i A_{ji} \times Maturity_i}$. The lower is the index, the higher is the degree of maturity mismatch. The weighted value of maturity mismatch index for non-bank financial institutions is 0.0956. Therefore, the maturity mismatch parameter λ is set at the value of 0.0956, implying that the maturity of asset is around

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$10.46(=1/0.0956)$ times that of the liability.[55]

For intermediary type j, I also construct its fire-sale cost index. The asset fire-sale cost index for intermediary type j is defined as the weighted average illiquidity of its asset holdings: $FCI_j = \frac{\sum_i A_{ij} \times Illiquidity_i}{A_j}$. The higher is the index, the higher is the fire cost. I use these fire-sale cost indices to split the whole sample into subsamples with low and high fire-sale cost in the subsample estimation.

The fire-sale cost index constructed above, however, can not directly be used to measure the fire-sale cost parameter since it has no cardinal meaning. I compare the model estimated fire-sale cost with the estimates of fire-sale cost based on the haircut estimates reported by BIS Committee on the Global Financial System.[56] The estimated fire-sale parameter implies a haircut rate of 42% in recession and 16% in normal periods. These values lie within the broad range of estimated haircut rates for nonbank financial institutions by BIS Committee on the Global Financial System.

B.3 Alternative Measurements for Asset Return Uncertainty

Since I only have annual data for ROA in 15 years, there is not enough data to construct a time-series of the idiosyncratic component of bank ROA using moving sample regression.[57] One possible estimate of uncertainty is based on the following two-step procedure proposed by Gilchrist, Sim, and Zakrajsek (2014) using stock price information of non-bank financial institutions. First, I remove the forecastable variation in daily excess returns using the standard (linear) factor model:

$$R_{it_d} - r^f_{t_d} = \alpha_i + \boldsymbol{\beta_i'} \boldsymbol{f_{t_d}} + u_{it_d} \tag{32}$$

where i index banks and t_d, $d = 1, 2, ..., D_t$ indexes trading days in quarter t. R_{it_d} denotes the daily return of bank i, $r^f_{t_d}$ is the risk-free rate, and $\boldsymbol{f_{t_d}}$ is a vector of observable risk factors. I employ a 4-factor model-namely the Fama and French (1992) 3-factor model, augmented with the momentum risk factor proposed by Carhart (1997).

In the second step, I calculate the quarterly firm-specific standard deviation of daily idiosyncratic returns, according to the following formula:

$$\sigma_{it} = \sqrt{\left[\frac{1}{D_t} \sum_{d=1}^{D_t} (\hat{u}_{it_d} - u_{it})^2\right]} \tag{33}$$

where $\hat{u}_{it_d}$ denotes the OLS residual- the idiosyncratic return-from equation (32) and $u_{it} = \frac{1}{D_t} \sum_{d=1}^{D_t} \hat{u}_{it_d}$ is the sample mean of daily idiosyncratic returns in quarter t. σ_{it} is an estimate of time-varying equity

[55] The (weighted) average maturity of existing loans (assets) at date t , assuming the bank neither defaults nor makes any adjustments on the current investment in loans, is $\sum_{s=0}^{\infty} \frac{(s+1)\lambda k_{t+s}}{k_t} = \sum_{s=0}^{\infty} (s+1)\lambda(1-\lambda)^s = \frac{1}{\lambda}$, as the residual loan outstanding at date $t+s$, $s \geq 0$, is $k_{t+s} = k_t(1-\lambda)^s$.

[56] There are very few empirical surveys of haircuts in the repo and securities lending markets. One of the more comprehensive was conducted by the BIS Committee on the Global Financial System in 2007 and 2009.

[57] For instance, if we use rolling window of 10 years to run the regression, we can at most obtain estimates of bank idiosyncratic component in 6 years.

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volatility for bank i. This measure is purged of the forecastable variation in expected returns. The cross-sectional average of σ_{it} can serve as an estimate for the uncertainty shock.

Figure B1 shows the comparison between the estimated asset return uncertainty and the one estimated using the method proposed by Gilchrist, Sim, and Zakrajsek (2014). Although the raw magnitudes are different for these two measures, they roughly follow the same pattern. The correlation between the two series is as high as 0.9846.

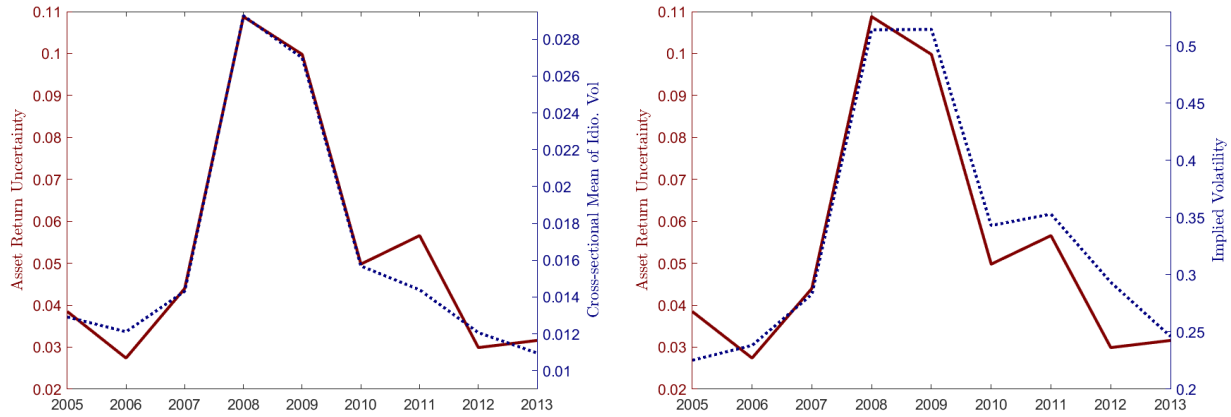


Figure B1: **Robustness Check: Alternative Measurements for Asset Return Uncertainty**. The left panel of this figure depicts the baseline measurement of asset return uncertainty (left axis) and an alternative measure of uncertainty (right axis) using the method proposed in Gilchrist, Sim, and Zakrajsek (2014), where uncertainty is measured as the cross-sectional mean of idiosyncratic volatility. The right panel of this figure depicts the baseline measurement of asset return uncertainty (left axis) and an alternative measure of uncertainty (right axis) using option prices information of non-bank financial institutions in the Option Metric database, where uncertainty is measured as the annual average of daily (365-day) implied volatility of at-the-money-forward call options following Alfaro, Bloom, and Lin (2019).

Note that I couldn't observe the daily stock price information for all banks in my sample. Thus this method is subject to sample limitation. Due to the data limitation, I choose to follow the standard approach as in Bloom (2009), Bloom, Floetotto, Jaimovich, Saporta-Eksten, and Terry (2019) and Arellano, Bai, and Kehoe (2018) to construct the uncertainty shock measures in the paper. The other advantage of the standard approach is that it provides a very tight link between the model and the data. The gross asset return rate is assumed to follow an AR(1) process, both in the model and in the data. And asset return uncertainty is measured as the cross-sectional standard deviations of the residual terms, both in the model and in the data.[58] This tight mapping provides much cleaner identification in the parameter estimation.

I also use option prices information of non-bank financial institutions in the Option Metric database and back out the implied uncertainty using Black-Scholes formula. Uncertainty is measured as the annual average of daily (365-day) implied volatility of at-the-money-forward call options following Alfaro, Bloom, and Lin (2019). The pattern of the estimated "perceived" uncertainty are also quite similar to the "realized" uncertainty. The correlation between these two measures is 0.9668. This is unsurprising. As shown in the literature (see Baker, Bloom, and Davis (2016) among others), many

[58] In both the data and the model, I fit the measured gross asset return rate into an AR(1) process. The asset return uncertainty is measured as the cross-sectional standard deviations of the residual terms.

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measures of uncertainty are highly correlated.

B.4 Write-down of Assets vs. Reduction in Asset Origination

In the section, I provide empirical evidence on the new asset issuance (investment) to explore whether the decline in banks' asset size comes from the reduction in the new asset issuance or is it just the case the banks have recognized the losses on their existing assets (asset write-down) during the crisis period.

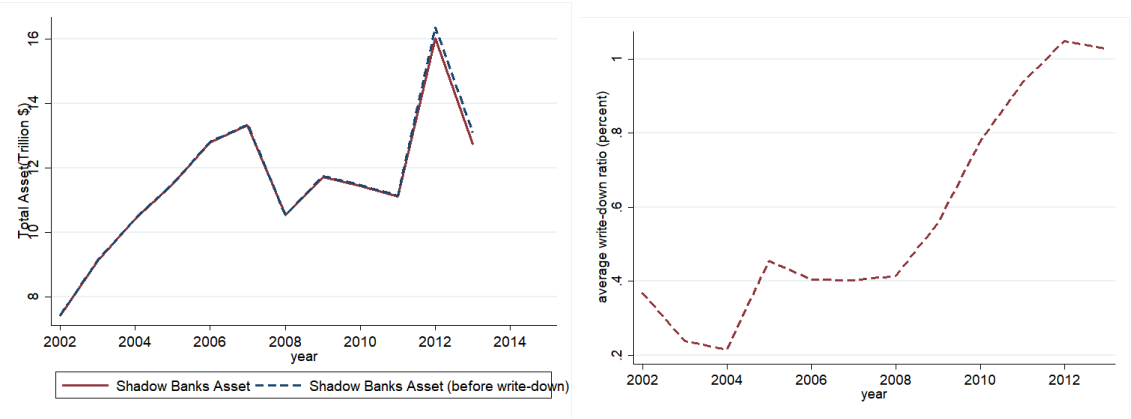


Figure B2: **Robustness Check: Taking Into Account Write-down of Assets**. The left panel illustrates the total asset value of the shadow banking industry before and after write-down. The right panel illustrates the change in write-down ratio of the shadow banking industry over time.

In the data, total assets is the book value of assets after writing down the impaired loan, and securities and other credit impairment charges: $\text{asset}_{t+1} = \text{asset}_t - \text{maturing asset}_t + \text{new asset issuance}_t - \text{writedown}_{t+1}$.[59] Thus, the changes in total assets capture both changes in new asset issuance and asset write-down. I calculate the total asset value before the write-down by adding back the value of impaired loans and securities and other credit impairment charges. The total asset value before the write-down is plotted together with the total asset after the write-down in Figure B2. The left panel of this Figure shows that total asset value before write-down is very close to its value after write-down. The right panel of Figure B2 shows that asset write-down accounts for less than 1% of the total asset value through the whole crisis period. This indicates that the dramatic asset contraction in the 2007-2009 financial crisis was mainly driven by the changes in new asset issuance. Shadow banks start to recognize the losses on their existing assets after the crisis. This is consistent with Laux and Leuz (2010) who find that extant rules have various safeguards and offer substantial discretion to banks, which allows them to avoid marking their asset to distorted market prices. They show that banks use this flexibility during the crisis and find that the claim that fair-value accounting exacerbated the crisis is largely unfounded.

[59] In the data, I could not directly observe what fraction of asset will mature in each period. I also do not have direct bank-level measure for asset issuance. I back out the measure for new asset issuance using data of assets and asset write-down (impaired loan, and securities and other credit impairment charges), assuming in each year λ_a fraction of assets matures: $\text{maturing asset}_t = \lambda_a \cdot \text{asset}_t$, where the annual asset maturing rate $\lambda_a = 1 - (1 - \lambda)^4$. The change in new asset issuance measures the increase or decrease in bank investment.

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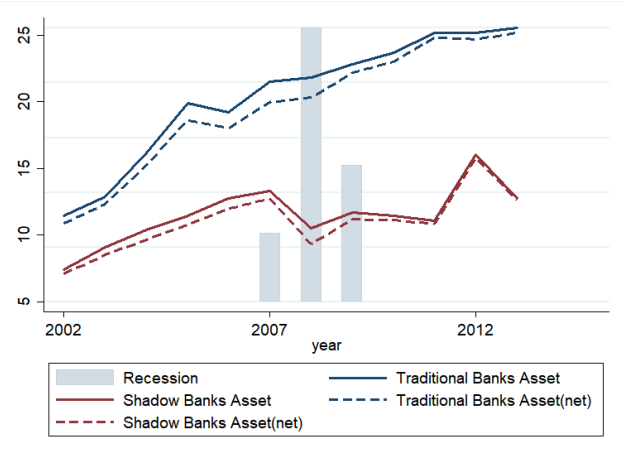


Figure B3: **Robustness Check: Netting Out Cross-holdings of Assets**. This figure illustrates the total asset value of the shadow banking industry after netting out cross-holdings of assets.

Since assets and liabilities of different financial institutions are not netted out during the aggregation, simply adding up the assets and liabilities of different banks might lead to double counting of asset cross positions. To mitigate this concern, I also calculate the total assets of shadow banking sector after netting out cross-holding of assets. The results are plotted in Figure B3. It can be seen that the changes in assets are similar after netting out cross-holdings of assets. From Figure 4 and Figure B3, one can conclude that the assets of the shadow banking sector shrank during the 2007-2009 recession and the total banking credit shrinkage mainly came from the contraction in the shadow banking sector.

B.5 Heterogeneity in Asset Contraction

It is worthwhile to explore if the asset contraction in the shadow banking industry is driven by a particular type of shadow banks. I calculate the total size of different types of shadow banks and calculate their asset growth rates. I find that there is cross-sectional variations in the degree of asset contraction across different types of shadow banks.

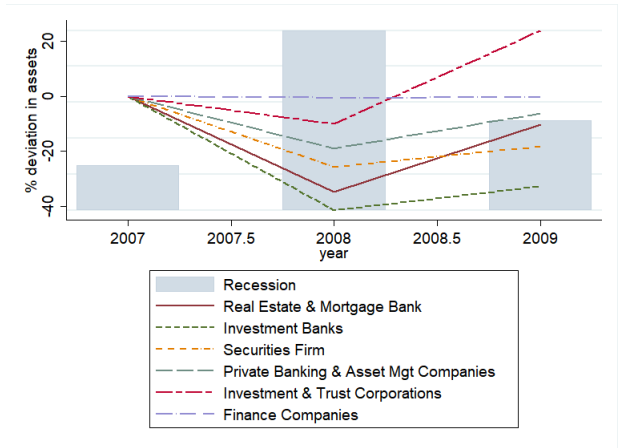


Figure B4: **Heterogeneity in Asset Contraction**. This figure illustrates the percentage deviation in asset sizes of different types of shadow banks during the 2007-2009 financial crisis.

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I normalize the total assets of different types of shadow banks by calculating the percentage deviations from their base values at the year 2007. From Figure B4, we can see that the assets of all types of shadow banks contract but by different proportion. Investment banks and real estate and mortgage banks decrease the most. Finance companies decrease less. But the contraction is not driven by only one type of institution. To check the cross-sectional fitness of the model, I sort the shadow banks based on the measures of fire-sale costs and maturity mismatch and split the full sample into subsamples. I reestimate the model based on these subsamples and report the average deleveraging and asset contraction in each subsample. The results are shown in Table 11. First, the actual data reported in columns 1 and 2 of Table 11 reveals that deleveraging and asset contraction are much stronger in bank subsamples with higher fire-sale costs and higher maturity mismatch. Next, I compare the model-generated moments with those observed in the actual data, and conclude that the model can fit the heterogeneous bank characteristics in each subsample. The over-identification tests fail to reject the baseline model in each subsample. The simulated moments are not statistically significantly different from the actual moments, so the baseline model is valid. Last, relying on the structural model, I compare the contribution of uncertainty shocks to deleveraging and the asset contraction in each subsample. I find that uncertainty shocks can account for a slightly larger portion of deleveraging and asset contraction in banks with higher fire-sale costs and higher maturity mismatch.

B.6 The Conversion of TED spread

The TED spread is calculated as the difference between 3-month LIBOR rate and 3-month T-Bill rate. It is a proxy for bank funding cost. The TED spread plotted in Figure 3 is the annualized rate which is available at daily frequency.

The data for interbank spread is converted from daily frequency to annual frequency in Figure 9. There are at least two ways to carry out this conversion. One way is to simply calculate the arithmetic mean of the TED spread shown in Figure 3 for all days in each year. The other way is as follow: first, convert the annualized daily-frequency LIBOR rate and T-Bill rate to daily rate for each day in each year:

$$r_d^{\text{daily libor}} = (1 + r_d^{\text{annualized libor}})^{1/360} - 1 \tag{34}$$

$$r_d^{\text{daily tbill}} = (1 + r_d^{\text{annualized tbill}})^{1/360} - 1 \tag{35}$$

where $r_d^{\text{daily libor}}$ and $r_d^{\text{daily tbill}}$ are the daily rate of LIBOR and T-bill for each day d, respectively.

Then for each year t, calculate the annual rate for LIBOR and T-bill based on the daily rates. The following equations hold:

$$\prod_{d=1}^{N_t} (1 + r_d^{\text{daily libor}}) = (1 + r_t^{\text{libor}})^{\frac{N_t}{360}} \tag{36}$$

$$\prod_{d=1}^{N_t} (1 + r_d^{\text{daily tbill}}) = (1 + r_t^{\text{tbill}})^{\frac{N_t}{360}} \tag{37}$$

where N_t is the number of daily observations in year t. r_t^{libor} and r_t^{tbill} are the annual rate of LIBOR and T-bill respectively at year t.

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The annual rate of TED spread at year t is calculated as the difference between these two rates:

$$r_t^{\text{TED}} = r_t^{\text{libor}} - r_t^{\text{tbill}} \tag{38}$$

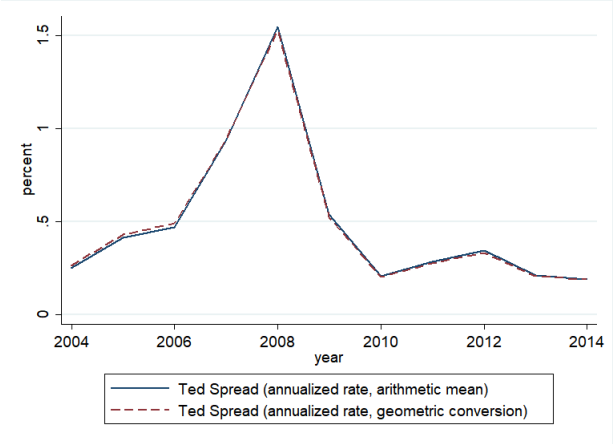


Figure B5: **Annual TED Spread**. This figure illustrates the annual rate of TED spread calculated based on two methods: simple arithmetic mean and geometric conversion.

As shown in Figure B5, both methods generate very similar changes in TED spread. Since interest higher than 3% only happened in very few days of 2008, taking the annual mean of the spread naturally will average out these changes. The simulated quarterly spread also fits the data very well. The spike in the quarterly simulate spread is higher since the rate is averaged at the quarterly frequency. In Figure 9 , I construct all the moments at the annual frequency to facilitate comparisons between the model and the data.

B.7 Alternative Measurement of Banking Funding Cost

I also construct the other proxy for bank funding costs. I first calculate the bank-level interest rate by dividing the total interest expenses of each bank by its total funding. Then I calculate the mean of these funding costs across banks for each year. Since the total interest expenses include the expenses spent on all funding sources, the average interest rate captures the average cost of short-term funding, long-term funding, and customer deposits. Figure B6 shows that the 3-month annualized LIBOR rate captures the average funding cost of the shadow banks quite well. This is well expected since short-term funding accounts for more than 75% of total funding in the shadow banking industry.

B.8 The Illustration of Policy Function Margin and Distribution Margin

Figure B7 illustrates the leverage for different asset quantiles of banks using the bank-level data. Shadow banks of different asset sizes on average choose a lower leverage during the crisis. This captures the policy function margin. Figure B7 also shows that bigger banks tend to have higher leverage; this positive correlation is robust for both types of banks. This result is consistent with Adrian and Shin (2010). As Adrian and Shin (2010) argued, if banks target a leverage ratio, the optimal

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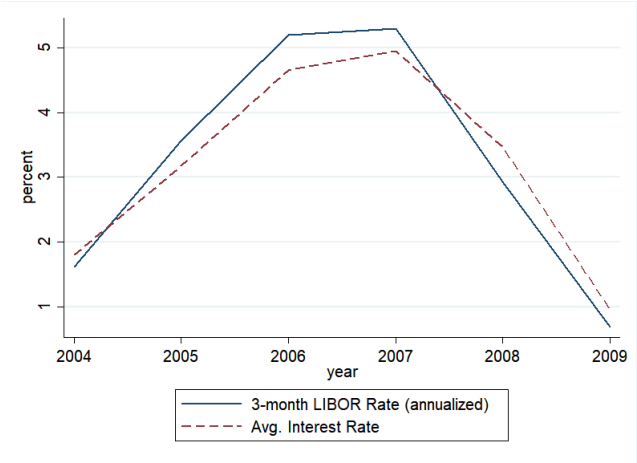


Figure B6: **3-month LIBOR Rate and The Average Funding Cost of the Shadow Banks**. This figure illustrates the 3-month LIBOR rate and the average funding cost of the shadow banking industry. To calculate the average interest rate. I first calculate the bank-level interest rate by dividing the total interest expenses of each bank by its total funding. Then I calculate the average of these funding costs across banks for each year.

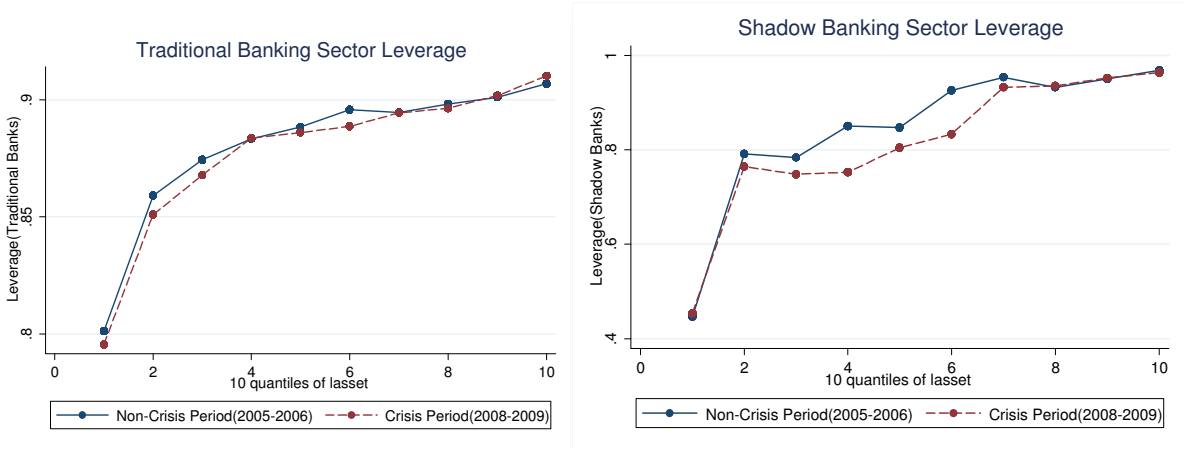


Figure B7: **Leverage Over Asset Quantiles**. This figure depicts the change in leverage ratio of shadow banks and traditional banks over 10 asset quantiles. The leverage ratio is defined as the total debt-to-total asset ratio within each asset quantile. Non-crisis periods are defined as from 2005 to 2006, whereas crisis periods are defined as from 2008 to 2009.

leverage will not increase with asset values. However, if banks target a level of risk exposure, leverage will be positively correlated with assets values. My finding supports the conjecture that banks target a certain level of risk exposure.

Figure B8 depicts the change in bank asset distribution. Shadow banks choose a smaller asset size and the whole asset distribution becomes more skewed toward the lower asset level region. This captures the distribution margin.

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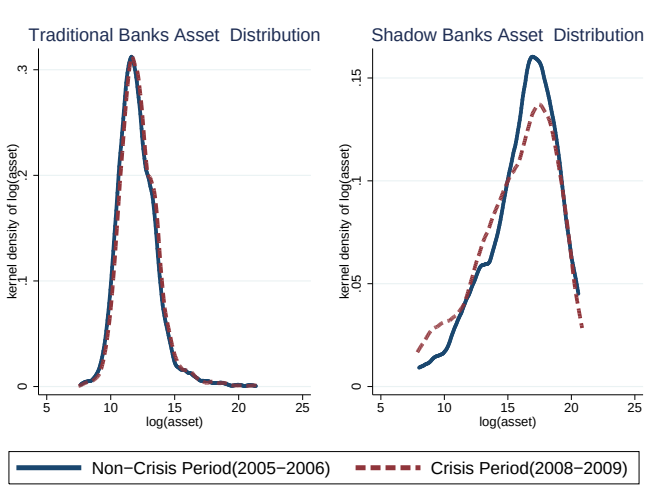


Figure B8: **Asset Distribution**. This figure depicts the change in asset distribution of shadow banks and traditional banks. The measure of small and medium-sized shadow banks increases during the recession whereas the distribution of traditional banks barely changes. Non-crisis periods are defined as from 2005 to 2006, whereas crisis periods are defined as from 2008 to 2009.

B.9 Cross-sectional Variation in Asset Contraction and Deleveraging

I also split the sample into three types based on maturity mismatch and fire-sale cost.[60] As shown in Table D1, there exists a certain degree of nonlinearity in the responses. For the shadow banks with medium maturity mismatch and fire-sale cost, the effect of uncertainty shocks lies in the middle and is slightly smaller than what its value should be if the relationship between the effect of uncertainty shocks and maturity mismatch or fire-sale cost is linear. To be more specific, I first calculate what the responses of medium type should be based on linear interpolation using data of high and low types, and then compare the actual responses with the linearly-predicted responses. For instance, if the relationship between asset contraction and firesale cost is linear, based on the firesale cost index and asset contraction of low and high types, the linear prediction of asset contraction for the medium type should be 19.11%. But its actual value is 16.48%, which is lower than its linearly-predicted value.

[60] Since there is no bank-level information on maturity, I use aggregate data to obtain the maturity mismatch and fire-sale cost estimates of shadow banks (see Appendix B.2 for more details).

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C Theory Appendix

C.1 The Timeline of the Baseline Model

The timing of the baseline model can be summarized in Figure C1. In each period, based on idiosyncratic state variables z, k, b and an aggregate state variable vector $\mathbf{s} = \{\sigma, x\}$, the incumbent bank would choose whether to default, as well as how much to borrow from the interbank market - b' , and how much risky asset to initiate - k'. Potential entrants decide whether to start a new bank based on the aggregate shocks and signals about their productivities.

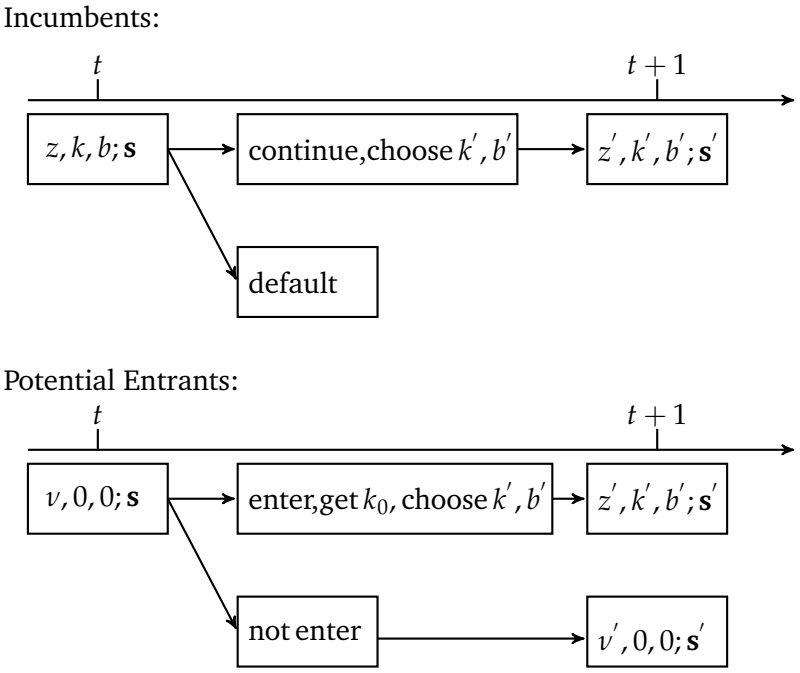


Figure C1: **Timeline from Period t to $t+1$.** This figure illustrates the timeline of the full model. In each period, based on idiosyncratic state variables z, k, b and an aggregate state variable vector $\mathbf{s} = \{\sigma, x\}$, the incumbent bank would choose whether to default, as well as how much to borrow from the interbank market - b', and how much risky assets to initiate - k'. Potential entrants decide whether to start a new bank based on the aggregate shocks and signal ν about their productivities. A potential entrant will start a new bank if the value of entry V^e is higher or equal to entry cost f_e. An entrant starts with start-up asset k_0 and decides the optimal k' and b'.

C.2 The Impact of Uncertainty Shocks in Models without Maturity Mismatch or Fire-sale Costs

To further check whether models without maturity mismatch or fire-sale costs can also generate similar impacts on interbank spreads, bank asset sizes and leverages, I also carry out two counterfactual experiments by re-simulating the baseline model under two alternative specifications: one without maturity mismatch in which I impose the maturity mismatch parameter $\lambda = 1$; one without fire-sale cost in which I impose the fire-sale cost parameter $\zeta = 1$. In both simulation, I fit in the sequence of asset return uncertainty shocks in the baseline model, and the values of other parameters are kept

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the same as in the baseline model, such that the differences in responses can only result from the differences in maturity mismatch or fire-sale costs. The simulations reveal what would have happened in counterfactual economies if there were no maturity mismatch or fire-sale costs. The simulated pathways of spread, asset, and leverage are shown in Figure C2. Uncertainty shocks in the model without maturity mismatch or fire-sale costs explain only around 30% of the change in the interbank spread, leverage, and asset of the U.S. shadow banking sector. This implies that maturity mismatch and firesale cost amplify the impact of uncertainty shocks.[61] I also re-estimate the model under these alternative specifications and carry out the event studies using these re-estimated parameters. The results of the event studies are quantitatively similar.

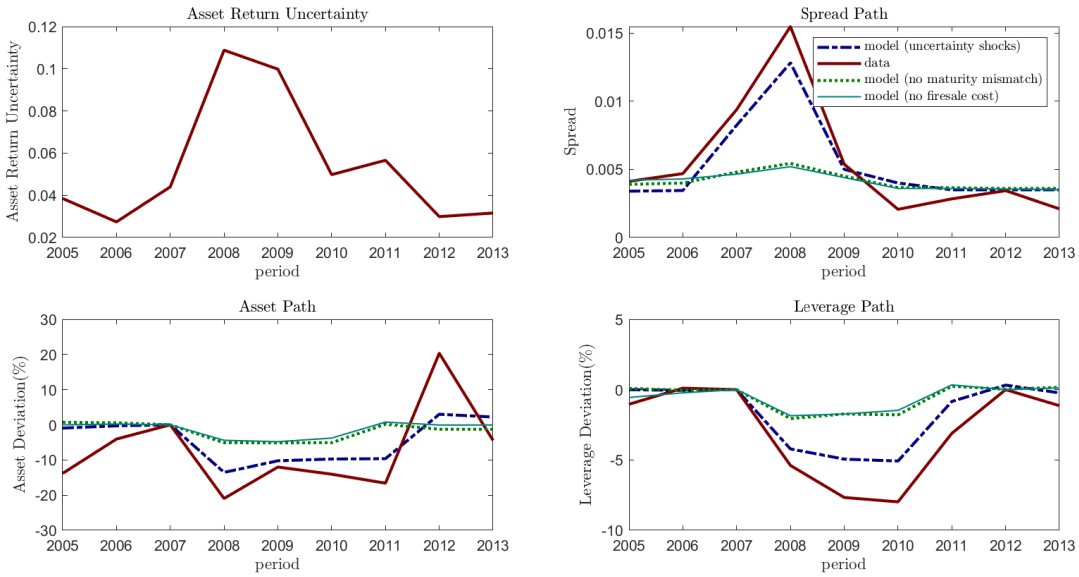


Figure C2: **Event Study: model(no maturity mismatch/no fire-sale cost) vs. data**. The figure depicts the simulated evolution paths of total assets, aggregate leverage ratio and mean interest spread of shadow banks in two alternative models (without maturity mismatch or without fire-sale cost) compared with the baseline model and their data counterparts. Leverage is defined as the total-debt-to-total-asset ratio.

C.3 Spread Schedule Comparison in the Baseline Model

The left panel of Figure C3 shows the spread schedule of risky debt for bank's with a smaller asset size under low and high asset return uncertainty in the baseline model, respectively. The interest spread is higher for banks with asset size that is 80% of average asset size. Asset return uncertainty's impact on spread is still positive. The right panel of Figure C3 shows the spread schedule of risky debt for banks with lower asset return under low and high asset return uncertainty. Interest spread is higher for banks with an asset return that is 80% as large as average asset return. The impact of asset return uncertainty on interest spread is still positive.

[61] In another related quantitative exercise, I shut down maturity mismatch or firesale cost in the model with purely first-moment shocks. Maturity mismatch and firesale cost amplify the impact of first-moment shocks by a similar proportion as they do to the uncertainty shocks.

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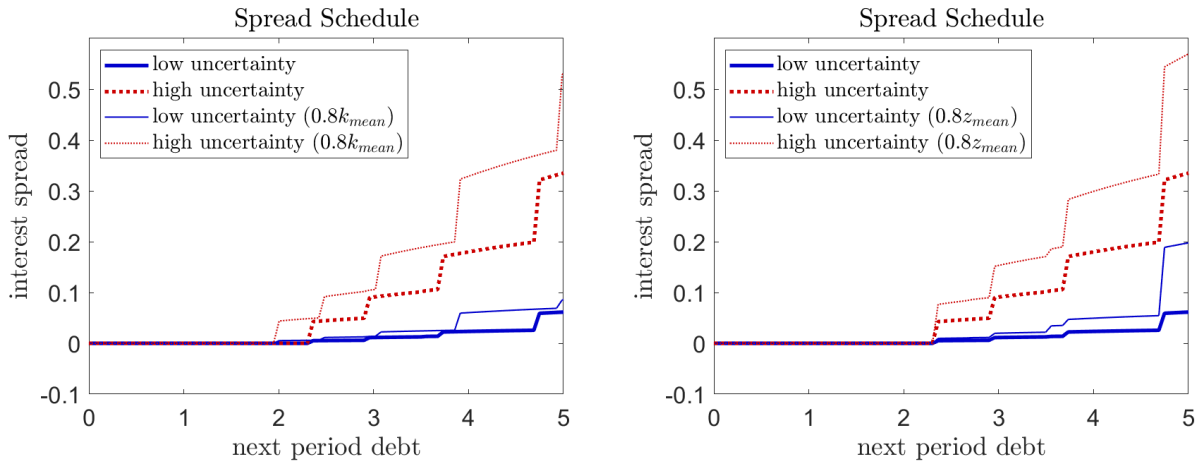


Figure C3: **Spread Schedule Comparison**. The left panel shows the interbank interest spread as a function of the debt issuance for a bank with mean permanent asset return component, stochastic asset return component and 80% of mean asset level. The right panel shows the interbank interest spread as a function of the debt issuance for a bank with mean permanent asset return component, asset level and 80% of mean stochastic shocks.

C.4 Policy Experiment

In the policy experiments, I consider the following three possible interventions:

(a) *TARP*: The government purchases a certain fraction of long term illiquid asset (5% of total assets for banks which are in distress (banks whose cash balance is negative)).

(b) *Debt Guarantee with exogenous bailout probability*: When a bank defaults, the government purchases its debt with a fixed probability $\bar{p}$.

(c) *Debt Guarantee with endogenous bailout probability*: When a bank defaults, the government purchases its debt with probability p, where p is a function of bank size and asset return.

In Table 10, the government interventions are chosen such that the total government resources used in each case are equal. Under a debt guarantee, two bailout probability strategies are chosen such that the expected bailout probabilities are the same in the two cases. Thus we have the following equation:[62]

$$\int \bar{p} b \mathbb{1}\{d_b(z,k,b;\mathbf{s}) = 1\}\mu_b(z,k,b)d(z \times k \times b) = 0.05 * \int \mathbb{1}\{c_a < 0\}k\mu_a(z,k,b)d(z \times k \times b)$$

where $\mu_a(z,k,b)$ and $\mu_b(z,k,b)$ is the stationary distribution of banks under policy intervention a and b, respectively; $d_b(z,k,b;\mathbf{s})$ is the default choice of banks under policy intervention b; c_a is the cash balance of banks under policy intervention a. The value of $\mathbb{1}\{.\}$ equals 1 if the condition in the brackets holds, and equals 0 otherwise. The left side is the amount of resources the government need to spend under debt guarantee with exogenous bailout probability $\bar{p}$. The right side is the amount of

[62] To ease notations, I use a, b, c in the subscript to denote the policy functions and distributions of banks under each scenario.

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resources the government need to spend under TARP. From this equation, I can numerically solve for the exogenous bailout probability $\bar{p}$.

Assume the endogenous bailout probability p is an increasing function of the bank's size and its asset return: $p = \kappa(zx)^{\chi}k^{1-\chi}$. Then we have the following two equations:

$$\int pb\mathbb{1}\{d_c(z,k,b;\mathbf{s}) = 1\}\mu_c(z,k,b)d(z\times k\times b) = 0.05 * \int \mathbb{1}\{c_a < 0\}k\mu_a(z,k,b)d(z\times k\times b) \quad (39)$$

$$\int p\mathbb{1}\{d_c(z,k,b;\mathbf{s}) = 1\}\mu_c(z,k,b)d(z\times k\times b) = \bar{p}\int \mathbb{1}\{d_b(z,k,b;\mathbf{s}) = 1\}\mu_b(z,k,b)d(z\times k\times b) \quad (40)$$

The first equation makes sure the amount of resources spent under TARP equals the amount of resources spend under debt guarantee with endogenous bailout probability p. The second equation ensures the expected bailout probabilities are the same in the two cases under debt guarantee. With these two equations, I can numerically solve for the values of two unknown parameters, κ and χ.

C.5 Additional Subsample Estimation

To test the fitness of the model over time, I also estimate the model on the early subsample of data from 1998 to 2005 and late subsample of data from 2006 to 2013. The estimation results are reported in Table D6 in the Appendix. For each subsample, I compare the model-generated moments with those observed in the actual data, and conclude that the model is able to fit the data well in each subsample. The over-identification tests fail to reject the baseline model in each subsample at the significance level of 5%.[63] The simulated moments are not statistically significantly different from the actual moments, reconfirming the validity of the baseline model.

C.6 Incorporating Endogenous Asset Portfolio Choices

In this section, I further tackle alternative extensions of the first-moment shock model to check if it is possible that alternative extensions of the first-moment shock model could fit the data as well as the baseline model. One possibility is that different banks have different exposure to certain types of risky assets, and when the return to this type of asset changes, this endogenously generates the increased asset return uncertainty. I find that this conjecture is not true.

I extend the baseline model by allowing the shadow banks to choose its asset portfolio endogenously. Assume the bank could choose two types of assets, a risky asset and a risk-free asset. In the spirit of Gamba and Triantis (2008), assume there is a debt adjustment cost such that the bank would hold risk-free assets and debt simultaneously. Specifically, following Gamba and Triantis (2008), I assume there is no direct cost associated with paying down debt. There is, however, a proportional cost on new debt issued. $\Phi(b,b')$ is the debt adjustment cost:

$$\Phi^b(b,b') = \begin{cases} \phi_b\,(b'-b) & \text{if } b' > b \\ 0 & \text{if } b' < b \end{cases} \quad (41)$$

[63] The J-statistics and p-values for the overidentification test are 2.2317 and 0.6932 in the early subsample, 2.1098 and 0.7156 in the late subsample.

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The debt adjustment cost creates an incentive for the bank to hold both debt and risk-free asset simultaneously since it can help increase the bank value. In times of low profitability, the bank needs to reduce net debt to avoid triggering financial distress. There are two ways to reduce net debt: reducing debt or increasing risk-free asset holding. With the debt issuance cost, the bank chooses to increase risk-free asset holding rather than paying down debt. This is because the bank may later wish to restore its net debt to a higher level to take advantage of interest tax shields. It will be better off paying out cash to shareholders at that time rather than issuing new debt and incurring issuance costs.

Denote $\pi \equiv (zx-1)k_1$ as the bank's profit from they risky asset. The shadow bank chooses the optimal asset portfolio and debt issuance to maximize its value:

$$V^c(z,k_1,k_2,b;\mathbf{s}) = \max_{\{b',k_1',k_2'\}} \left\{(1+\gamma_e 1_{\{e<0\}} - \tau_d 1_{\{e>0\}})e + \mathbb{E}_{z,s} m(\mathbf{s},\mathbf{s}')V(z',k_1',k_2',b';\mathbf{s}')\right\} \tag{42}$$

s.t.

$$e = \begin{cases} c + (1-\lambda)k_1 - k_1' - \Phi(k_1',k_1) - k_2' & \text{if } c \geq 0 \\ (1-\lambda)k_1 + \frac{c}{\zeta} - k_1' - \Phi(k_1',k_1) - k_2' & \text{if } c < 0 \end{cases} \tag{43}$$

$$c = (1-\tau_c)\pi + \lambda k_1 + (1+r(1-\tau_c))k_2 - b + qb' + \frac{\tau_c(b'-qb')}{1+r} - \Phi^b(b,b') - \xi \tag{44}$$

For the extended first-moment shock model with endogenous asset portfolio choice, the aggregate state variable $\mathbf{s} = x$. The asset return uncertainty is determined by a fixed parameter $\overline{\sigma}$, which is estimated to match the average asset return uncertainty. The debt adjustment cost parameter is estimated jointly with other parameters in the estimation. I add one more moment- the fraction of risky assets in the estimation, which helps me to pin down the value for the debt adjustment cost parameter. Increasing the value of this parameter will increase the incentive for banks to hold risk-free assets to avoid costly debt adjustment. The estimation results are shown in Table D10. I find that an extended first-moment shock model with asset portfolio choice cannot match the mean default rate, the mean interest spread, and the standard deviation of new asset issuance/asset ratio and interest spread. Thus, in the baseline analysis, I intentionally keep the model as parsimonious as possible while incorporating the crucial features of the shadow banking sector such as fire-sale cost and maturity mismatch. Instead of endogenizing the data-generating process of asset return, I focus on developing a tractable framework to study the impact of time-varying uncertainty and potential policy implications.

C.7 Incorporating Systemic Risk

The other extension concerns systemic risk. I redefine the asset return of each bank as $zx^{1+\theta}$, where θ follows an AR(1) process:

$$log\,(\theta_t) = (1-\rho_\theta)\,log\,(\mu_\theta) + \rho_\theta log\,(\theta_{t-1}) + \varepsilon_t,\ \varepsilon_t \sim N(0,\varphi_\theta^2) \tag{45}$$

The increase of θ_1 captures the increase in intraclass correlation of the bank asset returns. I add the mean, standard deviation, and persistent of systemic risk in the target moments.[64] These extra

[64]System risk is measured as the intraclass correlation of bank return on assets based on rolling window estimation.

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moments help pinning down the parameters governing the AR(1) process- μ_θ, ρ_θ, and φ_θ.

The results are shown in Table D11 in the Appendix. I find the estimate of the persistence is very low, and the variance of the innovation to the systemic risk process is both high and imprecisely estimated. The extended model with only first-moment shocks and systemic risk shocks still significantly underestimates the mean default rate, the mean interest spread, and the standard deviation of new asset issuance/asset ratio and interest spread. Although systemic risk undoubtedly varies over time, I do not find that adding a systemic risk shock to the model takes it as far as adding a second-moment shock.

C.8 Introducing Time Variations to the Volatility of Aggregate Productivity

To further check the robustness of the quantitative results, I introduce time variations to the volatility of aggregate productivity as well to check to what extent the impact of uncertainty shocks will be affected if there are changes in the volatility of aggregate productivity as well. The aggregate productivity x_t follows a Markov process with transition function $\pi_x(x_t|x_{t-1}, \sigma^x_{t-1})$, where σ^x_{t-1} is an aggregate shock to the standard deviation of aggregate state shocks. The aggregate shock σ^x_t follows a Markov process with transition function $\pi_{\sigma^x}(\sigma^x_t|\sigma^x_{t-1})$. Now the aggregate state variable vector $\mathbf{s} = \{\sigma, \sigma^x, x\}$. The aggregate productivity x_t follows an AR(1) process,

$$log\,(x_t) = \rho_x log\,(x_{t-1}) + \sigma^x_t \epsilon^x_t \tag{46}$$

where ϵ^x_t is an i.i.d standard normal variable, the variance of innovations σ^x_{t-1} move over time, generating periods of low and high macro-uncertainty. Following Bloom, Floetotto, Jaimovich, Saporta-Eksten, and Terry (2019), I assume macroeconomic uncertainty σ^x_t follows the same two-state Markov chain as the microeconomic uncertainty σ_t:

$$\sigma^x_t \in \{\sigma^x_L, \sigma^x_H\}\,, \text{ where } Pr(\sigma^x_{t+1} = \sigma^x_j | \sigma^x_t = \sigma^x_k) = \pi^\sigma_{k,j}. \tag{47}$$

I choose the average volatility of the aggregate productivity, $\sigma_x = 0.007$, following Lin and Zhang (2013). Following Bloom (2009), high volatility has twice the value of low volatility, so that σ^x_L and σ^x_H, are chosen to match the average volatility of the aggregate productivity. I reestimate the model and carry out the same event study. The results are overall robust, preserving the dynamics reported in Figure 9. The relative contributions of each shocks remain similar to the baseline estimates.

C.9 Incorporating a Traditional Banking Sector

We observe a much smaller deleveraging in the traditional banking sector. Multiple factors could contribute to this fact, including the differences in leverage, maturity mismatch, fire-sale cost, shock processes, and regulatory oversight. To decompose the contribution of each factor to the smaller deleveraging in the traditional banking sector, I also build and estimate a model incorporating a

$\text{SystemicRisk}_t = \sum_{i\neq j, t\in w_{t-4,t}} corr(\text{ROA}_{i,t}, \text{ROA}_{jt}) \cdot \omega_{ij}$, where $\omega_{ij} = \frac{\text{asset}_{it}+\text{asset}_{jt}}{\text{totalasset}_t}$, $w_{t-4,t}$ is rolling windows of 5 years from period $t-4$ to period t, i and j are any two different shadow banks within the window $w_{t-4,t}$.

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traditional banking sector and carry out a series of counterfactual experiments. The traditional banks issue long-term loans and risk-free assets and borrow using customer deposits, subject to capital requirements.

The traditional banks invest in long-term loans k_1 and a risk-free asset k_2. They finance their investment using customer deposits b. After investment and financing decisions have been made, the bank capital w' can be defined as:

$$w' = k_1' + k_2' - b' \tag{48}$$

Following Begenau and Salomao (2019) and Corbae and D'Erasmo (2018), I assume commercial banks are subject to Basel-III-type capital requirements. The Basel III accords stipulate that banks must back a specific percentage of risk-weighted assets with capital. Safe assets and risky assets have different risk weights. The capital requirement can be written as:

$$w' \geq \kappa(\omega_1 k_1' + \omega_2 k_2') \tag{49}$$

where κ is capital requirement parameter, ω_1 and ω_2 are the risk weights of risky assets and risk-free assets, w is the bank capital.

Following Corbae and D'Erasmo (2018), I set $\omega_1 = 1$, $\omega_2 = 0$, and $\kappa = 0.04$. These values are associated with regulation in place before the recent financial crisis. Given $\omega_1 > \omega_2$, holding risk-free assets helps the traditional banks relax the capital requirement constraint, but it might reduce the bank profitability and solvency. This creates an precautionary motive for the banks to hold the risk-free assets. In the equilibrium, the bank might choose a capital equity ratio higher than the required level.

The bank's problem is written recursively. Rewriting the capital requirement constraint as the following collateral constraint is helpful in stating the problem:

$$b' \leq (1-\kappa)k_1' + k_2' \tag{50}$$

The traditional bank's problem can be written as follows:

$$V(z, k_1, k_2, b; \mathbf{s}) = \max_{\{b', k_1', k_2'\}} \left\{(1 + \gamma_e 1_{\{e<0\}} - \tau_d 1_{\{e>0\}})e + \mathbb{E}_{z,s} m(\mathbf{s}, \mathbf{s}') V(z', k_1', k_2', b'; \mathbf{s}')\right\} \tag{51}$$

s.t.

$$e = \begin{cases} c + (1-\lambda)k_1 - k_1' - \Phi(k_1', k_1) - k_2' & \text{if } c \geq 0 \\ (1-\lambda)k_1 + \frac{c}{\xi} - k_1' - \Phi(k_1', k_1) - k_2' & \text{if } c < 0 \end{cases} \tag{52}$$

$$c = (1-\tau_c)\pi + \lambda k_1 + (1 + r(1-\tau_c))k_2 - b(1+r_d) + b' + \tau_c r_d b - \xi \tag{53}$$

$$b' \leq (1-\kappa)k_1' + k_2' \tag{54}$$

where r_d is the deposit rate. In the calibration, its value is set to 0.659% following Corbae and D'Erasmo (2018). $\pi \equiv (zx - 1)k_1$ is the bank's profit from the risky assets.

The empirical measurement for maturity mismatch and fire-sale cost reveals that traditional banks

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have much lower maturity mismatch and fire-sale cost. I set the maturity mismatch parameter to $\lambda = 0.2301$ based on the maturity mismatch index. The estimation reveals that asset return uncertainty in this sector is much smaller than that of the shadow banking sector. Also, due to the reserve requirements and lower asset return, this sector has much lower leverage. To estimate the contribution of each factor, I first feed in the uncertainty and asset return shocks processes of the shadow banking sector to the traditional banking sector, this generates a deleveraging that is only 21% as large as the deleveraging in the shadow banking sector. Then I impose the leverage ratio of the traditional banking sector to be the same as that of the shadow banking sector prior to the arrival of the shocks in the simulation by relaxing the reserve requirement. This generates a deleveraging that is only 32% as large as the deleveraging in the shadow banking sector. Then I further change the maturity mismatch and fire-sale cost parameters in the traditional banking sector to their values in the shadow banking sector. The traditional banks' deleveraging increases to 67% of its value in the shadow banking sector. The remaining 33% of difference in deleveraging can be attributed to the difference in the financing methods in these two sectors.

C.10 Additional Robustness Checks for the Impact of Financial Shocks and Firesale Cost Shocks

To further check the robustness of the results regarding the impact of financial shocks, I also test the robustness of the results under two alternative specifications. In the first alternative specification, I assume the financial shocks follow the same transition matrix as the business cycle (Cooley and Prescott (1995)); In the second alternative specification, I assume it follows the same transition matrix of the uncertainty shocks. The magnitudes of the quantitative results generated using these alternative specifications are slightly larger than the baseline since the shock processes are more persistent. In these two alternative specifications, the relative contributions of each component remain stable. Uncertainty shocks remain as the most important factor. To further check the robustness of the model, I also estimate a Pseudo General Equilibrium (Pseudo-GE) model in the spirit of Bloom (2009). I estimate the actual changes in interest rates that arise after an asset return uncertainty shock and directly feed them into the model with risk-neutral investors in an expectations consistent way. The relative contributions of each components in this alternative setting are quantitatively similar to the baseline extended model.

To further check the robustness of the results regarding the impact of firesale cost shocks, I also test the robustness of the results under three alternative specifications. In the first alternative specification, I assume the fire-sale cost follows the same transition matrix as the business cycle (Cooley and Prescott (1995)); In the second alternative specification, I assume it follows the same transition matrix of Gilchrist, Sim, and Zakrajsek (2014)'s capital resale shocks; In the third alternative specification, I assume it follows the same transition matrix of the uncertainty shocks. The magnitude of the quantitative results generated using these alternative specifications are slightly larger than the baseline. In all these alternative specifications, the relative contributions of each component remain stable. Uncertainty shocks remain as the most important factor. For instance, if I use the transition matrix of

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business cycle, fire-sale cost shocks and uncertainty shocks together can explain around 81% of asset contraction and 80% of deleveraging, whereas fire-sale cost shocks and first-moment shocks together can at most explain 32% of asset contraction and 30% of deleveraging. To further check the robustness of the model, I also estimate a Pseudo General Equilibrium (Pseudo-GE) model in the spirit of Bloom (2009). I estimate the actual changes in interest rates that arise after an asset return uncertainty shock and feed them into the model with risk-neutral investors in an expectations consistent way. Again, the relative contributions of each component remain stable in this alternative setting.

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D Additional Tables

Table D1: Cross-sectional Variation in Asset Contraction and Deleveraging

A: Split by asset fire-sale cost	FCI	deleveraging		asset contraction	
		actual	linear prediction	actual	linear prediction
high fire-sale cost	0.40	11.36%	-	34.46%	-
medium fire-sale cost	0.18	5.50%	6.57%	16.48%	19.11%
low fire-sale cost	0.12	5.25%	-	14.88%	-
B: Split by maturity mismatch	MMI	deleveraging		asset contraction	
		actual	linear prediction	actual	linear prediction
high maturity mismatch	11.72	9.01%	-	28.89%	-
medium maturity mismatch	9.67	7.02%	7.10%	21.53%	22.10%
low maturity mismatch	4.58	2.36%	-	5.24%	-

Notes: This table reports the cross-sectional variations of asset contraction and deleveraging in the shadow banking sector. I sort the shadow banks based on the measures of fire-sale costs or maturity mismatch and split the full sample into three subsamples. FCI is firesale cost index. MMI is maturity mismatch index. See Appendix B.2 for more details about measurement construction. For shadow banks with medium maturity mismatch and firesale cost, I report both their actual responses and the responses calculated based on linear interpolation using data of high and low types.

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Table D2: Moments Across Alternative Model Specifications

Target Moments	Actual Moments	Baseline Model	First-moment Shocks	No Fire-sale Cost: $\zeta = 1$	No Maturity Mismatch: $\lambda = 1$	No Adjustment Cost: $\phi = 0$
mean of leverage	0.932	0.930	0.974	0.998	0.999	0.989
mean of new asset issuance/asset ratio	0.437	0.432	0.427	0.428	1.111	0.506
stdv. of new asset issuance/asset ratio	0.152	0.143	0.050	0.058	0.062	0.211
autocorrelation of new asset issuance/asset ratio	0.876	0.865	0.926	0.858	0.823	0.807
mean default rate	1.10%	1.09%	0.19%	0.29%	0.47%	0.89%
mean of dividend/asset ratio	0.021	0.026	0.059	0.044	0.063	0.035
stdv. of dividend/asset ratio	0.015	0.017	0.005	0.006	0.005	0.026
autocorrelation of dividend/asset ratio	0.795	0.725	0.864	0.781	0.776	0.692
mean equity issuance/asset ratio	0.012	0.012	0.007	0.005	0.005	0.006
leverage-asset slope	0.043	0.048	0.049	0.050	0.049	0.054
mean entrant growth	0.242	0.216	0.291	0.296	0.290	0.299
entrant relative size	0.201	0.211	0.202	0.192	0.203	0.204
mean of interest spread	0.55%	0.53%	0.09%	0.14%	0.23%	0.45%
stdv. of interest spread	0.44%	0.41%	0.06%	0.11%	0.17%	0.37%
mean of asset return uncertainty	0.064	0.063	0.063	0.063	0.062	0.062
stdv. of asset return uncertainty	0.045	0.045	-	0.044	0.045	0.044
autocorrelation of asset return uncertainty	0.732	0.711	-	0.717	0.709	0.708
skewness of asset return uncertainty	0.931	0.984	-	1.109	0.998	0.989
J-statistics		2.5034	11.5786	17.1414	32.1257	9.4896
p-value		0.6440	0.0208	0.0042	<1e-5	0.0911

Notes: Calculations are based on an annual sample of shadow banks from the Bankscope and Worldscope databases. The estimation is done with SMM, which chooses model parameters by matching the moments from a simulated panel of banks to the corresponding moments from the data. J-statistics tests the over-identification constraint for the moment conditions.

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Table D3: Assessing The Impact of Uncertainty Shocks, First-moment Shocks, Financial Shocks, and Fire-sale Cost Shocks

	With Financial Shocks				With Fire-sale Shocks				With both Financial and Fire-sale Shocks			
	1	2	3	4	5	6	7	8	9	10	11	12
Uncertainty Shocks	Yes	Yes	No	No	Yes	Yes	No	No	Yes	Yes	No	No
First-moment Shocks	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No
Fire-sale Cost	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Maturity Mismatch	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Financial Shocks	Yes	Yes	Yes	Yes	No	No	No	No	Yes	Yes	Yes	Yes
Fire-sale Shocks	No	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Asset Contraction	92.54%	75.78%	25.42%	11.30%	95.88%	78.20%	27.35%	13.59%	98.15%	85.99%	35.89%	24.04%
Deleveraging	92.82%	76.57%	25.69%	11.56%	95.16%	77.41%	27.09%	13.37%	97.68%	85.52%	35.25%	23.71%
Δ Interest Spread	93.49%	77.78%	27.09%	12.24%	96.72%	79.47%	29.15%	14.58%	99.04%	86.08%	36.91%	25.39%

Notes: This table reports to what extent each extended model specification could explain the asset contraction, deleveraging, and changes in interest spread. The results for the extended model with time-varying financial shocks are shown in columns 1-4. The results for the extended model with time-varying fire-sale cost shocks are shown in columns 5-8. The results for the extended model with both time-varying financial shocks and time-varying fire-sale cost shocks are shown in columns 9-12. To study in a comprehensive manner the impact of alternative shocks that could have explained the contraction of credit in the shadow banking sector, for each extended model, using the estimated parameters, I gauge its explanatory power under four specifications: with both uncertainty and first-moment shocks, with only uncertainty shocks, with only first-moment shocks, and without uncertainty shocks or first-moment shocks. Column 1 reports the joint contribution of uncertainty shocks, first-moment shocks, and financial shocks. Column 2 reports the joint contribution of uncertainty shocks and financial shocks. Column 3 reports the joint contribution of first-moment shocks and financial shocks. Column 4 reports the pure contribution of financial shocks. Column 5 reports the joint contribution of uncertainty shocks, first-moment shocks, and fire-sale cost shocks. Column 6 reports the joint contribution of uncertainty shocks and fire-sale cost shocks. Column 7 reports the joint contribution of first-moment shocks and fire-sale shocks. Column 8 reports the pure contribution of fire-sale shocks. Column 9 reports the joint contribution of uncertainty shocks, first-moment shocks, financial shocks, and fire-sale cost shocks. Column 10 reports the joint contribution of uncertainty shocks, financial shocks, and fire-sale cost shocks. Column 11 reports the joint contribution of first-moment shocks, financial shocks, and fire-sale shocks. Column 12 reports the joint contribution of financial shocks and fire-sale shocks.

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Table D4: Summary Statistics

Variable	Mean	SD	p25	p50	p75
size (log(asset))	15.530	2.545	13.822	16.198	17.883
leverage	0.932	0.091	0.894	0.958	0.975
new asset issuance/asset ratio	0.437	0.152	0.293	0.404	0.543
equity issuance/asset ratio	0.012	0.019	0.001	0.008	0.018
dividend/asset ratio	0.021	0.015	0.016	0.020	0.031
short term funding fraction	0.744	0.112	0.689	0.736	0.806
fraction of risky asset	0.610	0.163	0.533	0.585	0.655
customer deposit funding fraction	0.043	0.025	0.024	0.030	0.063

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Table D5: Parameter Estimates for Alternative Model Specifications

First-Moment Shock Model

γ_e	ζ	ξ	k_0	ϕ	ρ_v	$\overline{\sigma}$	μ	A	f_e	ω
0.0608	0.6857	0.0066	0.0028	0.1580	0.9185	0.0318	3.3120	17.5217	0.0038	3.2155
(0.0091)	(0.0398)	(0.0012)	(0.0006)	(0.0125)	(0.0340)	(0.0127)	(0.3197)	(2.3539)	(0.0010)	(0.3120)

No Fire-sale Cost Model

γ_e	ξ	k_0	ϕ	ρ_v	$\pi^{\sigma}_{H,H}$	$\pi^{\sigma}_{L,L}$	σ_H	σ_L	μ	A	f_e	ω
0.0619	0.0074	0.0032	0.1554	0.9311	0.9442	0.9787	0.0676	0.0165	3.2809	17.1223	0.0035	3.1548
(0.0121)	(0.0016)	(0.0004)	(0.0124)	(0.0352)	(0.0121)	(0.0177)	(0.0104)	(0.0077)	(0.2701)	(1.9974)	(0.0008)	(0.2986)

No Maturity Mismatch Model

γ_e	ζ	ξ	k_0	ϕ	ρ_v	$\pi^{\sigma}_{H,H}$	$\pi^{\sigma}_{L,L}$	σ_H	σ_L	μ	A	f_e	ω
0.0621	0.6071	0.0071	0.0034	0.1325	0.9342	0.9456	0.9768	0.0672	0.0167	3.2401	17.4424	0.0034	3.1754
(0.0120)	(0.1221)	(0.0009)	(0.0008)	(0.0104)	(0.0360)	(0.0220)	(0.0258)	(0.0108)	(0.0062)	(0.4129)	(1.9678)	(0.0007)	(0.3356)

No Asset Adjustment Cost Model

γ_e	ζ	ξ	k_0	ρ_v	$\pi^{\sigma}_{H,H}$	$\pi^{\sigma}_{L,L}$	σ_H	σ_L	μ	A	f_e	ω
0.0609	0.6905	0.0069	0.0032	0.9379	0.9460	0.9763	0.0670	0.0166	3.2631	17.4097	0.0036	3.1685
(0.0099)	(0.1466)	(0.0008)	(0.0007)	(0.0371)	(0.0221)	(0.0257)	(0.0101)	(0.0062)	(0.4170)	(1.9432)	(0.0008)	(0.3208)

Notes: The estimation is done with SMM, which chooses model parameters by matching the moments from a simulated panel of banks to the corresponding moments from the data. This table reports the estimated parameters, with clustered standard deviation in the parentheses.

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Table D6: Parameter Estimates for Subsamples

High Maturity Mismatch Subsample

γ_e	ζ	ξ	k_0	ϕ	ρ_v	$\pi^{\sigma}_{H,H}$	$\pi^{\sigma}_{L,L}$	σ_H	σ_L	μ	A	f_e	ω
0.0619	0.6389	0.0069	0.0036	0.1912	0.9307	0.9458	0.9789	0.0673	0.0169	3.3485	17.2908	0.0035	2.8570
(0.0118)	(0.0395)	(0.0007)	(0.0008)	(0.0170)	(0.0361)	(0.0252)	(0.0275)	(0.0110)	(0.0018)	(0.6153)	(1.9106)	(0.0008)	(0.2965)

Low Maturity Mismatch Subsample

γ_e	ζ	ξ	k_0	ϕ	ρ_v	$\pi^{\sigma}_{H,H}$	$\pi^{\sigma}_{L,L}$	σ_H	σ_L	μ	A	f_e	ω
0.0609	0.8251	0.0070	0.0035	0.1829	0.9289	0.9447	0.9768	0.0666	0.0163	3.2450	17.5009	0.0037	3.0925
(0.0109)	(0.0595)	(0.0011)	(0.0008)	(0.0151)	(0.0340)	(0.0210)	(0.0258)	(0.0108)	(0.0011)	(0.6066)	(1.9760)	(0.0009)	(0.3309)

High Fire-sale Cost Subsample

γ_e	ζ	ξ	k_0	ϕ	ρ_v	$\pi^{\sigma}_{H,H}$	$\pi^{\sigma}_{L,L}$	σ_H	σ_L	μ	A	f_e	ω
0.0621	0.5901	0.0068	0.0035	0.1926	0.9306	0.9459	0.9771	0.0675	0.0170	3.3501	17.2551	0.0032	2.8752
(0.0120)	(0.0385)	(0.0008)	(0.0009)	(0.0186)	(0.0358)	(0.0247)	(0.0260)	(0.0108)	(0.0015)	(0.6203)	(1.9325)	(0.0006)	(0.2958)

Low Fire-sale Cost Subsample

γ_e	ζ	ξ	k_0	ϕ	ρ_v	$\pi^{\sigma}_{H,H}$	$\pi^{\sigma}_{L,L}$	σ_H	σ_L	μ	A	f_e	ω
0.0610	0.8052	0.0072	0.0034	0.1822	0.9301	0.9445	0.9765	0.0671	0.0166	3.2128	17.4638	0.0034	3.0501
(0.0101)	(0.0554)	(0.0011)	(0.0008)	(0.0128)	(0.0345)	(0.0208)	(0.0255)	(0.0108)	(0.0010)	(0.6081)	(1.9721)	(0.0007)	(0.3259)

Early Subsample

γ_e	ζ	ξ	k_0	ϕ	ρ_v	$\pi^{\sigma}_{H,H}$	$\pi^{\sigma}_{L,L}$	σ_H	σ_L	μ	A	f_e	ω
0.0615	0.7107	0.0081	0.0033	0.1889	0.9308	0.9456	0.9769	0.0681	0.0170	3.3320	17.4421	0.0035	3.2361
(0.0118)	(0.0423)	(0.0008)	(0.0007)	(0.0179)	(0.0348)	(0.0221)	(0.0259)	(0.0112)	(0.0021)	(0.6215)	(1.9586)	(0.0009)	(0.3581)

Late Subsample

γ_e	ζ	ξ	k_0	ϕ	ρ_v	$\pi^{\sigma}_{H,H}$	$\pi^{\sigma}_{L,L}$	σ_H	σ_L	μ	A	f_e	ω
0.0624	0.6971	0.0045	0.0035	0.1878	0.9311	0.9447	0.9780	0.0670	0.0165	3.2125	17.4435	0.0033	2.8969
(0.0121)	(0.0417)	(0.0009)	(0.0008)	(0.0172)	(0.0356)	(0.0218)	(0.0257)	(0.0097)	(0.0019)	(0.6124)	(1.9689)	(0.0007)	(0.3265)

Notes: The estimation is done with SMM, which chooses model parameters by matching the moments from a simulated panel of banks to the corresponding moments from the data. This table reports the estimated parameters, with clustered standard deviation in the parentheses.

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Table D7: Simulated Moments Estimation: Extended Model with Financial Shocks

A. Moments

Target Moments	Data	Model	t-statistics
mean of leverage	0.932	0.930	0.119
mean of new asset issuance/asset ratio	0.437	0.441	-0.142
stdv. of new asset issuance/asset ratio	0.152	0.152	-0.133
autocorrelation of new asset issuance/asset ratio	0.876	0.862	0.243
mean default rate	1.10%	1.12%	-0.130
mean of dividend/asset ratio	0.021	0.022	-0.052
stdv. of dividend/asset ratio	0.015	0.017	-0.367
autocorrelation of dividend/asset ratio	0.795	0.732	0.745
mean of asset return uncertainty	0.064	0.064	-0.003
stdv. of asset return uncertainty	0.045	0.045	-0.015
autocorrelation of asset return uncertainty	0.732	0.717	0.219
skewness of asset return uncertainty	0.931	1.012	-0.552
mean of equity issuance/asset ratio	0.012	0.012	0.113
stdv. of equity issuance/asset ratio	0.019	0.014	1.014
covariance (equity issuance, new asset issuance)	0.0008	0.0009	-0.158
leverage-asset slope	0.043	0.048	-0.324
mean entrant leverage	0.972	0.979	-0.140
mean entrant growth	0.242	0.210	0.403
entrant relative size	0.201	0.192	0.133
mean of interest spread	0.55%	0.56%	-0.114
stdv. of interest spread	0.44%	0.43%	0.096

B. Parameter estimates

ρ_γ	σ_γ	μ_γ	ζ	ξ	k_0	ϕ	ρ_v	$\pi^{\sigma}_{H,H}$	$\pi^{\sigma}_{L,L}$	σ_H	σ_L	μ	A	f_e	ω
0.686	0.281	0.061	0.713	0.006	0.003	0.161	0.915	0.944	0.978	0.067	0.017	3.254	17.25	0.003	2.931
(0.074)	(0.037)	(0.018)	(0.041)	(0.001)	(0.001)	(0.019)	(0.035)	(0.025)	(0.019)	(0.002)	(0.001)	(0.600)	(1.896)	(0.001)	(0.385)

Notes: Calculations are based an annual sample of shadow banks from the Bankscope and Worldscope databases. The estimation is done with SMM, which chooses model parameters by matching the moments from a simulated panel of banks to the corresponding moments from the data. Panel A reports the simulated and actual moments and the clustered t-statistics for the differences between the corresponding moments. Panel B reports the estimated parameters, with clustered standard deviation in the parentheses.

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Table D8: Simulated Moments Estimation: Extended Model with Fire-sale Cost Shocks

A. Moments

Target Moments	Data	Model	t-statistics
mean of leverage	0.932	0.931	0.054
stdv. of leverage	0.091	0.085	0.341
covariance (leverage, new asset issuance)	-0.0021	-0.0029	0.870
mean of new asset issuance/asset ratio	0.437	0.430	0.175
stdv. of new asset issuance/asset ratio	0.152	0.145	0.136
autocorrelation of new asset issuance/asset ratio	0.876	0.863	0.212
mean default rate	1.10%	1.08%	0.123
mean of dividend/asset ratio	0.021	0.020	0.112
stdv. of dividend/asset ratio	0.015	0.016	-0.206
autocorrelation of dividend/asset ratio	0.795	0.735	0.691
mean of asset return uncertainty	0.064	0.064	-0.011
stdv. of asset return uncertainty	0.045	0.045	-0.015
autocorrelation of asset return uncertainty	0.732	0.713	0.335
skewness of asset return uncertainty	0.931	1.001	-0.596
mean of equity issuance/asset ratio	0.012	0.012	0.112
leverage-asset slope	0.043	0.047	-0.299
mean entrant growth	0.242	0.220	0.288
entrant relative size	0.201	0.222	-0.211
mean of interest spread	0.55%	0.54%	0.110
stdv. of interest spread	0.44%	0.42%	0.157

B. Parameter estimates

ρ_ζ	σ_ζ	μ_ζ	γ_e	ξ	k_0	ϕ	ρ_v	$\pi^{\sigma}_{H,H}$	$\pi^{\sigma}_{L,L}$	σ_H	σ_L	μ	A	f_e	ω
0.891	0.182	0.701	0.060	0.005	0.003	0.150	0.921	0.947	0.979	0.0676	0.0167	3.052	17.60	0.003	2.945
(0.072)	(0.035)	(0.018)	(0.009)	(0.001)	(0.001)	(0.012)	(0.013)	(0.023)	(0.019)	(0.010)	(0.005)	(0.536)	(1.637)	(0.001)	(0.324)

Notes: Calculations are based an annual sample of shadow banks from the Bankscope and Worldscope databases. The estimation is done with SMM, which chooses model parameters by matching the moments from a simulated panel of banks to the corresponding moments from the data. Panel A reports the simulated and actual moments and the clustered t-statistics for the differences between the corresponding moments. Panel B reports the estimated parameters, with clustered standard deviation in the parentheses.

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Table D9: Simulated Moments Estimation: Extended Model with both Financial Shocks and Fire-sale Cost Shocks

A. Moments

Target Moments	Data	Model	t-statistics
mean of leverage	0.932	0.930	0.104
stdv. of leverage	0.091	0.0925	-0.366
covariance (leverage, new asset issuance)	-0.0021	-0.0025	0.658
mean of new asset issuance/asset ratio	0.437	0.441	-0.478
stdv. of new asset issuance/asset ratio	0.152	0.153	-0.232
autocorrelation of new asset issuance/asset ratio	0.876	0.861	0.286
mean default rate	1.10%	1.11%	-0.087
mean of dividend/asset ratio	0.021	0.024	-0.423
stdv. of dividend/asset ratio	0.015	0.016	-0.300
autocorrelation of dividend/asset ratio	0.795	0.743	0.620
mean of asset return uncertainty	0.064	0.064	0.009
stdv. of asset return uncertainty	0.045	0.045	-0.016
autocorrelation of asset return uncertainty	0.732	0.716	0.335
skewness of asset return uncertainty	0.931	0.982	-0.344
mean of equity issuance/asset ratio	0.012	0.012	0.112
stdv. of equity issuance/asset ratio	0.019	0.015	0.204
covariance (equity issuance, new asset issuance)	0.0008	0.0007	0.137
leverage-asset slope	0.043	0.040	0.211
mean entrant growth	0.242	0.221	0.257
entrant relative size	0.201	0.215	-0.139
mean of interest spread	0.55%	0.56%	-0.154
stdv. of interest spread	0.44%	0.42%	0.157

B. Parameter estimates

ρ_γ	σ_γ	μ_γ	ρ_ζ	σ_ζ	μ_ζ	ξ	k_0	ϕ	ρ_v	$\pi^\sigma_{H,H}$	$\pi^\sigma_{L,L}$	σ_H	σ_L	μ	A	f_e	ω
0.682	0.285	0.060	0.889	0.178	0.705	0.005	0.002	0.201	0.910	0.942	0.979	0.067	0.017	3.014	17.68	0.003	2.943
(0.068)	(0.036)	(0.013)	(0.078)	(0.034)	(0.023)	(0.001)	(0.001)	(0.019)	(0.024)	(0.031)	(0.050)	(0.019)	(0.009)	(0.635)	(1.864)	(0.001)	(0.297)

Notes: Calculations are based an annual sample of shadow banks from the Bankscope and Worldscope databases. The estimation is done with SMM, which chooses model parameters by matching the moments from a simulated panel of banks to the corresponding moments from the data. Panel A reports the simulated and actual moments and the clustered t-statistics for the differences between the corresponding moments. Panel B reports the estimated parameters, with clustered standard deviation in the parentheses.

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Table D10: Simulated Moments Estimation: Extended First-moment Shock Model with Asset Portfolio Choices

A. Moments

Target Moments	Data	Model	t-statistics
mean of leverage	0.932	0.943	-0.848
mean of new asset issuance/asset ratio	0.437	0.442	-0.152
stdv. of new asset issuance/asset ratio	0.152	0.082	1.593
autocorrelation of new asset issuance/asset ratio	0.876	0.863	0.212
mean default rate	1.10%	0.35%	3.016
mean of dividend/asset ratio	0.021	0.036	-1.265
stdv. of dividend/asset ratio	0.015	0.004	1.669
autocorrelation of dividend/asset ratio	0.795	0.851	-1.125
mean of asset return uncertainty	0.064	0.064	0.005
mean of equity issuance/asset ratio	0.012	0.009	0.126
fraction of risky asset	0.610	0.601	0.258
leverage-asset slope	0.043	0.049	-0.355
mean entrant growth	0.242	0.261	-0.290
entrant relative size	0.201	0.209	-0.091
mean of interest spread	0.55%	0.18%	3.121
stdv. of interest spread	0.44%	0.14%	2.869

B. Parameter estimates

γ_e	ζ	ξ	k_0	ϕ	ρ_v	$\overline{\sigma}$	μ	A	f_e	ϕ_b	ω
0.0660	0.6714	0.0067	0.0034	0.1425	0.9231	0.0318	3.2147	17.6814	0.0035	0.0033	2.9515
(0.0098)	(0.0402)	(0.0012)	(0.0012)	(0.0115)	(0.0304)	(0.0126)	(0.4222)	(2.3318)	(0.0009)	(0.0008)	(0.3252)

Notes: Calculations are based an annual sample of shadow banks from the Bankscope and Worldscope databases. The estimation is done with SMM, which chooses model parameters by matching the moments from a simulated panel of banks to the corresponding moments from the data. Panel A reports the simulated and actual moments and the clustered t-statistics for the differences between the corresponding moments. Panel B reports the estimated parameters, with clustered standard deviation in the parentheses. ϕ_b is the debt adjustment cost parameter.

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Table D11: Simulated Moments Estimation: Extended First-Moment Shock Model with Systemic Risk Shocks

A. Moments

Target Moments	Data	Model	t-statistics
mean of leverage	0.932	0.951	-1.325
mean of new asset issuance/asset ratio	0.437	0.447	-0.170
stdv. of new asset issuance/asset ratio	0.152	0.090	1.504
autocorrelation of new asset issuance/asset ratio	0.876	0.852	0.322
mean default rate	1.10%	0.31%	3.131
mean of dividend/asset ratio	0.021	0.040	-1.466
stdv. of dividend/asset ratio	0.015	0.005	1.325
autocorrelation of dividend/asset ratio	0.795	0.856	-1.297
mean of asset return uncertainty	0.064	0.064	-0.010
mean of systemic risk	0.0015	0.0011	0.169
stdv. of systemic risk	0.695	0.636	0.870
autocorrelation. of systemic risk	0.025	0.023	0.376
mean of equity issuance/asset ratio	0.012	0.033	-0.226
leverage-asset slope	0.043	0.050	-0.431
mean entrant growth	0.242	0.246	-0.140
entrant relative size	0.201	0.216	-0.150
mean of interest spread	0.55%	0.17%	3.322
stdv. of interest spread	0.44%	0.12%	3.011

B. Parameter estimates

γ_e	ζ	ξ	k_0	ϕ	ρ_v	$\overline{\sigma}$	μ	A	μ_θ	ρ_θ	φ_θ	f_e	ω
0.0662	0.6438	0.0089	0.0038	0.1380	0.9005	0.0320	3.0210	18.6004	0.0008	0.0354	0.5489	0.0043	2.9712
(0.0117)	(0.0321)	(0.0015)	(0.0005)	(0.0109)	(0.0231)	(0.0101)	(0.3999)	(1.9251)	(0.8621)	(3.3379)	(5.0125)	(0.0011)	(0.3577)

Notes: Calculations are based on an annual sample of shadow banks from the Bankscope and Worldscope databases. The estimation is done with SMM, which chooses model parameters by matching the moments from a simulated panel of banks to the corresponding moments from the data. Panel A reports the simulated and actual moments and the clustered t-statistics for the differences between the corresponding moments. Panel B reports the estimated parameters, with clustered standard deviation in the parentheses. μ_θ, ρ_θ, and φ_θ govern the mean, persistence, and standard deviation of the systemic risk process.

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